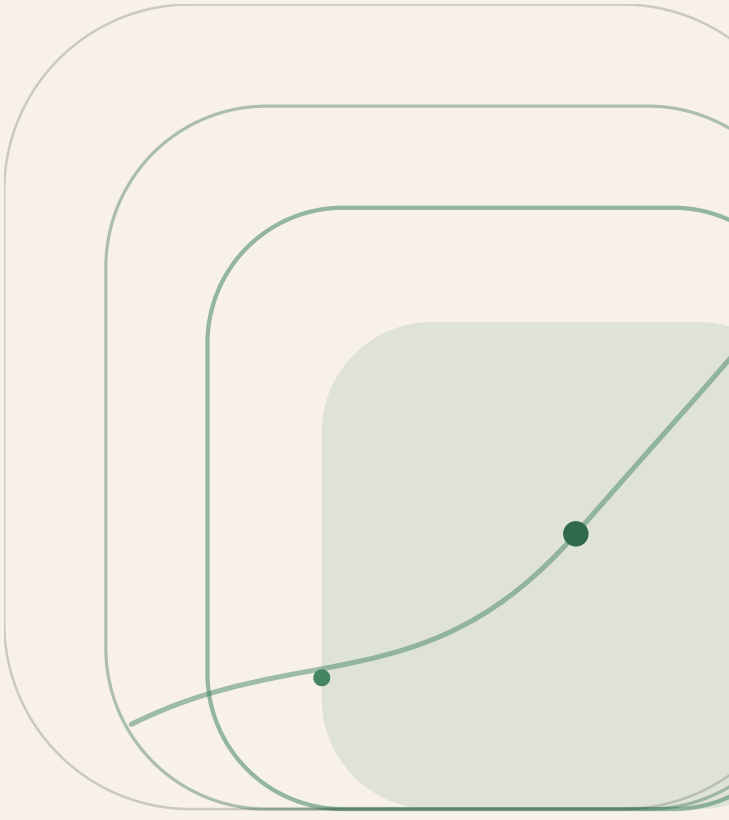
Greecon
SH.P.K

DEPLOYMENT GUIDE

Complete Deployment Guide

Hardware and software setup for every kind of site, from the first call to day 90, in one book.

FOR	The founder, the project manager and the electrician
PRINT	Keep one copy in the office. For a job, print the pack or guide that fits it (PDFs 01 to 09)
COVERS	Before you begin · platform go-live · the project manager's job · every hardware type · the four site packs · commissioning and handover · runbooks · forms, words, symptoms, shopping lists, contacts



Version 1.0 · 2 November 2026

Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

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PART 1

Before you begin

This book takes the platform from nothing to live, and a customer from the first call to day 90. Three people use it: the founder, the project manager and the electrician. Read this part first. It says who does what, how the pieces fit together, which pack goes with which site, and what must be ready before the first customer.

1.1 Who does what

Who	Does	Never does
The founder	Puts the platform on the internet, once (Part 2). Gives the project manager a login. Owns every incident about the platform or about security. Reads the morning check when the project manager is away. Meets the customer at day 60 and day 90	Hands out hardware, or installs anything at a customer, before the go-live checklist is ticked
PROJECT MANAGER	Surveys, prepares the customer in Greecon and the box at the office (the kit's numbers and the converter are set there too), installs the low-voltage side with the electrician, works down the setup list, commissions, hands over, and looks after the site (Parts 3 and 6)	Works inside a distribution board. Touches the solar panels' wiring, a tank that must be entered, wastewater or chemicals. Promises a saving. Says Greecon is a safety system
ELECTRICIAN	All the mains-voltage work: meters' voltage taps and current clamps, the pump's run contact, the 4 to 20 mA loops and their 24 V supply, anything inside a distribution board, under the customer's lock-out procedure	Works on a live conductor for Greecon's sake. Connects anything of Greecon's to a control circuit
CUSTOMER	Names the people who are told, gives access and a socket that stays on, starts and stops their own equipment, signs the handover	

The colours used for these people in every step table are the same all through the book: black for the electrician, green for the project manager, sand for the customer.

STOP

Five rules that never bend (section 3.1 has all six).

1. **Greecon only reads.** It never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure. The project manager watches and writes it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only.**
4. **Never touch the solar panels' wiring (DC),** a tank you must climb into, wastewater or chemicals.
5. **Greecon is not a safety system.** The float switches, relief valves and trips already in the customer's equipment stay the safety layer.

1.2 How the pieces fit together

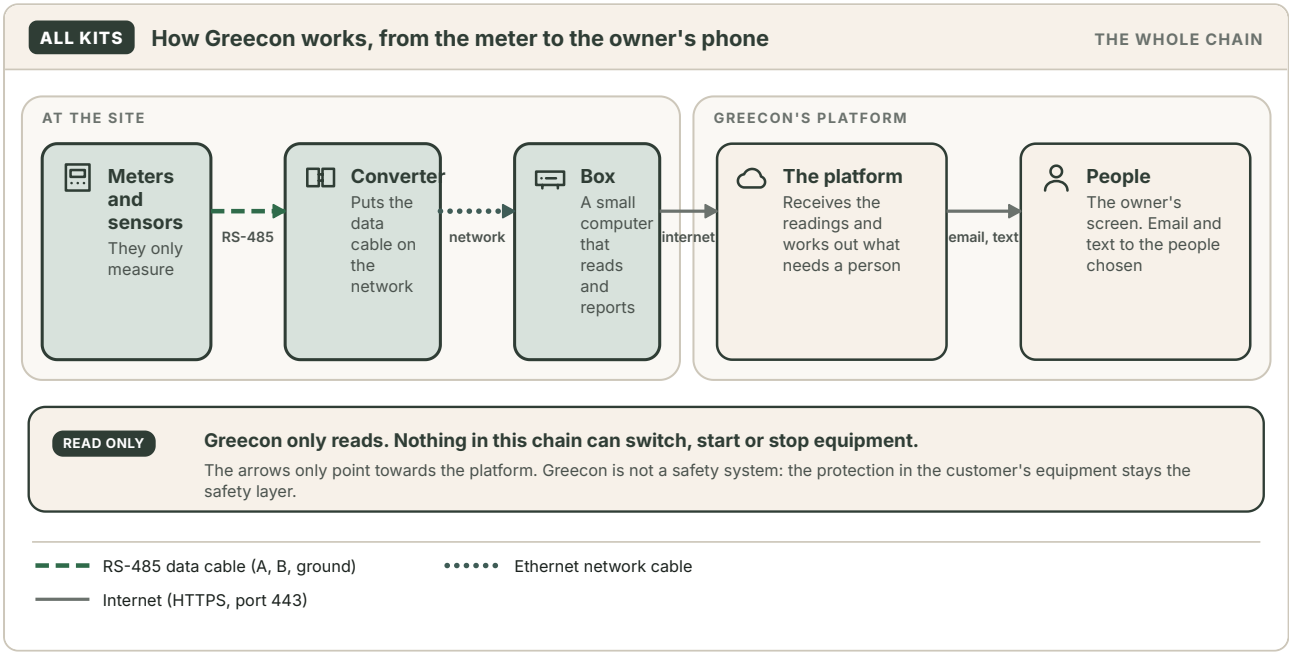


Figure 1-1. How Greecon works: meters and sensors, the converter, the box, the internet, the platform and the people who are told. Everything points one way.

Meters and sensors at the site are read by a small computer, the **box**. The meters and modules share one two-wire cable (RS-485); a **converter** puts that cable on the network so that the box can reach every device. Nobody types an address: when a person presses **Find equipment** on the setup list, the box looks at the site's network and the platform fills in where each device is, asking the person only where it is not certain. The box sends what it reads to the platform over the internet, and keeps about three days of readings if the internet is down. The platform turns readings into a picture of the site and, when something needs attention, tells the people on the list. Everything points one way: Greecon reads, and nothing it does can switch a pump, an inverter or a valve.

1.3 The path of a customer

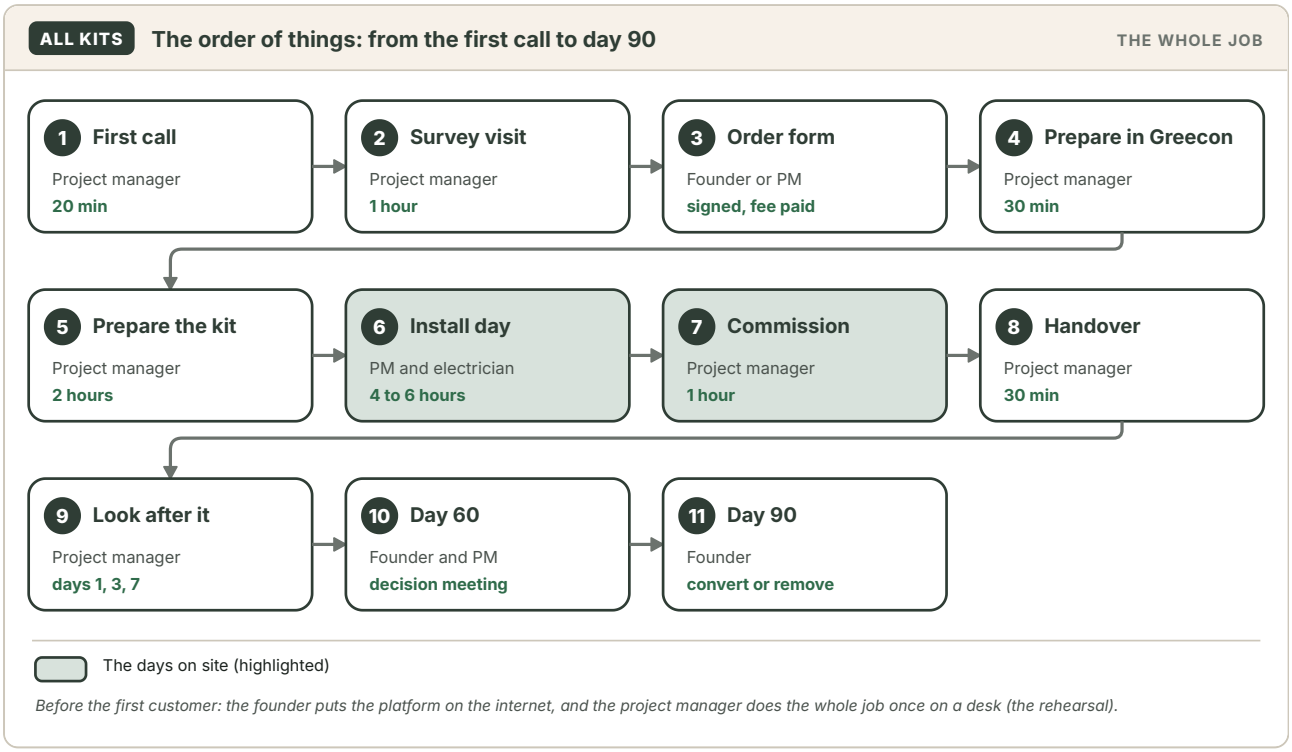


Figure 1-2. The order of things: eleven stages from the first call to day 90. The two highlighted stages are the days on site.

Section 3.3 has the table of the eleven stages with who does what and how long each takes. The two days on site are the install day and the commissioning; everything else is at the office, at the telephone, or a five-minute morning check.

1.4 Which pack for which site

If the site has	Use	Template in the platform	The pack is
A water tank and pump, solar on the roof and a grid connection (the usual first site)	**Kit H**: Hotel or guesthouse	Hotel or guesthouse	Part 5, PDF 04
Solar and the grid, with or without a battery	**Kit S**: Solar plant with grid connection	Solar plant with grid connection	PDF 05
Soil and weather at a field, with the water supply behind it	**Kit F**: Farm or greenhouse with irrigation	Farm or greenhouse with irrigation	PDF 06
A tank, a pump and the pipe it feeds	**Kit W**: Water tank and pump station	Water tank and pump station	PDF 07
None of these	Ask the founder: the platform has an empty template, to be used only when nothing else fits		

The Hardware Reference (Part 4) describes every part of every kit, and Appendix D has the shopping lists with room for three quotes.

1.5 Before the first customer

Tick these before the first customer is surveyed. Nothing here needs a technician.

- ☐ **The platform is live,** and the founder has ticked every line of the go-live checklist (Part 2, the last page).
- ☐ **The project manager has a login** of their own and has chosen their own password.
- ☐ **The rehearsal is done** (section 3.11, record A.5): the whole job on a desk, with a laptop standing in for the kit, and once more with the first real kit in its place. It finds the mistakes in the room instead of at a hotel.
- ☐ **One full kit has been bought and bench-tested.** Change the specification, not the kit, when something fails. Quotes for every part are in the shared drive.
- ☐ **The spares kit is complete,** and the project manager's tools are in the car (Appendix D).
- ☐ **An electrician is named, and a second one,** and the contact sheet is filled in (Appendix E).
- ☐ **The pilot order form and the discovery questions** are to hand (the Sales kit, document 04, is not part of this book).

1.6 How to read this book

The set. The same text is printed in ten PDFs, so that each reader prints only what they need. A part has the same number and the same words in every PDF it appears in.

PDF	For	Print
00 Complete Deployment Guide	Everyone	Once, to keep in the office
01 Founder Go-Live Guide (Part 2)	The founder	Once, before the first customer
02 Project Manager Field Manual (Parts 3 and 6, the runbooks a project manager uses, the appendices)	The project manager	Once, and the forms for each site
03 Hardware Reference (Part 4)	The project manager and the electrician	Once, for the spares kit
04 to 07 Site packs H, S, F and W (Part 5)	The project manager and the electrician at that kind of site	The whole pack, for each site
08 First Farm: Start Here	A project manager who has just joined: what the job is, who does what, the nine steps of a first farm, the electrician's brief and the farm pack	Once, for the first farm
09 Electrician's Brief: Farm	The electrician who fits a farm kit	Send it two or three days before install day

Marks. A box to tick is a real square, to be ticked in pen. Shaded boxes: **Do** (do this), **Check** (check this), **Stop** (stop, and call the electrician where it says so), **Tip** and **Say** (words for the customer). Every drawing is labelled with its kit (Figure H-2 is the wiring of kit H) and shows how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers.

Prices are planning estimates in EUR, excluding VAT and labour, from public shop prices seen in October 2026. They are not quotes. Replace them with real quotes in the first week.

Screen names follow the web app as redesigned in October 2026: Greecon's team has **Today**, **Clients**, **Requests**, **Team** and **Settings**; a customer has **Overview**, **Sites**, **Alerts**, **Reports** and **Settings**.

What this book is made from. The launch documents in `docs/launch/` and the platform's own code; it is rebuilt with `npm run docs:pdf`. If something in it does not match what you see, fix the source and rebuild, and do not mark up the PDF. The twelve setup-list step titles are checked against the platform at every build. A maker's terminal numbers, register tables and menus are in the maker's manual that came with the part.

PART 2

Platform go-live: the founder

For: the founder, once. **Takes:** three to four hours the first time, all of it clicking and pasting. **You need to know nothing technical.** Every step says what you should see when it worked, and what to do if you do not see it. Where a button has a different name today, look for the one that means the same.

When this is finished, a prospect can open the demo, a customer can sign in, a project manager can set a site up, a box at a hotel can report in, and a person's phone rings when a tank runs dry.

Read Launch decisions (document 27) first if you want to know why things are as they are.

What you need before you start

You need	Why	Where
A GitHub login that can see `greecon-tech/app`	The platform is built from it	github.com
A credit card for Railway	Hosts the platform and its database	railway.com
Access to where the `greecon.earth` addresses are managed (the registrar or DNS login)	Two web addresses and email proof need adding there	wherever you bought the domain
An email address that can receive mail from `@greecon.earth` (`info@greecon.earth`)	New requests for access, and replies to alert emails, go there	your mail
A password manager	Every password and key below goes in it, not in a note or a chat	1Password, Bitwarden or similar
A phone number you can receive texts on	To test that texts arrive	yours
About EUR 40	First month of hosting and EUR 20 of text-message credit	card

Keep one document (in the password manager) called **Greecon platform**. Every key you create below goes in it the moment you create it. If you lose one, nothing is broken: you make a new one and paste it in again.

Step 1 Three addresses, decided

These are decided so that you do not have to think about them. The second one matters most.

What	Address	Why
The web app (customers, staff, the project manager)	`https://app.greecon.earth`	What people type and bookmark. The public website, greecon.earth, already has a button to exactly this address
The platform's own address, which every box at every site reports to	`https://api.greecon.earth`	**A box remembers this address for as long as it lives.** If it ever changes, someone has to visit every box. Choose it once, here, and never change it.
Where alert emails come from	`alerts@greecon.earth`	Replies go to `info@greecon.earth`

Step 2 Railway: the project and its database

1. Sign in at railway.com with GitHub. Add the card.
2. **New Project** → **Deploy from GitHub repo** → choose `greecon-tech/app`. If it asks which branch, choose `main`. Railway will try to build something and probably fail the first time. That is expected: it does not know yet which of the two programs to build. Continue.
3. In the project, click **+ New** → **Database** → **Add PostgreSQL**. Railway creates the database.
   - This platform needs PostgreSQL 14 or newer (Railway's current default is newer than that).
4. **Region:** where Railway offers a choice (project or service **Settings**), choose a **European** region for the services and the database, so that customers' data stays in Europe. A database's region is fixed when it is made, so choose before you add it.

You should see: a project with two boxes, your repository's service and `Postgres`, with Postgres showing as active. Rename the repository's service to **api** (click it, Settings, Service Name). The names matter below.

Step 3 Railway: the API service

Click the service you named `api`.

Settings → **Build** → **Dockerfile Path:** `infra/docker/api.Dockerfile`

Settings → **Deploy** → **Healthcheck Path:** `/health` (this lets Railway see that a new version is alive before it switches to it).

Variables. Add these one at a time (use **Raw Editor** if Railway offers it and paste the block). The first column is the name, exactly as written.

Name	Value	What it does
`DATABASE_URL`	`${{Postgres.DATABASE_URL}}`	Type it exactly like that, with the braces. Railway fills in the database's address.
`JWT_SECRET`	a long random value, at least 32 characters	Signs everyone's session. Make one in your password manager ("generate password", 48 characters) and paste it. **If it is short or missing, the platform refuses to start, on purpose.**
`PORT`	`4000`	The door the platform listens on
`TRUST_PROXY`	`1`	Lets the platform see each caller's real address behind Railway
`PUBLIC_API_URL`	`https://api.greecon.earth`	The address written into every box's install command
`PUBLIC_WEB_URL`	`https://app.greecon.earth`	The links in each alert message: to the alert, and to say it has been seen
`BOOTSTRAP_ADMIN_EMAIL`	`eridon.manuka@greecon.earth`	Your own login, created on the first start. **Removed in step 6.**
`BOOTSTRAP_ADMIN_PASSWORD`	a password of at least 12 characters, not used anywhere else	Your first password. **Removed in step 6.**
`SALES_NOTIFY_EMAIL`	`info@greecon.earth`	Who is emailed when someone asks for access (the first ten requests of a day; every request is kept under **Requests**, and a day with more says so in one email)

Messages are added in steps 7 and 8. Do not set `ENABLE_API_DOCS`, `AUTO_MIGRATE` or the old `EDGE_INGEST_*` variables.

One setting is optional and best left alone: `RESTART_GRACE_MINUTES` (default `10`, from 0 to 60). After the platform itself restarts (a deploy, a crash), the boxes' last words are older than usual for a few minutes because the platform was not listening, not because they are silent. For this many minutes after it starts, nothing that comes from *missing* readings (a box silent, a reading out of date) is raised or messaged, and the screens say "Greecon was just restarted; boxes are reconnecting". A reading that is measured and dangerous (a tank nearly empty) is never held back by it. `0` switches the wait off.

Settings → Networking → Generate Domain gives the API a temporary Railway address (ending `.up.railway.app`). Use it only to check that the API is alive, below. Then **Custom Domain →** **`api.greecon.earth`**: Railway shows a record to add to your DNS (step 5).

You should see: after a minute or two, in the service's **Deployments → View logs**, a line such as `Database updated: 19 migration(s) applied, 0 recorded as already applied.` (on later deploys, `Database is up to date.`), then `Nest application successfully started`. Also, the first time:

```
Password set for eridon.manuka@greecon.earth from BOOTSTRAP_ADMIN_PASSWORD. Sign in, change it in Settings, then remove both BOOTSTRAP_ADMIN variables.
```

(On a database that never had your account the line says "Administrator … created from BOOTSTRAP_ADMIN_PASSWORD" instead; either means it worked.) Open `https://<the temporary address>/health` in a browser: it must show `"status":"OK"` and `"database":"connected"`.

Warnings in the log are information, not failures. The platform lists what is not set up yet (messages are off, and so on). The log line "The deployment's settings are not safe to start with" is a failure: it names the setting to fix. The usual cause is a `JWT_SECRET` that is too short.

Step 4 Railway: the web service

1. **+ New → GitHub Repo →** `greecon-tech/app` again, same branch. Name it **web**.
2. **Settings → Build → Dockerfile Path:** `infra/docker/web.Dockerfile`
3. **Variables:** one only.

Name	Value
`GREECON_API_URL`	`http://${{api.RAILWAY_PRIVATE_DOMAIN}}:4000`

   (If you did not name the first service `api`, put its name in place of `api`.) This is the private road between the two services. The API does not need to be reachable by the web app through the internet.
4. **Settings → Networking → Generate Domain**, then **Custom Domain →** **`app.greecon.earth`**.

You should see: the web service deploys and its temporary address shows a sign-in page titled "Greecon Platform".

Step 5 DNS: the two addresses

At wherever `greecon.earth` is managed, add the two records Railway showed you (one for `api`, one for `app`). They are usually `CNAME` records pointing at an address Railway gives. Railway shows a green tick when it sees them, and then issues the padlock (HTTPS) by itself. This can take from a few minutes to an hour.

You should see: `https://api.greecon.earth/health` shows `"status":"OK"` and `"database":"connected"`, and `https://app.greecon.earth/login` shows the sign-in page with a padlock.

If it never turns green: the record was added to the wrong name (it must be `api` and `app`, not `api.greecon.earth.greecon.earth`), or the DNS host puts a proxy in front of it. Switch any proxy or "orange cloud" off for these two names.

Step 6 First sign-in, and removing the starting password

1. Open `https://app.greecon.earth/login`. Click **Greecon team**. Enter `eridon.manuka@greecon.earth` and the starting password. **Continue.**

2. You land on **Today**, Greecon's own start page. With no customer yet it says *No customer site is live yet.* If you see it, the platform works end to end.
3. Open **Settings**, then its **Account** tab, and under **Change your password** use **Change password** to choose your own, long and not used anywhere else. Put it in the password manager.
4. **Remove the two `BOOTSTRAP_ADMIN_` variables** from the API service in Railway. Leaving them is not dangerous (they only fill a gap and never replace a password), but there is no reason to leave a password in a settings page.
5. Give the project manager a login of their own: open **Team**, and under **Greecon team** enter their name and email and press **Add to the team**. A temporary password is shown **once**. Tell it to them by phone or in person, not by email. They sign in with **Greecon team**, and are asked for a password of their own straight away.

Only the owner (you) can add or remove team members. A project manager can see the team and cannot change it.

Step 7 Email (Resend)

Email is how nearly every message reaches a person: Act now, Check today, the morning message, and the note that tells you someone asked for access.

1. Sign up at resend.com (free plan: 100 emails a day, 3,000 a month, one domain: enough for the first five sites).
2. **Domains → Add Domain →** `greecon.earth` (or a subdomain such as `alerts.greecon.earth`, which keeps this mail apart from your own). Resend shows several DNS records (they prove the mail really comes from you, which is what keeps it out of spam). Add every one of them where `greecon.earth` is managed, then click **Verify**. It can take up to an hour.
3. **API Keys → Create API Key →** "Sending access", for that domain. Copy it (it begins `re_`) into the password manager.
4. In Railway, on the **api** service, add:

Name	Value
`RESEND_API_KEY`	the key
`ALERT_FROM_EMAIL`	`alerts@greecon.earth` (or `alerts@alerts.greecon.earth` if you used the subdomain)
`ALERT_FROM_NAME`	`Greecon`
`ALERT_REPLY_TO`	`info@greecon.earth`

   Railway redeploys the service.
5. **Test it:** sign in, open **Settings** and its **Who is told** tab, add yourself with your email, **Save the list**, then **Send a test message**. It should say "Accepted for delivery" and the email should arrive within a minute. Look in spam too. If it is in spam, mark it "not spam" and ask the Resend support page about your domain's setup (a missing record is the usual reason).

When the free plan is not enough: a hundred emails a day is reached at about fifteen sites. The paid plan is about USD 20 a month for 50,000. Nothing else changes.

Step 8 Text messages (Twilio)

Text messages are for **Act now** only: a tank nearly empty, a pump running with no water moving (where there is a flow meter). The first person is texted ten minutes after the email if nobody has acknowledged the alert (by pressing **Acknowledge** in Greecon: replying to the text does nothing), and the next person is emailed and texted thirty minutes after the alert began, and so on down the list.

1. Sign up at twilio.com. Verify your own phone. Add about EUR 20 of credit (a text to Albania costs a few cents; check Twilio's price page for Albania).
2. You need something to send *from*. Two ways; choose one:
   - **A name:** `Greecon`. Twilio lists Albania as a country where a sender name can be used without registering first, and a name is what Albanian recipients are used to seeing. A text sent from a name cannot be replied to. Some Albanian operators have been reported to replace a name with a generic one.
   - **A number:** buy a Twilio number that can send to Albania. Replies can be received, and the sender is the same everywhere. **Test before you decide**, with phones on all four networks (One, Vodafone, Telekom and Plus, whichever your customers use).
3. In Railway, on the **api** service, add:

Name	Value
`TWILIO_ACCOUNT_SID`	from the Twilio dashboard (begins `AC`)
`TWILIO_AUTH_TOKEN`	from the Twilio dashboard. Treat it like a password.
`TWILIO_FROM`	`Greecon`, or the number in international form, such as `+1...`

4. **Test it:** on **Who is told** (in **Settings**), add yourself with your mobile number *with the country code* (`+355 69 ...`), save, and **Send a test message**. A text should arrive within a minute. Ask three people on other networks to try it too.

If the text does not arrive: Twilio's console shows each message and why it was refused (**Monitor → Logs → Messaging**). The commonest reasons are a number not written with its country code and a sender name a network does not accept. **What was sent**, under **Who is told**, shows what Greecon asked Twilio to do.

You can launch without texts. Then only email carries Act now. Say so to customers, and do it quickly: texts are the reason a manager wakes up at night. The setup checklist will not let a site be commissioned until at least one of the two ways works.

Step 9 Greecon's own morning check

Greecon must hear that a customer's box has gone quiet before the customer does.

1. Sign in as yourself (**Greecon team**), open **Settings**, then **Who is told**. This is Greecon's own list.
2. Add yourself and the project manager, each with an email address. **Save the list.**
3. Leave the morning message on. It is sent at 07:30, Albania time (change it under the list if you prefer).

Each morning you get one email: “Greecon, Mon 19 Oct: Stable. All 3 sites are reporting and nothing needs doing.” or “Attention needed. 1 of 3 sites need a look.” with the reasons in the line for each site. The same list is on **Today**, under **Needs a look**, worst first; the sites that are fine are folded away below it, and each live site shows how many days it has been live (*Day N*). The runbook (Part 7) says what to do.

Step 10 The demo

Two demos, for two uses.

The one that is always there, with no sign-in: `https://greecon-tech.github.io/app/`. A frozen picture of the product with simulated data, one page per kind of user. To switch it on once: on GitHub, in `greecon-tech/app`, **Settings → Pages → Build and deployment → Source: GitHub Actions**. After the next change to `main`, it publishes by itself. Use this for the link in an email.

The live one, for a meeting: sign in at `https://app.greecon.earth/login` with **Client login**, email `demo@greecon.earth`, password `demo123`. It shows three simulated sites, a farm, a water site and a solar-plus-battery site (there is no hotel in it), that move as the day goes (the solar picture follows the sun). It holds nothing real, it can only look and try, and it cannot send anyone a message. This login is an **operator**: it has no **Who is told**, and its **Reports** shows **Trends** only, with no **Downloads**, so those two are shown on your own rehearsal client instead (the Sales kit (document 04, not part of these guides), section 4). The password is public on purpose; do not give it to a customer as their own login.

Step 11 Watching it

Do this	How	Why
An outside check that the platform is up	Make a free account at uptimerobot.com or betterstack.com. Add a monitor for `https://api.greecon.earth/health` every 5 minutes that looks for the word `connected` (a "keyword" monitor), alerting your email and phone: it then also notices a database that has gone. Add one for `https://app.greecon.earth/login`.	If the whole platform is down, nothing inside it can tell you. This is the only thing that can.
Backups	In Railway, open the Postgres service's **Backups** tab and turn on a daily one (what your plan allows: Railway's scheduled backups may need the paid Pro plan, which is worth it once a customer is live). In addition, once a week, take a copy off Railway: Part 7, "Backups and restoring".	A backup that only lives next to the database does not survive losing it.
Calendar reminders	Today plus 6 months: re-check the legal list (document 22). 30 September 2027: renew `security.txt` (it carries an expiry date).	Things that expire quietly
Look at the cost	After a week, Railway's **Usage** page. Expect about EUR 20 to 40 a month.	No surprises

Step 12 The rehearsal, then the checklist

Before the first real customer, the project manager does the whole job once on a desk, with a laptop standing in for the kit: Part 3, "Rehearsal". It takes an afternoon, and it finds the mistakes in the room instead of at a hotel.

If something goes wrong

What you see	What it means	What to do
The API's log says "The deployment's settings are not safe to start with"	A setting is missing or weak	Read the line above it: it names the setting. Usually `JWT_SECRET` shorter than 32 characters.
The web app shows an error on every page	The web app cannot reach the API	Check `GREECON_API_URL` on the **web** service, and that the API service is running
Sign-in says "Invalid email or password"	Wrong password, or the account was turned off	Check the password manager. If it was lost: set `BOOTSTRAP_ADMIN_EMAIL` and `BOOTSTRAP_ADMIN_PASSWORD` again **only if the account has no password**; otherwise ask another owner to reset it, or see Part 7, R14 (its last point).
Everyone is suddenly sent to the sign-in page	`JWT_SECRET` was changed, which signs everyone out	Nothing is broken. They sign in again.
The install command on a site's setup page starts with `http://` and a private name instead of `https://api.greecon.earth` (the API's log also warns of it at start-up)	`PUBLIC_API_URL` is missing	Set it (step 3), redeploy, then make a new code
Emails say "not sent: email is not set up"	`RESEND_API_KEY` is missing or the sender is wrong	Step 7. The API's log names the reason.
Railway shows the service crashed and restarting	Something in the log	Open the logs and read the last red line. Part 7, "The platform is down".

Changing things later

- **Rotate the signing secret** (if you think it leaked, or when a team member leaves who knew it): change `JWT_SECRET` on the API. Everyone is signed out and signs in again. Nothing else changes.
- **Go back to the previous version:** Railway, the service, **Deployments**, the earlier one, **Redeploy**. The database is not rolled back; new versions only add to it, so the old version still works with it.
- **Hold every box on the software it has** (while a problem is investigated): set `EDGE_AUTO_UPDATE` to `false` on the API. Boxes then stop taking new versions until it is removed.
- **Turn the live demo's simulated feed off:** set `DEMO_FEED` to `false`. Only the demo account changes.
- **Change how long Greecon waits after a restart before it calls a box silent** (ten minutes unless you say otherwise, so that a deploy does not alarm anyone while the boxes reconnect): set `RESTART_GRACE_MINUTES` to a whole number of minutes from 0 to 60; 0 switches the wait off. Leave it unset unless there is a reason. What the wait does: Part 7, R7 (point 6).
- **Move to Google Cloud:** when this outgrows Railway, see Deployment on Google Cloud (document 10). The services are ordinary containers; nothing here ties them to Railway.

What not to do

- Do not give a customer the demo login, and do not share your own login. Everyone has their own.
- Do not change `api.greecon.earth` once a box exists.
- Do not put a key or a password in an email, a chat or a document that is not the password manager.
- Do not hand out hardware or install anything at a customer before the checklist above is ticked.

TEAR OFF AND KEEP

Go-live checklist

Tick each one yourself; none needs a technician.

- [ ] `https://api.greecon.earth/health` shows `"status":"OK"` and `"database":"connected"`, and `https://app.greecon.earth/login` shows a padlock
- [ ] I signed in with **Greecon team** and saw **Today**
- [ ] The two `BOOTSTRAP_ADMIN_` variables are gone, and my password is in the password manager
- [ ] The project manager signed in with their own login and chose their own password
- [ ] A test email reached my inbox (not spam), and says it is from `alerts@greecon.earth`
- [ ] A test text reached my phone and the phones of three people on other networks (or I have decided to launch with email only, and said so)
- [ ] **Who is told** in Greecon's own account has me and the project manager; tomorrow morning I get the morning check
- [ ] `https://greecon-tech.github.io/app/` opens the demo
- [ ] The outside check is watching both addresses and has my phone number
- [ ] A backup exists (the Backups tab shows one, or I took one by hand and put it in the shared drive)
- [ ] `https://app.greecon.earth/privacy` opens without signing in, and says what Greecon keeps (the lawyer reads it before the first customer signs)
- [ ] I sent myself a "Request access" from the sign-in page and it arrived at `info@greecon.earth` and appeared under **Requests** (and in **Requests waiting** on **Today**)
- [ ] The rehearsal was done, and its notes are fixed or written down

Signed: _____ Date: _____

PART 3

The project manager's job, first call to day 90

For: the project manager who surveys, installs, commissions and hands over each site, and then looks after it. **Read with:** Part 4 (what to buy and how it is connected) and Part 7 (when something is wrong).

You are not expected to be an electrician or a programmer. You are expected to be careful, to write things down, and to say "I do not know yet" instead of guessing. Greecon is built so that nearly everything you do is a form with a plain-words check beside it, and so that the platform tells you when something is not done.

For the equipment in Greecon's kit you type no IP address, no port, no unit number and no register: the kit is made of approved models, their numbers are set at the office, and the box finds them on the network. What you give is what only a person can know: the range printed on a label, who signed a paper, who the electrician was, who is told. Equipment Greecon does not sell (a soil probe, a weather sensor, a battery) is still entered by hand, and section 3.8 says where.

3.1 The rules that never bend

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit. Nothing in this platform can command equipment, and nothing you install may be made able to.
2. **Anything inside a distribution board is the electrician's work.** Clamping current transformers around live conductors, taking voltage for a meter and connecting a pump's run contact are done by a qualified electrician, under the customer's lock-out procedure, with you watching and writing it down. You do the low-voltage part: the box, the network, the converter, the data cable (you run it to the board; the electrician connects its ends at the meters, because the meters' terminals are inside the board), the labels, and the 12 or 24 volt supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with the 24 V supply that powers them, as the module's manual draws it.
3. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater, or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
4. **Greecon is not a safety system.** Say so every time, in the words in the sales kit. The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
5. **Never promise a saving, a percentage or "AI".** Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim.

6. **Write everything down at the time.** Photographs of every meter's label and display, the wiring before you close the board, the converter's settings, the box's label.

3.2 What you work with

Thing	What it is
Your login	Your own, from the founder. Sign in at `https://app.greecon.earth/login` and choose **Greecon team**. You are asked for a password of your own the first time. Never share it.
Today and **Clients** (the console)	**Today** is where you start: the sites that need a look, the requests waiting and the sites being set up. **Clients** is where you onboard a customer and open their account to set it up. **Requests** lists who has asked for Greecon, and **Team** the people at Greecon who can open a customer's account.
Search	**Ctrl+K** (**⌘K** on a Mac), or **/**, or the **Search** button at the top of the left bar (on a phone, the magnifier at the top right) opens one box. Type a few letters of a client's name to open it, of a request to see who asked, or of a page; inside a customer's account, of a site. **Enter** opens the first result, **Esc** closes.
The setup list	One page per site (`Set up <site name>`) that shows up to twelve steps (a step that does not apply to the site is left out), what is done, what is next, and what to do. Most steps turn **Done** by themselves when the platform sees them work. A few only a person can check, and some of those ask you to write who and when: the papers, who is told, the direction of the grid meter, the limits, the electrician and commissioning.
The install card	One page for each site, linked at the top of the setup list (**The install card**): what to ask the customer and look at before the day, what to bring, what each device is set to and its number on the cable, how it is wired, the converter's settings, the cable rules, what the customer's network is asked for, the visit in order, and three messages to copy to the customer. Print it for the visit.
Approved equipment	The meters and modules in a kit are approved models: Eastron SDM630 and SDM230 meters, a Waveshare Modbus RTU Analog Input 8CH for the 4 to 20 mA transmitters, and a Waveshare Modbus RTU IO 8CH for the pump's run contact. Greecon knows their registers and settings, so nobody types them.
The box	A small computer at the site. It reads the meters and sensors and reports to Greecon. One pasted line installs it. It is also what looks for the equipment on the network when you press **Find equipment**.

Thing	What it is
The converter	A small unit that puts the meters' cable (RS-485) on the network so that the box can reach them. Greecon sets it at the office.
The handover sheet	One printable page per site: where everything is, who is told, who signed the papers, who the electrician was, who commissioned it and when.
The pretend kit	A program for your laptop that pretends to be a whole kit (two meters, the 4 to 20 mA module and the pump contact module), for practice: Rehearsal (section 3.11).

3.3 The whole path of a customer

Stage	What happens	Who	Time
1. First call	Discovery questions (the Sales kit (document 04, not part of these guides)). Decide: worth a visit?	You	20 min
2. Survey visit	Look at the site, fill in the survey form, take photographs, agree what is measured	You (and the electrician if the plan needs one)	1 hour
3. Order form	The customer signs the pilot order form (the Sales kit (document 04, not part of these guides)) and the data terms are agreed; the pilot fee is paid. Nothing goes into the customer's building before both papers are done	Founder or you	
4. Prepare in Greecon	Onboard the customer, make the site from the template, write down that the papers are signed, add the box (section 3.5 below)	You, at the office	30 min
5. Prepare the kit	Install the software on the box at the office, set the unit numbers and the converter, test the kit on the desk, label it, pack it	You	2 hours
6. Install day	Mount, wire, connect, power up; check the labels; press **Find equipment**	You and the electrician	4 to 6 hours

Stage	What happens	Who	Time
7. Commission	Give the ranges, check every number against the real world; confirm the steps only a person can (the electrician's name and licence number among them); commission	You	1 hour
8. Handover	Print the sheet, show the owner the morning message and an alert, sign	You	30 min
9. Look after it	Day 1, 3 and 7 checks, then the morning check (Part 7)	You	5 min a day
10. Day 60	Decision meeting with the owner, against the three written goals	Founder and you	1 hour
11. Day 90	Convert (order hardware, start the monthly fee) or remove the kit	Founder	

3.4 Survey visit

Bring: this manual, the survey form below, a phone for photographs, a tape measure, a torch, a notebook. No tools are needed to survey; do not open anything.

Ask, in this order (the answers decide what is measured and what it costs):

1. Who decides, and who gets called at night? (Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address: see 3.8.8.)
2. What went wrong in the last twelve months that cost money or a guest? (Write it in their words. It is the reason they buy.)
3. The last twelve electricity bills (price per kWh, kWh bought each month) and the water bills. (The platform shows what solar kept on site at *their* price; without a price it shows kWh only.)
4. Is there solar? What brand is the inverter and where is its AC output? Is it allowed to send energy to the grid?
5. Where does the supply enter the building? Is it three-phase?
6. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Has the starter a spare "running" contact? (Without one there is no pump contact module, and "the pump is running but no water is flowing" cannot be raised.)
7. Is there a network socket in the plant room, or cellular signal? (Look at your phone there.)
8. Is there a power socket for the box, and is it on a circuit that stays on?
9. Who is the customer's electrician, and can they be there on install day?

Look at, and photograph:

- The supply entry and the main board (closed): where a meter could be fitted with the clamps around the incoming phases
- The inverter and its AC output cable
- The tank, the pump, the starter, any existing level or pressure gauge and flow meter
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable)

Go or no-go. Go if: there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called. **No-go, politely, and write why,** if: the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

Survey form (one per site; a page in the shared drive):

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Template: Hotel / Solar and grid / Farm / Water station	
Supply: single or three phase; main breaker size (A)	
Inverter brand, size (kW), AC output cable location	
Tank: height (m), water source, pump (kW), starter location, spare running contact on the starter	
Meters needed (grid, solar, load, pump) and where	
Sensors needed (level, pressure, flow, pump running) and where	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	

3.5 Prepare in Greecon (at the office)

You can do all of this before you set foot on site, and it makes the install day shorter.

1. **Sign in** with **Greecon team**. You land on **Today**.

2. **If the customer asked through the web page:** their request is waiting under **Requests waiting** on **Today**, and on the **Requests** page. Press **Onboard** (it opens **Clients** with the form filled in from their request). **Otherwise,** open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email. Press **Onboard client**.
3. A box appears: **Client onboarded — save this password now.** That is the owner's temporary password. It is shown **once**. Write it on the handover page, or tell it to the owner by phone. Do not send it by email. The owner signs in at `https://app.greecon.earth/login` with **Client login**, and chooses a password of their own straight away.
   - If a message says someone already has an account with that email, the person is already a customer. Do not use a different address to get round it: add the new site to the customer they already have. (The rehearsal in section 3.11 is the one exception.)
4. In the customer's row on **Clients**, press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page. A banner appears at the top: *You are in as Greecon staff, setting it up.* Everything you do from now is recorded under your own name in their audit trail. You can see their people but cannot add people to their account or reset their passwords while inside; their owner does that. **The setup session lasts four hours.** After that the page sends you to the sign-in page: sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there; only what you were in the middle of typing is lost. On install day, open it again when you start commissioning.
5. On **Sites**, press **Add a site** (top right; in the middle of the page while there is no site yet). The form, *New site from a template*, makes the site. Choose **What kind of site**:
   - **Hotel or guesthouse**: water tank and pump, solar, grid. The usual first site.
   - **Solar plant with grid connection**: solar and grid, with an optional battery and load meter.
   - **Farm or greenhouse with irrigation**: soil probe and weather, with the water supply behind it.
   - **Water tank and pump station**: tank, pump and the pump's electricity.
   - **Empty site**: nothing assumed. Use only when none of the above fits.

   The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and click **Create the site and start setting it up**.
6. You land on **Set up** . It says `0 of N steps done`, and highlights what comes next: **Order form signed and data terms agreed** (3.8.1). Do it now, with the signed papers in front of you: write who signed for the customer and on what date, and press **The papers are signed**. At the top of the page is **The install card**: open it and print it. It is your list for the visit (see install day, below).
7. Click **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` ending in a code like `GRC-XXXX-XXXX-XXXX` . **Copy it** (there is a copy button). It is shown once. The code works **once**, for **seven days**. Click **I have copied it**.
   - The code is the box's key. Anyone with the line can install a box as this site until it is used or expires. Treat it like a password: not in email or chat; paste it straight into the box.
   - Lost it, or the seven days passed? On the same step, press **Install a box, or replace this one**. It asks you to confirm, then shows a new line. A new code replaces the old one (the old line no longer works), and the box that uses the new line takes the place of the box now installed, which stops working at that moment.

3.6 Prepare the kit at the office

Do the install of the box's software, and set the kit's numbers and the converter, at the office, on your desk, so that the site visit is only plugging in and checking labels.

The box. Installing the operating system and pasting the install line are still done in a terminal (a keyboard and screen, or SSH); everything after that is done on the setup page.

1. Install the operating system on the box: **Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit**, with the SSH server on (you will use it from your laptop on the site's network; nothing can reach it from the internet), one user, a strong password, and the time zone set to Europe/Tirane. (Part 4 says which box to buy.)
2. Connect the box to your office network and open a terminal on it (a keyboard and screen, or SSH).
3. Paste the install line from section 3.5, item 7. It begins with `curl` and ends with the code. **Paste it exactly.** It runs in seven numbered steps and tells you each one as it goes: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with:

   ```
   Done. Greecon can see this box. On the site's setup page, give each device its address;
   ```

   (That last line is older than **Find equipment**: on the setup page, giving each device its address is now that button, in step 4 of the setup list.)
4. In Greecon, refresh the setup page. **Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not click anything to make it so: the platform saw the box.
5. Label the box with the site name and the date, and pack it.
   - The box keeps its credential when it is moved. Unplug it, carry it to the site, plug it in: it reports from the new place without anything being typed again.

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, **Giving this box its credential**, fix the cause and run the same line again: the code has not been used yet. From step 5 on the code is used up (it works once), and the same line would stop with "Greecon did not give the software to this box". Then run the line again **with no code at the end**: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are: the box cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), or a code already used or older than seven days. A clock that is only minutes out does not stop the install: it prints a WARNING, and the setup page later says "its clock is N minutes fast".

The numbers and the converter

Every kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office. On site you only check the labels.

Device	Make and model	Its number	How it is set
Grid meter	Eastron SDM630	1	On its own display
Solar meter	Eastron SDM630	2	On its own display
Water I/O module (the 4 to 20 mA module)	Waveshare Modbus RTU Analog Input 8CH	3	With the maker's setting tool
Load meter or pump meter, if the site has one	Eastron SDM630 or SDM230	4	On its own display
Pump contact module, if the site has one	Waveshare Modbus RTU IO 8CH	5	With the maker's setting tool
Soil probe (kit F)	The maker's own	6	By the maker's instructions, before it is buried
Weather sensor (kit F)	The maker's own	7	By the maker's instructions, before it is mounted

A battery is the customer's: it is addressed by hand (for example 8). Everything on one cable is set to the same speed: **9600 baud, 8 data bits, no parity, 1 stop bit**.

- **The meters.** Set each meter's number on its own display, one meter at a time (a meter is delivered as unit 1, and two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page (Part 4, section 4.5). Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains yourself.
- **The two Waveshare modules.** Each is delivered as address 1, and its address is kept in a register: there is no switch. Set it with the maker's setting tool and a USB-to-RS-485 adapter, one module at a time: **3** on the 4 to 20 mA module and **5** on the pump contact module. Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default **0 to 20 mA** mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- **The soil probe and the weather sensor** (kit F): 6 and 7, set with the USB-to-RS-485 adapter and the maker's instructions, before the probe is buried or the sensor mounted. A second probe takes the next free number.
- **The converter.** In its web page: mode **Modbus TCP to RTU (gateway)** (not "transparent", not "raw TCP", not "Modbus RTU over TCP"), serial 9600, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site, as install day says: a fixed one is best, and Greecon finds the converter either way.
- **Label everything** with its number (the converter with its settings), and photograph the labels.

Test the kit on the desk. The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (section 3.11, step 10). Anything that differs from this manual is written down and reported to Greecon.

3.7 Install day

Before the day. Two or three days before, send the customer the first two messages on **The install card**: the first to their contact, the second to whoever looks after their network. Each is in English, with an *Albanian draft* folded under it: have someone who writes Albanian every day read the draft before you send it (the interface and Greecon's own messages are English only). The day before, check the signed papers, the electrician's number and the kit, packed from the install card's list.

Before anyone touches anything: the signed order form and the data terms are in your hand, the electrician is there, and the customer's lock-out is agreed.

Order of work. The electrician and you work at the same time, in this order, so that nobody waits.

	Electrician	You
1	Walk the site with you; agree where each meter and sensor goes; apply the lock-out	Walk with them; photograph before
2	Fit the grid meter and the solar meter in the board: voltage connections with their fuses as the meter's manual says, the current clamps around the incoming phases, **arrow towards the site**	Mount the box, the converter and the 12 or 24 V supply on a shelf or rail; run the data cable from the converter to the board (the electrician connects it at the meters)
3	Connect the pump's run contact to input 1 of the pump contact module (a dry contact, never the relay outputs); fit the level and pressure transmitters (a wet part: only the electrician or the customer's plumber) and wire their 4 to 20 mA loops with the 24 V supply in series, as the module's manual draws it: level on input 1, pressure on input 2, flow meter on input 3	Check each device's number on its label against the install card (grid meter 1, solar meter 2, water module 3, load or pump meter 4, pump contact module 5) and photograph each label. They were set at the office: do not change them
4	Remove the lock-out only when you both agree the work is closed up	Check the converter's label (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502); connect it and the box to the network; power up
5	Stay until commissioning is done	Open the customer for setup again and go down the setup list: press **Find equipment**, give the ranges, and write down the electrician's name and licence number

Numbers are set at the office, and only checked here. The numbers and the speeds were set before the kit left (section 3.6). If a label is missing, or two devices of a kind carry the same number, **Find equipment** says **Not sure which**: do not guess, and do not change a number on site. For what to do, see Appendix C and the runbook R15.

Meter settings are read from the meter's own display, never assumed. Different makers, and different versions of the same meter, are delivered with different defaults (the Eastron SDM630 is 9600 baud, and the SDM230 can be 2400). Greecon sets every meter to the same speed, and to its own number, at the office; on site you read them again from the display, and photograph it.

The network, whichever way it comes:

Situation	What you do
The customer's cable and router (best)	Plug the box and the converter into the router. Ask the customer's IT person (or the router's page) to give the converter a **fixed address** that the router will not change (a "DHCP reservation" on the converter's MAC address, or a fixed address outside the router's range). Write it down. You do not type it anywhere: **Find equipment** finds the converter, and finds it again if its address ever changes, but a fixed address does not change under it. The box can take any address.
A fixed address you set yourself	Set the converter's own address in its web page, on the same range as the router.
No usable internet (cellular)	Fit the cellular router from the kit with a SIM, plug the box and the converter into **its** sockets. Greecon supplies the SIM, charged at cost. Check the signal where it sits.
A firewall	The box needs only **outgoing HTTPS (port 443)** to `api.greecon.earth` and, when it is installed, to `nodejs.org`, plus the time service (port 123 UDP, or the box's own). Nothing needs to come in. When someone presses **Find equipment**, the box looks at the site's own network for about a minute: it tries port 502 (or the port the person gives) on the private addresses of that network, such as 192.168.x.x and 10.x.x.x, and only reads; it never writes to anything and never looks outside the site's network. If the customer's network administrator asks, the second message on **The install card** says it in their words.

Before leaving the board: photograph the meter's label (model and serial number), its wiring, its display showing the address and speed, and the converter's settings page.

3.8 The setup list, step by step

Open the site's setup page. Each step is a card. A step opens to show its detail, and stays as you leave it. A step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**; the banner at the top, **Do next: …**, names the step to do next. The platform works most of them out by itself; you only press buttons for what a person must check or say. There are up to twelve steps, always in this order (a step that does not apply to the site is left out): the papers, the prices, the box, finding the equipment, the ranges, the devices answering, the readings, who is told, the direction of the grid meter, the limits, the electrician, and commissioning.

3.8.1 Order form signed and data terms agreed

(the first step: nothing goes into a customer's building without it)

Greecon does not put equipment in a customer's building without a signed order form and agreed data terms (what Greecon reads, who sees it and how long it is kept). Greecon keeps the two papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date**, for example *Signed for the hotel by Besa Krasniqi on 19 October*. The box cannot be left empty. Press **The papers are signed**. The step turns **Done**, *Signed and agreed.*, with your name, the time and what you wrote, and the same goes on the handover sheet. If a paper turns out to be wrong, press **Reopen**: if the site was commissioned, that is taken back too. Do it at the office, before anything else, and bring copies on install day. Commissioning stays grey until this step is done.

3.8.2 Prices from the customer's bills

(optional, but it is what turns kWh into euros)

In this step, open **Enter prices and baseline** (once something is entered it is called **Prices and baseline**; the same form is on the site's page). Enter: currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (blank if nothing); the water price per m³; "a typical month before": the kWh bought and the m³ of water from last year's bills (this is what the platform compares with; **without it no comparison is shown**, which is correct); where those numbers came from; whether the site **cannot send energy to the grid** (tick it if the plant is zero-export); and the time zone (`Europe/Tirane`). **Save.**

3.8.3 Install the box

Done by itself once the box reports (section 3.6). If the box says **Late** or **Offline**, it is not reaching Greecon: see Appendix C.

3.8.4 Where each device is on the network

Nobody types an IP address, a port or a unit number here. Press **Find equipment**. (The button is grey until the box is online and the site has devices; while the box is looking, pressing it again does nothing.)

- The box looks at the site's own network: only private addresses (such as 192.168.x.x and 10.x.x.x), port 502, about a minute at most. It asks each unit number 1 to 10 a few read-only questions, and it never writes to anything. After you press, the page says *The box is looking at the network now.* and follows the box by itself, so you do not reload it. The box picks the request up within about half a minute.
- The platform recognises what answered and matches it to the devices the site is meant to have. It fills in at once only what cannot be a guess: a device with no address yet, found at the unit number the kit gives it, as the model planned. Everything else waits until you press **Use this address**.

Each device gets a line in plain words:

- **Set**: found, and its address is filled in. Where an SDM230 was planned and the meter also answers as a three-phase one, the line adds *It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630: check the model on its label, and if it is one, remove this device and add the three-phase meter from "Add equipment Greecon already knows".*

- **Found**: found, and waiting for you to say yes: at an address other than the one it has, at a unit number other than the kit's, or a meter of the other model than the one planned. Press **Use this address** (or *Use these N addresses*, when there are several) when the line makes sense. The line says where and what, for example *Found at 192.168.1.60, unit 3: a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA.* At another number it adds *It is at unit 4, not 1: if it is this site's, use this address, and the install card gives 1 so that the next site looks like this one.* If a single-phase SDM230 is found where a three-phase SDM630 was planned, it adds *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* The line then keeps saying *Greecon reads it as a single-phase SDM230; the site's plan had a three-phase SDM630.*
- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* For an optional device the line adds *If the site does not have one, remove it from the list below.*
- **Not sure which**: two of one kind answer at the same number on different converters, or two of a kind answer at numbers that are not the kit's; or something answers but not clearly enough to tell the model (*Something answers at 192.168.1.60, unit 1, but not clearly enough to tell whether it is the three-phase SDM630 planned. Look again; if it is the same, check the speed and the A and B wires.*); or every unit number answers with the same words (*Units 1, 2 at 192.168.1.50 all answered with exactly the same words*), which means the converter is not in Modbus TCP to RTU (gateway) mode. Greecon refuses to guess, because a wrong guess could swap the grid meter and the solar meter, and every figure with it. Read the label on each device to see which is which, then use **Enter the address by hand** under each device, with the number on its label. Numbers are set at the office and are not changed on site: see Part 7, R15.
- **Enter by hand**: equipment Greecon cannot recognise on a network: the soil probe (6), the weather sensor (7) and a battery. Under the device open **Enter the address by hand**, fill in the **IP address** (the converter's, or the battery's own), the **Port** (502) and the **Unit number**, and press **Save**. (Once a device has an address the link reads **Change the address by hand**.)

When it worked, the step is **Done** (*Every device has an address.*), the panel says *Every device has the address it was found at.*, and the button now reads **Look again**.

Also on this step:

- **A device the site does not have.** An optional one (the pump contact module, the load meter, the pump meter, a battery) shows **The site does not have this: remove it**. Press it and confirm: the device and its readings go. If you removed one by mistake, add it back with **Add equipment Greecon already knows**, at the bottom of the step (the meters and the two modules).
- **The converter is on another part of the network**, or uses another port, or the box is not on your network (a desk rehearsal): open **I know the converter's address**, type the converter's **IP address**, each part without a leading zero (192.168.1.50, not 192.168.001.050), and the **Port** only if it is not 502 (the field can stay empty), and press **Look at this address**. Only private addresses are accepted, and the box does not look at itself (127.0.0.1).
- **Anything else that answered** is listed under *Also answering, and not part of this site's equipment*. Ask what it is before you go on.
- **If the converter's address changes later** (the router gave it a new one) the readings stop. Press **Find equipment** again, then **Use this address**: Part 7, R17.
- On the box, `greecon-edge find` does the same looking and prints what answers: for a desk, and for support.

- Pressing **Find equipment**, and every change of a device's address or model (old and new), is written in the customer's audit trail under your name.

If it does not work: *No Modbus converter answered on this network. Check that the converter is switched on and plugged into the same router as the box, then find the equipment again. If the converter is on a different part of the network, give its address (I know the converter's address).* Or *A converter answers at 192.168.1.50, but nothing answers behind it. Check each meter's address, the speed on the converter and the meters, and the A and B wires.* Or *The box did not pick this up. Check that it shows as online in the step above, then press Find equipment again.* The button stays grey while the box is not online (*The box has to be online first: it is the one that looks.*) or the site has no devices yet.

3.8.5 Which register holds which reading

- For equipment Greecon knows (the Eastron meters and the pump contact module), nothing is typed: the registers are already filled in. Check them against the maker's manual only if a reading looks wrong.
- For the 4 to 20 mA module (the **Water I/O module**) the page asks, for each transmitter, only for the range printed on its label, in the label's own words:
  - **Level transmitter** (input 1): the **Range on the label** (for a 0 to 5 m transmitter, 5), the **Unit on the label** (metres, or bar for a pressure sensor hung at the bottom of the tank; the page then says what depth it is, for example *0.5 bar holds up 5.0985 metres of water*) and **How tall the tank is inside** (say 4, always in metres): from the bottom to the top of the water when the tank is full. Greecon shows the level as a percentage of the tank, so a full tank reads 100.
  - **Pressure transmitter** (input 2): the **Top of the range** in bar (10 is usual).
  - **Flow meter** (input 3): the **Top of the range** in litres a minute (the form says lpm).
  - A comma works as a decimal point (2,5 is 2.5).
  - A transmitter the site does not have (no flow meter): tick **Not fitted at this site**. When you save, a box asks *Take away pump flow from this site? The reading and its history are removed. This cannot be undone.* Say yes only if the site really has no such transmitter; the reading is then gone, so that nothing waits for it.

  Press **Save the ranges**. It says *Saved. The box picks it up within a minute.*, and under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (a 0 to 5 m transmitter in a tank 4 m tall: 5 ÷ 4 × 100 = 125, so that a full tank, at 16.8 mA, reads 100). A transmitter shorter than the tank is refused, in words: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* To change a range, give it again; the form does not show the level's answer back, so type both of its numbers.

- For equipment Greecon does not know (the soil probe, the weather sensor, a battery): one form per reading, from the maker's register table. Under the reading's name (for example **Soil moisture**) press **Choose the register**, and fill:
  - **Register kind**: **Input** for a manual's 3xxxx, **Holding** for 4xxxx (the form starts on Holding).
  - **Address**: counted from zero, so a manual's **30053 is address 52**, and 40001 is address 0.
  - **Kind of number** and, for two-register numbers, **which half comes first** (the usual is the high half first).
  - **How it becomes the reading**: **Multiply and add** (a meter that reports watts for a reading in kW: multiply by 0.001), or **A straight line, such as 4-20 mA**, where **Signal at the bottom / top** are what the register holds at the two ends of the sensor's signal and **Reading at the bottom / top** are the reading's own values at those two ends (the maker's table says what).
  - **Which part of the value**: "Only when positive" and "Only when negative, as a positive number" split a meter that reports import as positive and export as negative into two readings (the templates do this for you).
  - **Save the register.**
- **Test any register before relying on it**, from the box:

  ```
  greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
  float32
  greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
  ```

  The first reads a meter, the second the pump contact module (`--type discrete` reads a digital input). Each prints the raw numbers and the value. If it prints `Could not read`, the problem is the wiring or the settings, not Greecon.
- A reading the site does not have holds the list at this step: for a transmitter, tick **Not fitted at this site**; for an optional device, press **The site does not have this: remove it** in step 4 of the setup list.

3.8.6 Every device answers the box

Done by itself. Each device shows **Done** ("Answering (N readings)") or **Needs attention** with the reason in words ("no answer from 192.168.1.50:502 (unit 2)"); on the box, `greecon-edge status` lists each device as **ok** or **NOT OK**. Look in this order: the converter's address (the router may have given it a new one: press **Find equipment** again, then **Use this address**), the number on the device's label against the install card, the **speed** at the converter, **A and B swapped** on the data cable, the converter not in gateway mode.

3.8.7 Readings arrive and look right

Done by itself when each reading has arrived, in a believable range. A reading that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter that is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank, so the reading comes out below zero), or a range or a tank height that was given wrongly: Part 7, R18. A tank level may read up to 5 points over 100 or under 0 without being called impossible. **Pump running** reads 1 while the pump runs and 0 while it is stopped.

3.8.8 Someone is told when the site needs attention

Open **Who is told** (the step has the link):

- **Add a person** for each name from the survey, **in the order they should be told.** Each needs an email, a mobile number **with its country code** (`+355 69 123 4567`), or both. The first person hears first.
- **Save the list.** Then, beside each person, **Send a test message.** It says what happened to the email and to the text ("Accepted for delivery to +355 ••• ••• 567. Ask Besa whether it arrived."). Tests are limited for the whole account: six an hour (a person with both an email and a number uses two of the twelve test messages allowed). If you reach the limit the page says "Test messages are limited to 6 an hour. Try again later."
- **Ask each person** whether it arrived: in the inbox, not in spam; on the phone.
- **Tell each person on the list how to say they have seen an alert.** Every alert message has an *I have seen this* link: one tap on the phone, then one button, no sign-in, good for 72 hours; it stops the chain of messages. Pressing **Acknowledge** on the **Alerts** page does the same, for a person who can sign in. Replying to an email or a text does not. So nobody needs a login for this; give a login of their own (the Operator role is enough) to the people who should also look at the screens: open **Settings**, then **People**, and look at who has a sign-in. If someone who should have one has none, the customer's owner adds them under **Add a person** with **Add person** (a Greecon support session can see the people but cannot add them), and the new person is given a temporary password, shown once. An Owner or an Administrator can press **Acknowledge** too; a Viewer or an Auditor cannot (the link in a message works for anyone it was sent to).
- **Give everyone an email address as well as a number.** The note that a colleague has acknowledged an alert, and the note that it has cleared, go by email only. A person who has only a phone number is texted the urgent alert and then hears nothing more: they may still set off for something that is already being dealt with.
- Back on the setup page, write what you saw if you like, and press **I checked this.** It is greyed out until the list has someone on it and a test has been sent, because a site nobody is told about must not be commissioned.

How alerts reach people, for the owner (say this at handover, in these words): **Act now** goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the *I have seen this* link in the message (no sign-in, good for 72 hours) or **Acknowledge** on the **Alerts** page (for a person with a login). **Check today** is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody **acknowledges** the alert (who took it), one when the problem has stayed clear for 15 minutes (**Resolved**), and one **morning message** at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.

3.8.9 Grid meter direction checked

(only on sites with a grid meter) Greecon cannot see which way the meter's clamps point. A reversed clamp swaps what the site buys with what it sells, and every figure with it. **With the sun down or the solar switched off, switch on a known load** (a kettle or a heater). **Grid import must go up.** If **Grid export** goes up, the clamps are the wrong way round: the electrician turns them (with the lock-out on), you check again. Then press **I checked this**, with a note of what you saw. If a clamp is left reversed, Greecon notices on its own at night or in cloud (energy cannot go out of a plant that nothing is producing for): the energy picture says so, and after five minutes a **Check today** alert, *The grid meter may be fitted the wrong way round*, opens. At midday it cannot tell, which is why this check stays.

3.8.10 Alert limits agreed with the customer

The step lists each reading that has limits, with what applies. The standard ones: tank **Critical below 15 percent** (an Act now alert), **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. Irrigation advice, **frost** and solar surplus have no standard value, and the platform says nothing about them rather than guess. (Frost: an **Act now** alert opens when the air at the field reaches the grower's own **Frost alert at or below** value and stays there for two minutes. Ask the grower at what temperature the crop is in danger and in which weeks; enter it on the weather sensor's air temperature.) **Go through them with the customer.** To change one, open the **Standard** drop-down beside the reading in this step (it says **1 set**, **2 set** once changed), type the numbers, press **Save limits**, and when they are agreed press **I checked this.** The same drop-down is in the **Limits** column under **Readings** on the site's **Equipment and readings** page. If the pump's running contact reads the wrong way round, the no-flow alert opens while the pump is stopped: the "Pump running" check in section 6.1 finds that. Agree numbers the customer would act on: an alert nobody acts on trains people to ignore the next.

3.8.11 Electrical work done by a licensed electrician

Fitting meters in a board and wiring transmitters and contacts is the licensed electrician's work, under the customer's lock-out. It is not yours. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number**, for example *Arben Hoxha, licence E-1234*, or that no electrical work was needed. The box cannot be left empty. Press **The electrical work is done**. The step turns **Done**, *Done by a licensed electrician.*, with your name, the time and what you wrote, and the same goes on the handover sheet. The platform does not check the licence: what you write is the record. Do not press it while the work is open or the lock-out is still on. Commissioning stays grey until it is done.

3.8.12 Commissioned

Greyed out until everything above is done (the prices are optional and do not hold it; the papers and the electrician's name and licence number do). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the checks for the site's kind in section 6.1. Press **Commission this site**, with a note. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

3.9 The four kinds of site

A. Hotel or guesthouse (the usual first site)

What it is: water tank and pump, solar on the roof, and the grid. **What is measured:** grid import and export (one three-phase meter at the supply entry), solar production (one three-phase meter on the inverter's AC output), tank level, line pressure, pump flow, pump running.

Bring: two Eastron SDM630 meters with current clamps, an RS-485 to Ethernet converter, a Waveshare 4 to 20 mA module (Modbus RTU Analog Input 8CH) and a Waveshare pump contact module (Modbus RTU IO 8CH) with their 24 V supply, a level transmitter and a pressure transmitter (and a flow meter if the owner

wants dry-run protection), the box, a small UPS, cable, DIN rail and enclosure (Part 4). The numbers of the meters and the modules were set at the office. **Ask the customer for:** twelve electricity bills, the water bills, the people to tell.

Wiring in one picture:

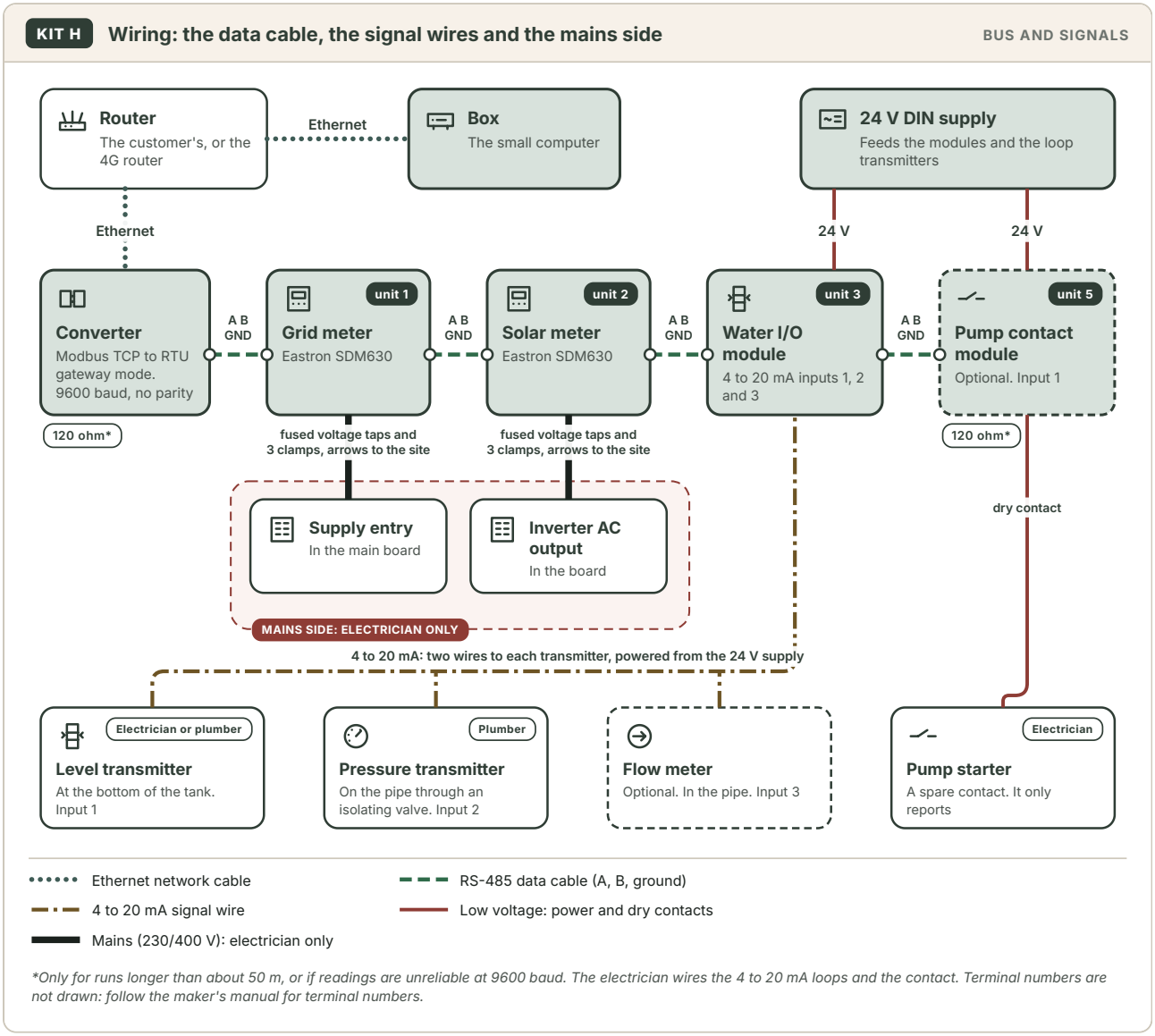


Figure H-2. Kit H. Wiring: one RS-485 cable from the converter through the grid meter, the solar meter and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module. Mains work is inside the dashed box.

Set up in Greecon: template **Hotel or guesthouse** (four devices: the grid meter, the solar meter, the water I/O module and the pump contact module, which is optional). Press **Find equipment**; the registers of all four are known. Then give the ranges (3.8.5): the level transmitter's range and the tank's height, the pressure transmitter's range and, if there is one, the flow meter's. The pump's running state comes from the pump contact module. Prices (3.8.2) so that "your solar kept N kWh on site, worth about X at your price of Y" can be said.

Agree with the owner: the tank limits (the standard 15 and 35 percent are a start: a tank that feeds a fifty-room hotel on a hot day should alert earlier), the no-flow level if the pump normally runs at a trickle (the standard is 0.2 litres a minute), who is told for water and who for electricity, and what "Act now" at 02:00 means for them.

Typical problems: the solar meter reading zero at midday (a closed clamp on the wrong cable, or the inverter is zero-export limited); the level reading wrong (a vented cable blocked, the transmitter not at the tank's bottom, or the range or the tank's height given wrongly); the pump-running contact inverted (reads running when stopped), which also opens the no-flow alert, an Act now, while the pump is stopped (3.8.10).

B. Solar plant with grid connection (with an optional battery)

What is measured: solar production (inverter AC output), grid import and export, optionally the site's own consumption (a load meter) and a battery's charge and power. It works **with any inverter brand**, because it measures at the AC cable with its own meter and not through the inverter maker's portal.

Bring: two or three SDM630 meters (grid, solar, optional load), converter, box, UPS. For a battery: the maker's Modbus register table (without it the battery is left out; do not guess registers). The numbers of the meters were set at the office; a battery is addressed by hand.

Ask the customer: the electricity price, **what is paid for energy sent to the grid, if anything**, and whether the plant is **allowed to send energy to the grid** (if not, tick *This site cannot send energy to the grid* in prices). Twelve bills.

Set up: template **Solar plant with grid connection**. Press **Find equipment**, and take off the battery and the load meter if they are not fitted (3.8.4). To take one off, press **The site does not have this: remove it**. A battery is not recognised on a network: its address and its two readings are entered by hand, from the maker's table. Do the direction check with care: on a plant that exports at noon, do it at night or with the inverter off.

Commissioning: at midday, solar production in Greecon agrees with the inverter's own display within about 5 percent, and grid export is the difference between production and what the site is using.

C. Farm or greenhouse with irrigation

What is measured: soil moisture (per zone) and air temperature and humidity at the field, and the water supply behind it (tank, pressure, flow, pump).

Bring: a soil probe and a weather sensor that speak Modbus RS-485 and that document their registers (without documented registers the sensor cannot be mapped), the 4 to 20 mA module for the tank and pressure and the pump contact module if there is a pump to watch, the box, a cellular router if there is no network in the pump house. The probe's number (6) and the sensor's (7) are set before the probe is buried and the sensor mounted. The box cannot recognise a probe or a sensor, so their addresses and their readings are entered by hand (steps 4 and 5 of the setup list), from the makers' manuals; the water side is as in the hotel.

Ask the grower: which crop, which soil, how deep the roots are, where in the field is typical (not the wettest or driest spot). **Greecon gives no irrigation advice until the grower's own "start irrigating below" value is entered,** because that number depends on the soil, the crop and the probe's depth and nobody can set it for them. After the first week of readings, ask the grower what reading means "time to water" and enter it as the limit. **Ask, too, at what air temperature frost damages the crop and in which weeks of the year.** Enter it as the **Frost alert at or below** limit on the air temperature (the number depends on the crop and how far on it is, so Greecon has none of its own, and says nothing about frost until it is set). A frost alert is an **Act now** message in the night: agree who is woken. **Map the air temperature as a signed number** (int16): below zero, an unsigned register reads a frosty morning as a hot afternoon.

Write in the handover: the probe's depth and where it is. Soil readings are only as good as the contact between probe and soil.

D. Water tank and pump station

What is measured: the tank's level, the line pressure, the pump's flow, whether it runs, and optionally the pump's electricity use (a single-phase meter such as the SDM230).

Ask: what the pump does when the tank is low and when it is full, and who is told today. The pipe's pressure range decides the pressure transmitter's range.

Set up: template **Water tank and pump station** (three devices: the water I/O module, the pump contact module and the pump meter, the last two optional). Press **Find equipment**, give the ranges, and take off what the site does not have (the pump meter, the pump contact module) with **The site does not have this: remove it**. This is the simplest site and a good first rehearsal.

3.10 After install

When	Do	Look for
Day 1	Open **Today**; phone the owner: did the morning message arrive?	Box online; readings current; the message reached them
Day 3	Same; look at one alert if one happened	No alert nobody understood
Day 7	Sit with the owner for 15 minutes: **was every alert useful? what would they change?** Adjust limits.	Alert count at most one a day; every one with a clear action
Every morning	The morning check email and **Today** (Part 7); each live site shows *Day N*	A silent box, or a site that needs a look
Day 30	Short call; ask what they have done because of Greecon	One thing, in their words, for the sales story
Day 60	The decision meeting, against the three written goals: data current at least 95 percent of the time; the owner says they read the morning message on at least 5 days in 7 (Greecon does not track opens: ask, and write the answer down); at least one real event caught	Convert, extend by up to 30 days once, or remove
Day 90	Convert (order hardware at the agreed price; the monthly fee starts at day 91) or take the kit away within 14 days	Data exported to the customer either way

3.11 Rehearsal: the whole job on a desk

Do this once before the first real site, once more with the first real kit, and again when anything in this manual changes. It takes an afternoon and needs no hardware except the box (or a spare computer running Debian, Ubuntu or Raspberry Pi OS) and your laptop.

1. **On your laptop** (Node.js 22 or newer): get the repository, then run:

   ```
   node scripts/pretend-meter.mjs --kit --port 5020
   ```

   It listens for any machine on the network and answers as a whole hotel kit: unit 1 the grid meter, unit 2 the solar meter, unit 3 the 4 to 20 mA module (the tank at 12 mA, the pressure at 8.96 mA, the flow at 5.12 mA) and unit 5 the pump contact module, with its input 1 closed. Leave it running: it prints each question it is asked, so that you can watch the box look.

2. In Greecon, onboard a client named **Rehearsal**, open it for setup, and make a site from the **Hotel or guesthouse** template. As the owner use **a second address of your own** (a plus-address such as `you+rehearsal@yourmail.example`, if your mail allows it): your own team address already has an account, an address can have only one, and Greecon refuses it. This is the one case where a different address is right.

3. On the setup list write that the papers are signed (for the rehearsal, *Rehearsal: signed by me today*) and press **The papers are signed**. Add the box, copy the line, and paste it on the box (section 3.6). Watch **Install the box** turn **Done**.

4. **Find the kit.** The kit listens on port 5020, not 502, so under **Where each device is on the network** open **I know the converter's address**, type **your laptop's address on the network** (the one the box can reach, such as 192.168.1.60; not 127.0.0.1, which the box takes for itself) and the port **5020**, and press **Look at this address**. After a minute at most the grid meter, the solar meter, the water module and the pump contact module are each **Set**, with lines such as *Already set, and answering at 192.168.1.60, unit 1: an Eastron meter, reading 1.80 kW at 231 V.* (your laptop's address in place of 192.168.1.60). The solar meter reads 5.20 kW; the water module is *a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA*; the pump contact module is *a digital input module, reading input 1 is closed*. The step is **Done**. You typed no unit number and no register.

5. **Give the ranges.** Under **Which register holds which reading** type **Level transmitter** range 5 and tank 4, **Pressure transmitter** 10, **Flow meter** 200, and press **Save the ranges**. Within a minute the readings arrive: **Tank level 62.5 %**, **Line pressure 3.1 bar**, **Pump flow 14** litres a minute and **Pump running 1**, with **Grid import** 1.8 kW and **Solar production** 5.2 kW. They come from the kit's signals and your ranges: the kit's 12 mA is the middle of the level transmitter's range, 2.5 m of 5 m, which in a tank 4 m tall is 62.5 percent of the tank ((12 − 4) ÷ 16 × 5 ÷ 4 × 100); 8.96 mA is 3.1 of 10 bar ((8.96 − 4) ÷ 16 × 10); 5.12 mA is 14 of 200 litres a minute ((5.12 − 4) ÷ 16 × 200). If a number differs, look at the ranges you typed.

6. Watch every step turn **Done**. Then add yourself under **Who is told**, send a test message, write for the electrician that no electrical work was needed (the box cannot be left empty), and confirm the steps that need a person. **Commission** the rehearsal site.

7. Stop the pretend kit. After about five minutes **Today** (under **Needs a look**) says the readings are out of date and the box says a device does not answer. Start it again. Then rehearse an alert: in the limits step open the tank's limits, give **Low below** 80 and **Critical below** 70 (both above the tank's 62.5 percent), and press **Save limits**: **Tank level is critical** is an **Act now**. It arrives by email at once and, if you added your number, by text ten minutes later unless you **Acknowledge** it on the **Alerts** page, which then tells the people who were told who took it.

8. Print the handover sheet. Download the readings from **Reports** (**Downloads**). Open the file in a spreadsheet.

9. When finished, remove the rehearsal site's equipment and then the site (on the site's **Equipment and readings** page: **Remove** beside each piece of equipment, then **The site itself**: a site that still has equipment cannot be deleted), or suspend the client from **Clients** (**More**, then **Suspend this client**, and confirm), so that it does not appear in the morning check. If you will use the Rehearsal client to show **Who is told** and **Reports** in a sales demonstration (the Sales kit (document 04, not part of these guides), section 4), keep it and its site until then, and start the pretend kit before the meeting: with it stopped the site shows as out of date.
10. **With the first real kit,** do steps 1 to 6 again with the real converter and devices on your desk in place of the pretend kit, and write down anything that differs from this manual (a number, a speed, a reading, a sentence on the screen). The registers of the two Waveshare modules have only been tried against a pretend module.
11. **To practise the by-hand form** (equipment Greecon does not sell), use the older pretend meter, which is still there:

    ```
    node scripts/pretend-meter.mjs --units 6,7 --port 5021
    node scripts/pretend-meter.mjs --units 6,7 --port 5021 --tank 12 --watts -1800
    ```

    It answers as units 6 and 7 with plain registers: a power at input registers 52 and 12, and at registers 0 to 3 the tank level in tenths of a percent, the pressure in hundredths of a bar, the flow in tenths of a litre a minute and 1 or 0 for the pump running. The second line makes the tank read 12 percent and a plant export. On a second rehearsal site from the **Farm or greenhouse with irrigation** template, give the **Soil probe** your laptop's address, port 5021 and unit 6 with **Enter the address by hand**, and map **Soil moisture** with **Choose the register**: **Input**, address 0, **Multiply and add**, multiply by 0.1. It reads 62 percent with the first line.

If a step in this manual does not match what you see, fix the manual (it is a file in `docs/launch/`), or tell the founder. It is meant to be corrected by the person who uses it.

PART 4

Hardware reference

This part describes every kind of hardware in a Greecon kit: what it is, what it must be, what it costs, how it is set up, how it is fitted, how to check it and what goes wrong. It is written from the hardware and wiring document (docs/launch/03) and is the place to look when a pack says “follow the maker’s manual”. The four site packs (Part 5) use these parts in four combinations.

TIP

Where this part is silent, the maker’s manual decides. The drawings show how the parts connect, not terminal numbers: terminal numbers differ between makes and versions. Every drawing says so. Part numbers named here are examples of a kind of part; the specification is what matters. Prices are planning estimates from public shop prices seen in October 2026, not quotes: replace them with real quotes in the first week and keep the quotes in the shared drive.

4.1 Rules of thumb

The aim is the cheapest way to run many sites for many years: the same few parts everywhere, prepared at the office, installed in an afternoon, and replaceable by anyone because nothing about a box is special. The parts that talk to the platform are **approved models**: the two Eastron meters (SDM630 and SDM230) and the two Waveshare modules (Modbus RTU Analog Input 8CH and Modbus RTU IO 8CH). Greecon knows their registers, their serial settings and the way to recognise them on a network, so nobody on site reads a datasheet or types a register. The rest of a kit (the box, the converter, the supply, the transmitters, the enclosure) is bought by specification and not by brand. Greecon speaks two standard languages, Modbus over the network and a wire signal of 4 to 20 mA read through a Modbus module, so a part that is not an approved model can be swapped when a maker stops selling it.

1. **One of each first.** Buy one full kit and bench-test it (the rehearsal, Part 3, section 3.11) before buying five. Change the specification, not the kit, when something fails.
2. **Read-only, always.** No relay output, no control wire, no write. Everything here only measures. (The pump contact module has relay outputs: they are never used.)
3. **Approved models for what is read over the cable, a published register table for anything else.** A kit is made of the approved models and no others. A soil probe, a weather sensor or a battery is not an approved model and is mapped by hand: if the maker will not give the register table, do not buy it, because it cannot be mapped honestly.
4. **Everything powered from one clean place,** on a UPS, in one enclosure, labelled.
5. **Pre-stage at the office:** the box has its software installed and tested, every device has its number and its speed, and the converter is set, before the kit leaves.
6. **Same settings everywhere:** unit numbers (grid meter 1, solar meter 2, 4 to 20 mA water module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7), speed 9600, 8 data bits, no parity, 1 stop bit, the converter on port 502. A new site then looks like the last one.

STOP

Greecon only reads, and only an electrician works on mains. Nothing in this part tells anyone to work on a live conductor. Everything inside a distribution board, every current clamp and voltage tap, and the pump's run contact are the electrician's work, done under the customer's lock-out procedure. The project manager does the low-voltage and software side. Greecon is not a safety system: the float switches, relief valves and trips already in the customer's equipment stay the safety layer.

4.2 The standard kits

Four kits cover the four kinds of site. Costs are planning estimates in EUR, excluding VAT and installation labour. A part marked optional is not always fitted. The meters and the two modules are approved models: the model named is the model bought. The first table is every part once; the tables after it are the four kits, with quantities.

Every part, what it does and which kits use it

Kits in brackets use the part only sometimes.

Part	What it does	Kits	Planning estimate (EUR)
Box	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.	H S F W	180 to 300
Converter	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).	H S F W	30 to 60 (basic) or 250 and up (industrial)
Three-phase meter	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.	H S	90 to 180 each
4 to 20 mA input module	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.	H F W	30 to 60
Pump contact module	Reports whether the pump is running, from a spare contact of its starter (a dry contact on input 1), as unit 5. It only reports; it never switches anything.	H (F) W	30 to 60
Level transmitter	Measures how full the tank is. It hangs at the bottom of the tank.	H F W	120 to 400
Pressure transmitter	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.	H F W	40 to 150
Flow meter	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.	(H) (F) (W)	150 to 500

Part	What it does	Kits	Planning estimate (EUR)
24 V DIN power supply	Powers the two modules, the loop transmitters and the converter from one place.	H S F W	25 to 50
UPS	Keeps the box and the converter up through short power cuts and long enough to say that the grid is down.	H S W	60 to 150
Enclosure and fittings	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.	H S F W	80 to 150
Cable and labels	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.	H S F W	40 to 80
Cellular router	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.	(H) (S) F (W)	150 to 200
Single-phase meter	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).	(W)	35 to 60
Soil moisture probe	Measures soil moisture at the probe's depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker's register table.	F	40 to 120 each
Temperature and humidity sensor	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker's register table.	F	20 to 80

Kit H: hotel or guesthouse (water, solar, grid)

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *One at the grid connection, one on the inverter's AC output.*	2	90 to 180 each
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200

About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.

Kit S: solar plant with grid connection

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *Grid and solar; a third on the site's own consumption if the owner wants it measured.*	2, or 3	90 to 180 each
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet; not in the hardware guide's list for this kit, see the network rules.*	0 or 1	150 to 200

For a battery add nothing but the maker's Modbus register table (the battery speaks Modbus itself on the site's network or on its own RS-485 port; if the maker will not publish the table, leave the battery out). **About EUR 600 to 1,200.**

Kit F: farm or greenhouse with irrigation

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router**	4G, DIN rail, data SIM. *The likelier case in a field.*	0 or 1	150 to 200
☐	**Soil moisture probe**	RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each
☐	**Temperature and humidity sensor**	RS-485 Modbus, with a published register table	1	20 to 80
☐	**Pump contact module** *(optional)*	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *If there is a pump to watch.*	0 or 1	30 to 60
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *If there is a pump to watch.*	0 or 1	150 to 500

The standard list for this kit has no UPS row, although rule 4 in section 4.1 says everything is powered on a UPS: decide at the survey whether this site gets one, and write the decision down. A cellular router is the likelier case in a field. **About EUR 700 to 1,500.**

Kit W: water tank and pump station

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200
☐	**Single-phase meter** *(optional)*	Eastron SDM230, on the pump's feed, to measure its electricity. *To measure the pump's electricity.*	0 or 1	35 to 60

About EUR 650 to 1,300.

The spares kit (one for each project manager)

One of each, so that a failed part is swapped on the day:

- ☐ A converter
- ☐ A 4 to 20 mA module
- ☐ A pump contact module
- ☐ A meter
- ☐ A 24 V supply
- ☐ A transmitter (pressure)
- ☐ A spare box, pre-installed, with no code, ready to take one
- ☐ Network cables
- ☐ A loop calibrator
- ☐ A multimeter
- ☐ Fuses
- ☐ A spare SIM

4.3 The box

The box runs one small program and holds a queue of readings. It needs very little, and it must survive power cuts, which is the hard part.

WHAT IT DOES	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.
SPECIFICATION	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input
PLANNING ESTIMATE (EUR)	180 to 300. Not a quote.
USED IN	Kits H, S, F, W

What the box needs

Needs	Why
Linux on a processor of the x86-64, 64-bit ARM or 32-bit ARM kind (the installer checks)	The software is a Node.js program
Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS (64-bit), with systemd	What the installer supports
4 GB of memory or more, 32 GB of storage or more	The program asks for about 400 MB; the queue is on disk
A real disk (SSD or eMMC), not an SD card	An SD card is the commonest reason these boxes fail: they corrupt on power cuts and wear out under constant writing
A real-time clock with a battery, and a BIOS setting of “power on after power loss” (often “Restore on AC Power Loss: Power On”)	After a cut the box must come back by itself, with the right time
9 to 36 V DC input, or a plain 12 V one	So that it can sit behind a DC UPS and share the 24 V supply
No fan, 0 to 50 °C or better	A plant room is hot and dusty
One Ethernet port (two is nicer)	One to the converter’s network, one spare

The kit list buys 8 GB and 128 GB or more; this table is the least the software will run on.

Options

Option	When	Planning estimate (EUR)	Note
Fanless x86 mini PC (an Intel N100-class machine)	The default. Most sites.	180 to 300	Check the real-time-clock battery and the power-on setting in the BIOS before buying many.
Raspberry Pi 5 (4 or 8 GB)	A bench box, or a small site with a proper SSD on NVMe or USB 3 and an RTC battery. Never an SD card.	120 to 200 with SSD and case	Starts by itself when power returns.
Industrial PC (Advantech UNO, OnLogic, Seeed reTerminal DM class)	Hot, dusty or vibrating sites; customers who insist on industrial parts	450 to 900	The same software runs on it. Not needed to begin.

Preparing the operating system (at the office, twenty minutes)

1. Install Debian 12 from the “netinst” image: only “SSH server” and “standard system utilities”. No desktop.
2. One user with a long password (in the password manager, against the site’s name). Host name `greecon-<site>`.
3. Time zone Europe/Tirane; `timedatectl set-ntp true`.
4. Automatic security updates: `apt-get install -y unattended-upgrades` and answer yes when it asks.
5. BIOS: power on after power loss; no sleep; boot from the disk only.
6. Then paste the install line (Part 3, section 3.6).

What the installer changes

So that nobody wonders: `/opt/greecon` (the software), `/etc/greecon` (the box’s credential, readable by the service only), `/var/lib/greecon-edge` (what it remembers), one service called `greecon-edge`, one user called `greecon`, and one command, `greecon-edge`. Nothing else.

What the box does by itself

- It restarts if it fails.
- It keeps about three days of readings if the internet is down, for a site of up to about thirteen devices (a site with more devices keeps fewer days), and sends them in order when it returns. A long outage keeps one reading a minute for each reading, so that three days fit.
- It takes new versions of its software from the platform and goes back to the version before if a new one fails to start three times.
- It reports every 30 seconds so that Greecon knows it is alive.

Commands the project manager uses

```
greecon-edge status                       how the box is: heard by Greecon? each device ok? the
queue
journalctl -u greecon-edge -n 50 --no-pager   what it has been saying
greecon-edge find                         look at the network and say what answers (reads only)
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
sudo systemctl restart greecon-edge       restart it
```

CHECK **Before the box leaves the office:** the install line has run to “Done. Greecon can see this box.”, the setup page shows the install step done with the box’s name and “Heard from N seconds ago”, the box is labelled with the site name and the date, and the BIOS power-on setting has been tried by pulling the power and restoring it.

4.4 The converter and the meter cable

The meters and modules speak Modbus RTU on a two-wire cable (RS-485). The box speaks Modbus TCP on the network. The converter sits between them.

WHAT IT DOES	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).
SPECIFICATION	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode
PLANNING ESTIMATE (EUR)	30 to 60 (basic) or 250 and up (industrial). Not a quote.
USED IN	Kits H, S, F, W

STOP

The setting that matters most: the converter must be in **Modbus TCP to RTU gateway mode** (sometimes called "Modbus gateway", "Modbus TCP server" or "protocol conversion"). Not "transparent" or "raw TCP" mode, and not "Modbus RTU over TCP": in those, the box's requests arrive on the cable as garbage and nothing answers.

The settings in the converter's web page

Greecon sets them at the office, before the kit leaves. They are here in case the converter ever has to be set again.

Setting	Value
Mode	Modbus TCP to RTU (gateway)
Serial: speed (baud)	9600 (the same as every meter on the cable)
Serial: data bits, parity, stop bits	8, none, 1
TCP: port	502
Address	A fixed address on the site's network (a reservation in the router, or set in the converter)

The box keeps one TCP connection per converter and asks one thing at a time, so any converter that does the above will do.

Wiring RS-485

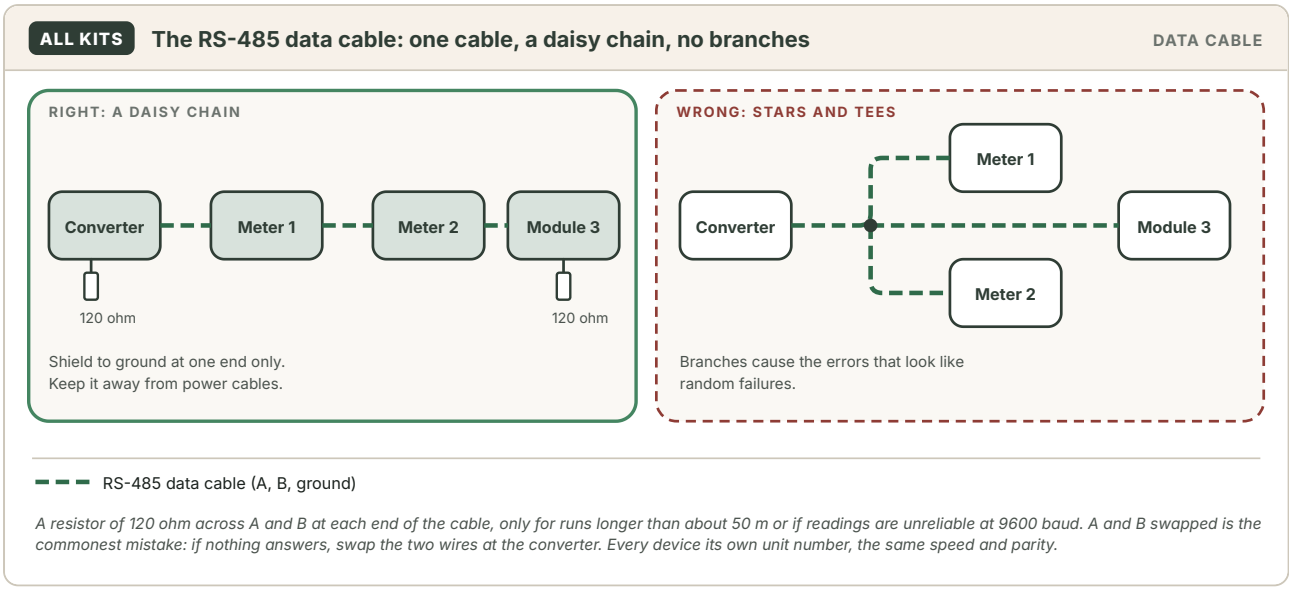


Figure 4-1. The RS-485 data cable: one cable and a daisy chain, never branches.

- **One cable, daisy chain,** from the converter to the first device, on to the next, and so on. No branches: stars and tees cause the errors that look like random failures.
- Use **shielded twisted pair,** the shield joined to ground at one end only. Keep it away from power cables.
- **A and B swapped** is the commonest mistake. If nothing answers, swap the two wires at the converter.
- **Every device its own unit number,** the same speed and parity. The meters and the Waveshare modules are delivered as number 1, so each is numbered at the office before it joins the cable: two with the same number would answer over each other.
- A 120 ohm resistor across A and B at each end of the cable is only for runs longer than about 50 m, or if readings are unreliable at 9600 baud.

The number each device has on the cable

Device	Number	How it is set, and when
Grid meter (Eastron SDM630)	1	On its display, at the office
Solar meter (Eastron SDM630)	2	On its display, at the office
4 to 20 mA module (Waveshare Modbus RTU Analog Input 8CH)	3	With the maker's setting tool, at the office
Load meter (SDM630) or pump meter (SDM230)	4	On its display, at the office
Pump contact module (Waveshare Modbus RTU IO 8CH)	5	With the maker's setting tool, at the office
Soil probe	6	The maker's way, before it is buried
Weather sensor	7	The maker's way, before it is mounted
A battery	By hand, for example 8	From the battery maker's manual

Every device on one cable has the same speed: 9600 baud, 8 data bits, no parity, 1 stop bit. On site you check the label on each device against the install card and do not change it.

Cable and labels

WHAT IT DOES	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.
SPECIFICATION	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels
PLANNING ESTIMATE (EUR)	40 to 80. Not a quote.
USED IN	Kits H, S, F, W

Label the unit number on the converter and on every device, the wire number at both ends, and the site name and date on a sticker inside the lid of the enclosure.

4.5 The meters

Eastron SDM630 (three-phase) at the grid connection, where the supply enters, and on the inverter's AC output. Eastron SDM230 (single-phase) for one pump or a small load.

Three-phase meter

WHAT IT DOES	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.
SPECIFICATION	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers
PLANNING ESTIMATE (EUR)	90 to 180 each. Not a quote.
USED IN	Kits H, S

Single-phase meter

WHAT IT DOES	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).
SPECIFICATION	Eastron SDM230, on the pump's feed, to measure its electricity
PLANNING ESTIMATE (EUR)	35 to 60. Not a quote.
USED IN	Optional in kit W

The battery (kit S, optional)

A battery is not bought from Greecon's list: it is whatever the customer has, and it is not an approved model. Greecon can read it only if the maker publishes the battery's Modbus register table. The battery speaks Modbus itself, on the site's network (its own address) or on its own RS-485 port (then it joins the cable with a

number of its own, by hand, for example 8). The box cannot recognise it on a network, so its address is entered by hand. If the maker will not publish the table, leave the battery out: do not guess registers. Two readings are mapped by hand from the maker's table: Battery charge (percent) and Battery power (kW, positive while charging). If there is no battery, remove the template's battery device (the project manager's step in the pack for kit S).

What Greecon reads

Total active power: register 52 on the SDM630 (address counted from zero, as a manual's 30053) and register 12 on the SDM230, 32-bit floating point, high half first, in watts; Greecon shows it in kW. Positive power is energy coming in, negative is energy sent out. That is why one grid meter gives both Grid import and Grid export, and why a clamp fitted the wrong way round swaps them (the direction check, setup step 9).

The Eastron meters in the templates already have their registers filled in. The unit scale applied is only watts to kilowatts. Greecon reads what the meter shows.

Settings made at the office

These differ between makes and versions, so they are read from the meter's own display, never assumed. Greecon sets them at the office, before the kit leaves; on site you only check them.

Setting	Value
Unit number	A different one on each meter: grid meter 1, solar meter 2, load or pump meter 4. Set on the meter's display
Speed	SDM630: 9600 as delivered, so check it. SDM230: depends on the version (the V1.6 manual says 2400); set it to 9600 on the meter's display, on the Baud Rate page
Parity and stop bits	None, and as the manual says (the converter uses 8 data bits, no parity, 1 stop bit)
CT ratio	On a meter with 5 A current transformers, set the ratio in the meter's menu (for example 250 for a 250/5 A transformer) so that the meter shows the real power. It depends on the transformers fitted, so it is set on site. Greecon reads what the meter shows

A meter's display works only when its voltage terminals have power, and powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and the project manager never wires mains.

Fit the model the install card lists. Greecon reads an SDM630 by its register 52 and an SDM230 by its register 12. If the box finds a single-phase SDM230 where a three-phase SDM630 was planned, **Find equipment** says so, and reads it as an SDM230 once you press **Use this address**. An SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only: **Find equipment** says it may be a three-phase meter, and the commissioning comparison with the meter's own display finds it too.

Fitting: the electrician’s work

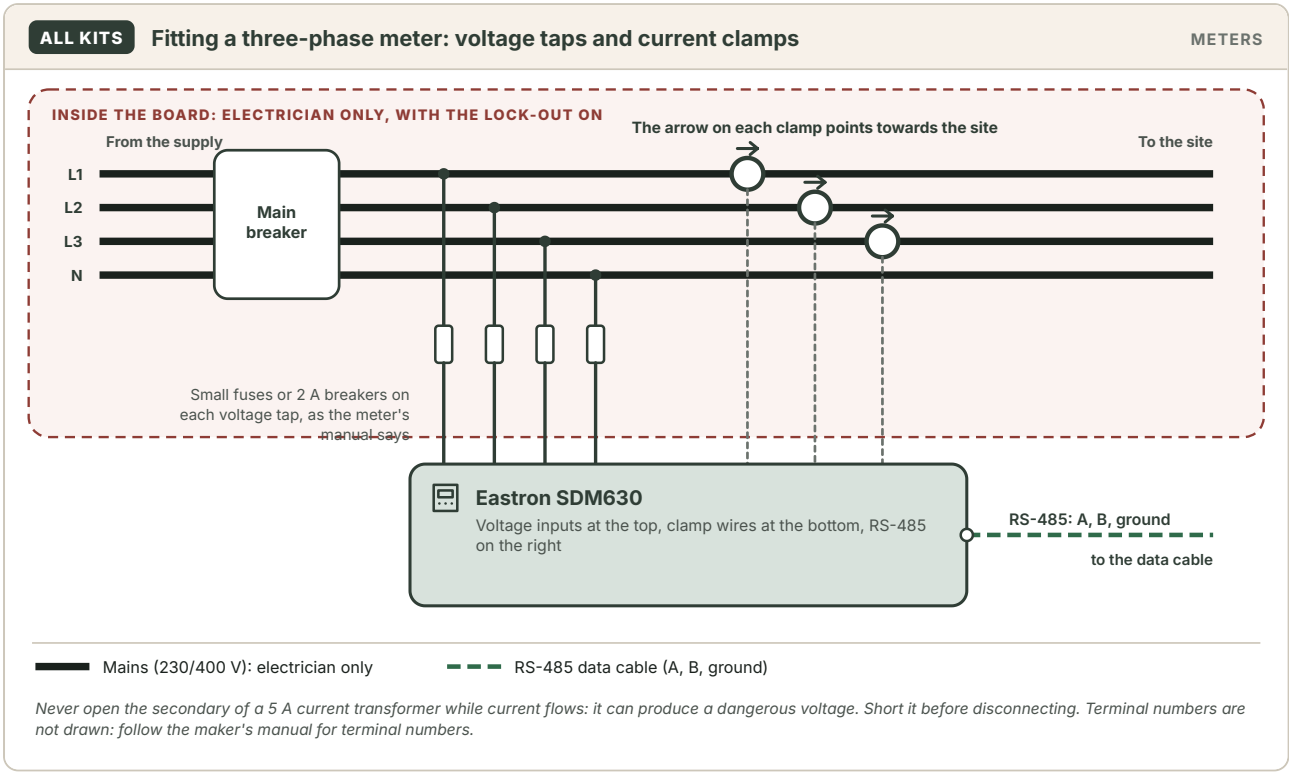


Figure 4-2. Fitting a three-phase meter: fused voltage taps and three current clamps, arrows towards the site. The electrician's work.

STOP

Inside a distribution board is the electrician’s work, with the lock-out on. The project manager watches, photographs and writes it down.

- **Clamps point towards the site.** On the grid meter this makes import positive. On the solar meter, the clamps point away from the inverter, towards the board (towards the site), so that production is read. The platform takes the size of the power for solar, so a reversed solar clamp does not read zero, but fit it the right way anyway.
- **Never open the secondary of a 5 A current transformer while current flows:** it can produce a dangerous voltage. Short the secondary before disconnecting. This is the electrician’s rule; it is repeated here because it is the one that hurts.
- Voltage taps have small fuses or 2 A breakers on each tap, as the meter’s manual says.
- **Photograph** the wiring, the label (model and serial number) and the display with its settings.

CHECK

Check on the day: with the sun down, switch on a known load: grid import must go up. At night solar reads zero or a few watts. At midday the solar meter agrees with the inverter’s own display within about 5 percent.

4.6 Water: tank, pressure, flow, pump

Two modules read the water side, both on the same RS-485 cable as the meters, and both approved models: the **4 to 20 mA module** (unit number 3) reads the level, pressure and flow transmitters, and the **pump contact module** (unit number 5) reads whether the pump is running. Greecon sets their numbers at the office; on site you check the labels.

The 4 to 20 mA module

WHAT IT DOES	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.
SPECIFICATION	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, F, W

Its inputs are left in their default 0 to 20 mA mode, on purpose. A 4 to 20 mA transmitter then reads **4000 to 20000 microamps**, and a wire that has come off reads **0**, which is below the 4 mA of an empty tank, so Greecon can see it and say so. (In the module's own 4 to 20 mA mode a broken wire might read like an empty tank.) Use the version that reads current, not the one that reads voltage.

Each transmitter has its own input: the level transmitter on **input 1**, the pressure transmitter on **input 2** and the flow meter on **input 3**. Greecon asks for the range printed on each label (below) and works out the register and the scale: nobody types a register or a conversion.

A 4 to 20 mA loop

A two-wire transmitter takes its power from the loop, so the loop needs a 24 V supply in series with the module's input. The electrician wires it as the module's manual draws it. The maker does not say that this module supplies the loop, so the loop has a supply of its own.

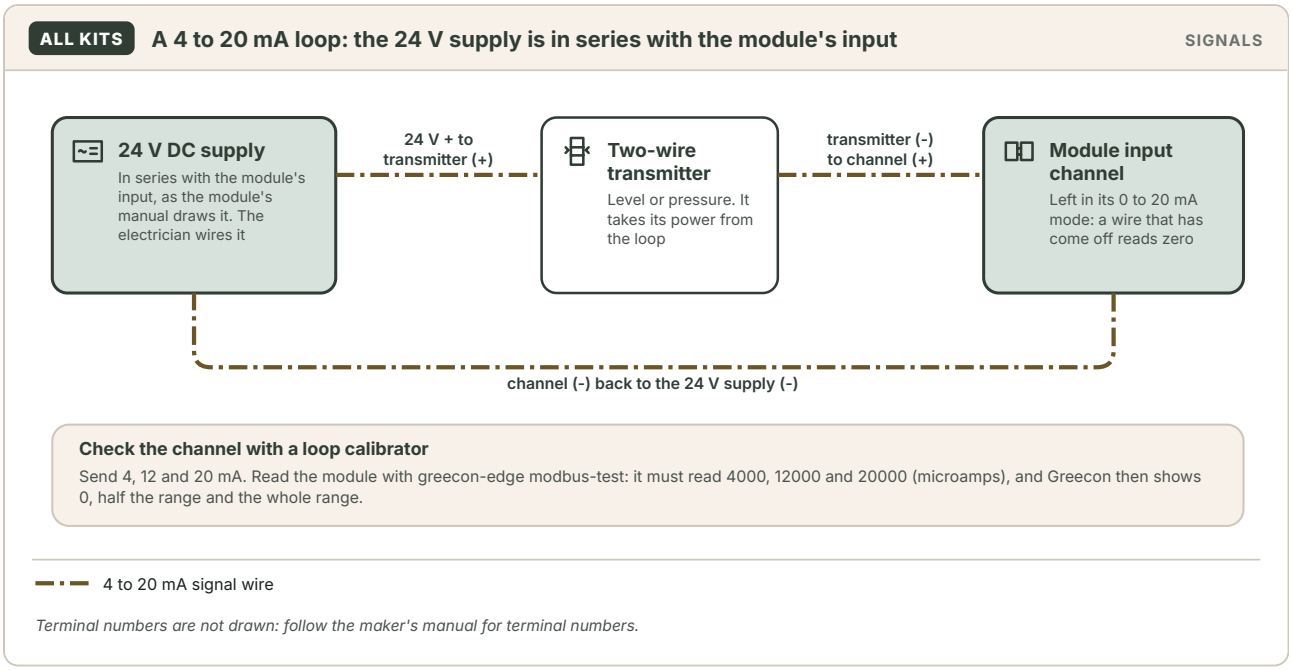


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

Ranges from the labels

The range of a transmitter is printed on its label, and the setup list asks for nothing else, in the label's own words:

- **Level transmitter:** the range on the label (a 0 to 5 m transmitter: 5), its unit (metres, or bar for a pressure sensor hung at the bottom of the tank: a bar holds up about 10.2 m of water) and how tall the tank is inside, from the bottom to the top of the water when it is full (say 4, in metres). Greecon shows a percentage of the tank: 0 when empty, 100 when full. A 0 to 5 m transmitter in a tank 4 m tall reads 100 at 16.8 mA and 125 at the top of its range (5 ÷ 4 × 100). One shorter than the tank is refused: it cannot see the top of the water.
- **Pressure transmitter:** the top of the range in bar (0 to 10 bar is usual).
- **Flow meter:** the top of the range in litres a minute.
- A transmitter the site does not have: tick **Not fitted at this site**. Greecon asks first: the reading and its history are removed.

Check the module with a loop calibrator (a hand-held that sends 4, 12 and 20 mA in place of a transmitter). Read the module from the box (address 0 is input 1, 1 is input 2, 2 is input 3):

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

It must read 4000, 12000 and 20000, and Greecon then shows 0, half the range and the whole range. On the tank's input the whole range reads more than 100 when the transmitter is longer than the tank (125 in the example), and Greecon may call that impossible while you test: that is the test, not a fault.

The pump contact module

WHAT IT DOES	Reports whether the pump is running, from a spare contact of its starter (a dry contact on input 1), as unit 5. It only reports; it never switches anything.
SPECIFICATION	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, W; optional in kit F

A spare auxiliary contact of the pump's starter is wired by the electrician to **input 1** as a dry contact, as the module's manual draws it (the module's DI COM is left unconnected). Never to a control circuit, and **never to the module's relay outputs**: Greecon only reads. **Check it by eye:** with the pump running, Pump running must read 1; stopped, 0. If it reads the other way round, the contact is inverted: the electrician uses the other contact of the starter. The module is optional: a pump whose starter has no spare running contact has none, the site then removes it from the list, and the handover says that "the pump is running but no water is flowing" cannot be raised.

The level transmitter

WHAT IT DOES	Measures how full the tank is. It hangs at the bottom of the tank.
SPECIFICATION	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)
PLANNING ESTIMATE (EUR)	120 to 400. Not a quote.
USED IN	Kits H, F, W

- The transmitter hangs at the bottom of the tank, so that zero is empty, its vented cable fixed so that it cannot move.
- **Only a transmitter with wetted parts approved for drinking water goes into a drinking-water tank.** A cheap sensor of unknown material is not worth a contaminated tank. If unsure, use a non-contact ultrasonic sensor above the water instead.
- The electrician or the customer's plumber fits it. If the tank must be entered, the customer's own staff do that and the project manager does not: if the only place for the sensor is inside, say it is not possible and write why.
- **In Greecon** give the range on the label and how tall the tank is inside (from the bottom, where the transmitter hangs, to the top of the water when it is full). Greecon works out the percentage: a 0 to 5 m transmitter in a tank 4 m tall reads 100 when the tank is full (16.8 mA), and 125 at the top of its range. Wired to input 1.

The pressure transmitter

WHAT IT DOES	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.
SPECIFICATION	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts
PLANNING ESTIMATE (EUR)	40 to 150. Not a quote.
USED IN	Kits H, F, W

On the pipe through an isolating valve, so that it can be removed without draining. A plumber does the wet connection. The pipe's pressure range decides the transmitter's range, and the top of that range, as the label says it, is what Greecon asks for. Wired to input 2.

The flow meter

WHAT IT DOES	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.
SPECIFICATION	4 to 20 mA output. Needed for "the pump is running but no water is flowing"
PLANNING ESTIMATE (EUR)	150 to 500. Not a quote.
USED IN	Optional in kits H, F, W

An electromagnetic or turbine meter in the pipe with a 4 to 20 mA output, installed with the straight run its manual asks for, and wired to input 3. Greecon asks for the top of its range, in litres a minute; where there is no flow meter the site ticks **Not fitted at this site**. Without it, or without the pump's running contact, "the pump is running but no water is flowing" cannot be raised, and the site should say so in its handover.

4.7 Farm sensors

A soil moisture probe and a temperature and humidity sensor that speak RS-485 Modbus RTU go on the same cable as everything else, each with its own unit number: soil probe 6, weather sensor 7 (set it before the probe is buried or the sensor mounted). They are not approved models: Greecon does not know their registers, and the box cannot recognise them on a network. So their address is entered by hand, and each reading is mapped by hand with **Choose the register**, from the maker's register table, which decides the register, the kind of number and the scale.

The soil moisture probe

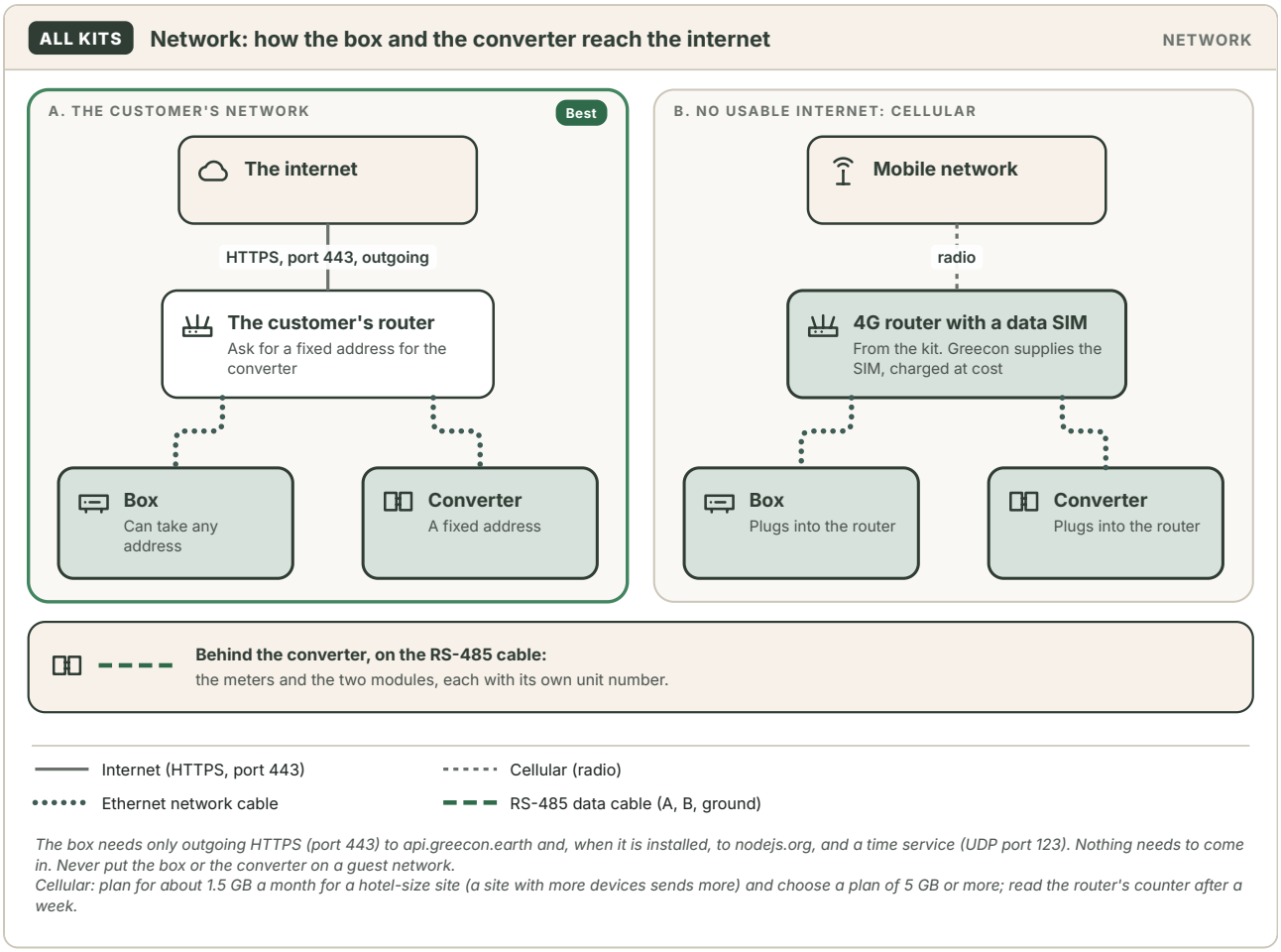
WHAT IT DOES	Measures soil moisture at the probe's depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker's register table.
SPECIFICATION	RS-485 Modbus, with a published register table, one for each zone
PLANNING ESTIMATE (EUR)	40 to 120 each. Not a quote.
USED IN	Kit F

Write down the probe's depth and where it is. Soil readings mean little without them, and they are only as good as the contact between the probe and the soil. Greecon gives no irrigation advice until the grower's value for "start irrigating below" is entered: after the first week of readings, ask the grower what reading means "time to water".

The temperature and humidity sensor

WHAT IT DOES	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker's register table.
SPECIFICATION	RS-485 Modbus, with a published register table
PLANNING ESTIMATE (EUR)	20 to 80. Not a quote.
USED IN	Kit F

4.8 The network



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure 4-5. The network: the customer's router (best), or a 4G router with a data SIM.

Situation	What to do
The customer's network	Box and converter on the router. Ask for a fixed address (a reservation) for the converter. The box needs only outgoing HTTPS (443) to `api.greecon.earth` (and to `nodejs.org` at install) and a time service (UDP 123). Nothing needs to come in. When someone presses **Find equipment**, the box looks at the site's own network for about a minute: it tries port 502 (or the port the person gives) on the private addresses of that network, such as 192.168.x.x and 10.x.x.x, only reads, and never writes to anything or looks outside the site's network. The install card has the words for the customer's network administrator.
Cellular	A 4G router with a data SIM; box and converter plug into its sockets. Plan for about 1.5 GB a month. That assumes a hotel-size site (four devices, seven readings) sending one request every ten seconds, about 5 KB each with its headers (about 1.3 GB), plus a heartbeat every 30 seconds (about 0.15 GB); a site with more devices sends more. It is an estimate: read the router's counter after a week and write the real figure down. Choose a plan of 5 GB or more.
No internet at all	Not sold. The box keeps about three days (for a site of up to about thirteen devices), which is for outages, not for a site that is never connected.

The cellular router

WHAT IT DOES	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.
SPECIFICATION	4G, DIN rail, data SIM
PLANNING ESTIMATE (EUR)	150 to 200. Not a quote.
USED IN	Kit F; optional in kits H, S, W

Keep the box and the converter on separate addresses from the customer's own equipment where you can, and never on a guest network.

4.9 Power

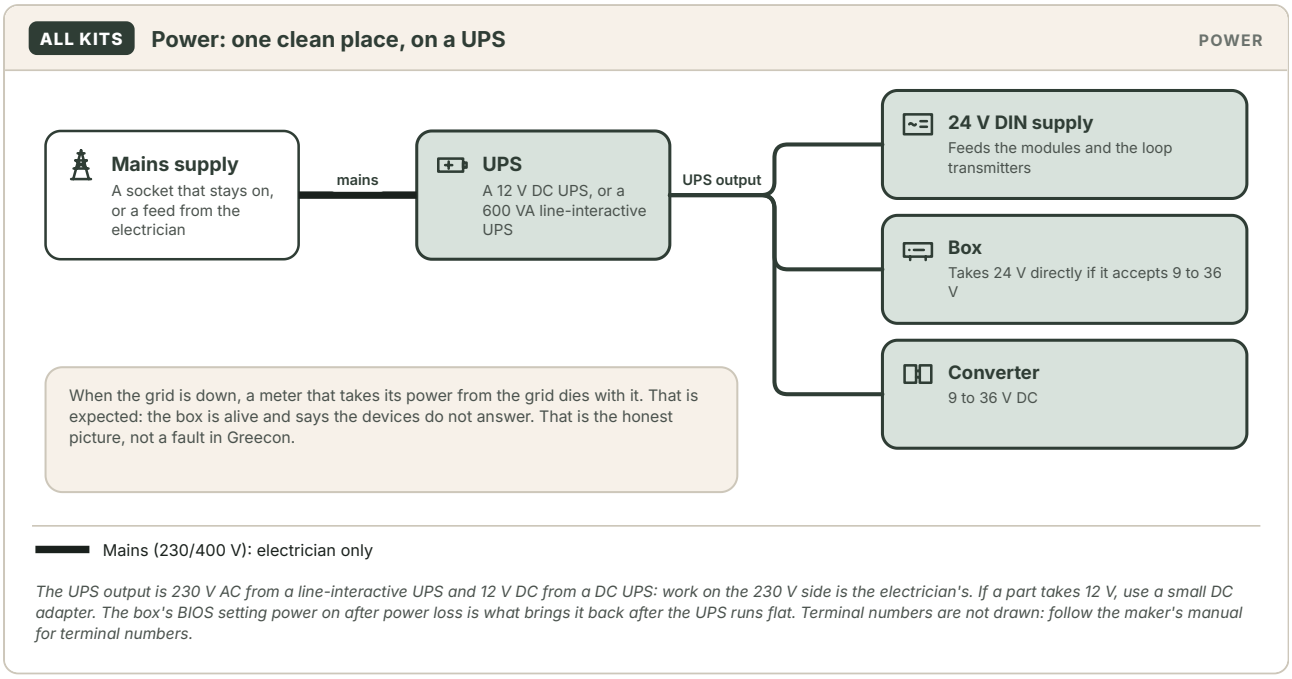


Figure 4-4. Power: everything from one clean place, on a UPS.

The 24 V power supply

WHAT IT DOES	Powers the two modules, the loop transmitters and the converter from one place.
SPECIFICATION	30 to 60 W
PLANNING ESTIMATE (EUR)	25 to 50. Not a quote.
USED IN	Kits H, S, F, W

The UPS

WHAT IT DOES	Keeps the box and the converter up through short power cuts and long enough to say that the grid is down.
SPECIFICATION	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS
PLANNING ESTIMATE (EUR)	60 to 150. Not a quote.
USED IN	Kits H, S, W

- One 24 V DIN supply feeds the two modules, the loop transmitters and, through a small DC adapter if the box takes 12 V, the converter. The box takes 24 V directly if it accepts 9 to 36 V.
- A UPS keeps the box and the converter up through short cuts and long enough to say that the grid is down. **When the grid is down, a meter powered from the grid dies with it.** That is expected: the box is alive and says the devices do not answer, which is the honest picture (“we cannot tell”), not a fault in Greecon.
- Auto power-on in the BIOS (section 4.3) is what brings the box back after the UPS runs flat.

4.10 Enclosure and labelling

The enclosure

WHAT IT DOES	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.
SPECIFICATION	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands
PLANNING ESTIMATE (EUR)	80 to 150. Not a quote.
USED IN	Kits H, S, F, W

One IP54 box (IP65 if outside) holds the converter, the two modules (the 4 to 20 mA module and, where there is one, the pump contact module), the supply, the fuses, the terminal blocks and, if there is room, the box. Label everything: the unit number on each device (set at the office) and the settings on the converter, the wire numbers at both ends, and the site name and the date on a sticker inside the lid. **Never write a code or a password on the box.** Photograph the inside before closing.

4.11 The project manager's tools

The project manager brings:

- [ ] A laptop
- [ ] A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- [ ] A loop calibrator (or a 4 to 20 mA source)
- [ ] A multimeter
- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's.

4.12 Buying

1. Ask three suppliers for a quote. For the approved models (the two Eastron meters and the two Waveshare modules) the model is named, and a look-alike is not the same part; for everything else it is the specification, not the brand. Keep the quotes. (The Appendices have a shopping list for each kit with room for three quotes.)

2. Buy one kit, bench-test it (the rehearsal), and fix the specification where it failed.
3. Then buy five. For each part keep: the make and model, the supplier, the date, the serial number or the batch, and the invoice. A spreadsheet is enough: the hardware register.
4. **Do not buy** a meter or a module that is not an approved model, a probe or a sensor whose register table you have not seen, a transmitter not rated for the fluid, or anything with a wireless link to the board that you cannot inspect.

Where the planning figures come from

Seen in October 2026, not quotes: Waveshare RS-485 to Ethernet converters, about EUR 25 to 60, and a look-alike at USD 45 (Electrokit, Botland); Eastron SDM630MCT V2 from about GBP 75 to 150 (Bimble Solar); a Waveshare 8-channel Modbus analog input module, about USD 25 to 36 (Waveshare); a Teltonika RUT241 4G router, about EUR 150 to 200 (Pricerunner); WIKA pressure transmitters, GBP 129 and up (RS); level transmitters from about USD 28 (unverified quality) to over GBP 900 (RS); fanless N100 mini PCs from about USD 112 in bulk to over USD 400 retail, with the real-time clock not stated by the listings: ask the vendor (Newegg). The Eastron register tables and defaults are in the maker's own manuals (the SDM630 Modbus protocol document and the SDM230 user manual V1.6): read the one that came with the meter before relying on this page, because versions differ. The planning figure for the pump contact module (30 to 60) is the earlier figure for a pump run input: it has not been checked against a shop price for the Waveshare Modbus RTU IO 8CH, and the registers of both Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk and anything that differs is reported to Greecon.

PART 5

The four site packs

There is one pack for each kind of site Greecon is sold for. A pack is complete on its own: it says what the site suits and measures, what to buy, what to ask at the survey, how everything is wired and connected (a system, a wiring, a network and an enclosure drawing for that kit), what to do at the office and on install day step by step with a box to tick and a name against every step, the numbers, addresses and ranges to check and write down, the platform's twelve setup steps in order, the commissioning tests with room for the numbers, the handover record, and the three problems most likely to meet you.

Pack	The site	Template in the platform	Planning estimate	Printed on its own as
H	Hotel or guesthouse: Water, solar and grid	Hotel or guesthouse	About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.	PDF 04
S	Solar plant with grid connection: Solar and grid, with an optional battery	Solar plant with grid connection	About EUR 600 to 1,200.	PDF 05
F	Farm or greenhouse with irrigation: Soil and weather, with the water supply behind it	Farm or greenhouse with irrigation	About EUR 700 to 1,500.	PDF 06
W	Water tank and pump station: Tank, pump and the pipe it feeds	Water tank and pump station	About EUR 650 to 1,300.	PDF 07

The four packs follow the same chapters in the same order, so that you find the same thing in the same place at every site. What they have in common is the same everywhere: the approved models, the unit numbers (grid meter 1, solar meter 2, 4 to 20 mA water module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7), set at the office, the data cable at 9600 baud with 8 data bits, no parity and 1 stop bit, the converter on port 502 in gateway mode, and Greecon only reading. The reasons for each part, its settings and the mistakes to avoid are in the Hardware Reference (Part 4).

A site that fits none of the four is not forced into one. The platform has an empty template (nothing assumed); use it only when none of the four fits, and ask the founder first.

5.1 Site pack H: Hotel or guesthouse

Water, solar and grid. Planning estimate: About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours. This pack is also printed on its own as PDF 04 (04-Site-Pack-H-Hotel-or-Guesthouse.pdf).

At a glance

WHO IT SUITS
A hotel or guesthouse with a water tank and a pump, solar on the roof and a grid connection. The usual first site.

WHAT IT MEASURES
The grid (import and export), the solar production, the tank, the line pressure, the pump's flow (with a flow meter) and whether the pump runs.

PLANNING ESTIMATE
About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.

TIME
Survey about 1 hour. Install day 4 to 6 hours with the electrician. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Hotel or guesthouse

STANDARD SETTINGS
Unit numbers: grid meter 1, solar meter 2, water module 3, pump contact module 5 (if the starter has a spare contact). 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode. All set at the office.

The order of the work: survey, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the labels, and the 12 or 24 V supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with their 24 V supply, as the module's manual draws it.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

HOW TO READ THIS PACK

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, the pump's run contact, and the 4 to 20 mA loops.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure H-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Hotel or guesthouse** makes four devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Grid import and **Grid export**	Grid meter: an Eastron SDM630 where the supply enters, unit 1. One meter gives both: positive power is energy coming in, negative is energy sent out	kW	None
Solar production	Solar meter: an Eastron SDM630 on the inverter's AC output, unit 2	kW	None
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow	Water I/O module, unit 3: the flow meter's 4 to 20 mA (optional)	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running	Pump contact module (optional), unit 5: the pump starter's spare contact	on or off	None

Without the flow meter, or without the pump's running contact, "the pump is running but no water is flowing" cannot be raised, and the handover should say so. Solar surplus has no standard limit: the platform says nothing about it rather than guess.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *One at the grid connection, one on the inverter's AC output.*	2	90 to 180 each
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200

About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in Part 4.

THE PROJECT MANAGER BRINGS

- [ ] A laptop
- [ ] A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- [ ] A loop calibrator (or a 4 to 20 mA source)
- [ ] A multimeter
- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the electrician if the plan needs one. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

ASK, IN THIS ORDER

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or a guest? Write it in their words. It is the reason they buy.
3. The last twelve electricity bills (price per kWh, kWh bought each month) and the water bills. The platform shows what solar kept on site at their price; without a price it shows kWh only.
4. Is there solar? What brand is the inverter, and where is its AC output? Is it allowed to send energy to the grid?
5. Where does the supply enter the building? Is it three-phase?
6. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Has the starter a spare "running" contact?
7. Is there a network socket in the plant room, or cellular signal? (Look at your phone there.)
8. Is there a power socket for the box, and is it on a circuit that stays on?
9. Who is the customer's electrician, and can they be there on install day?

LOOK AT, AND PHOTOGRAPH

- The supply entry and the main board (closed): where a meter could be fitted with the clamps around the incoming phases.
- The inverter and its AC output cable.
- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.

- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable.
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT H · HOTEL OR GUESTHOUSE

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Prices from the bills: electricity per kWh; paid for energy sent to the grid; water per m^3	
Twelve electricity bills and the water bills collected	☐ Yes ☐ Later
Supply: single or three phase; main breaker size (A)	
Inverter: brand, size (kW), where its AC output cable is; allowed to send to the grid?	
Tank: height (m), water source; pump (kW); starter location; spare running contact on the starter; pipe pressure (bar)	
Meters needed: grid, solar (and where)	
Sensors needed: level, pressure, flow, pump running (and where)	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit H and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers. Mains work is inside the dashed red box and is the electrician's.

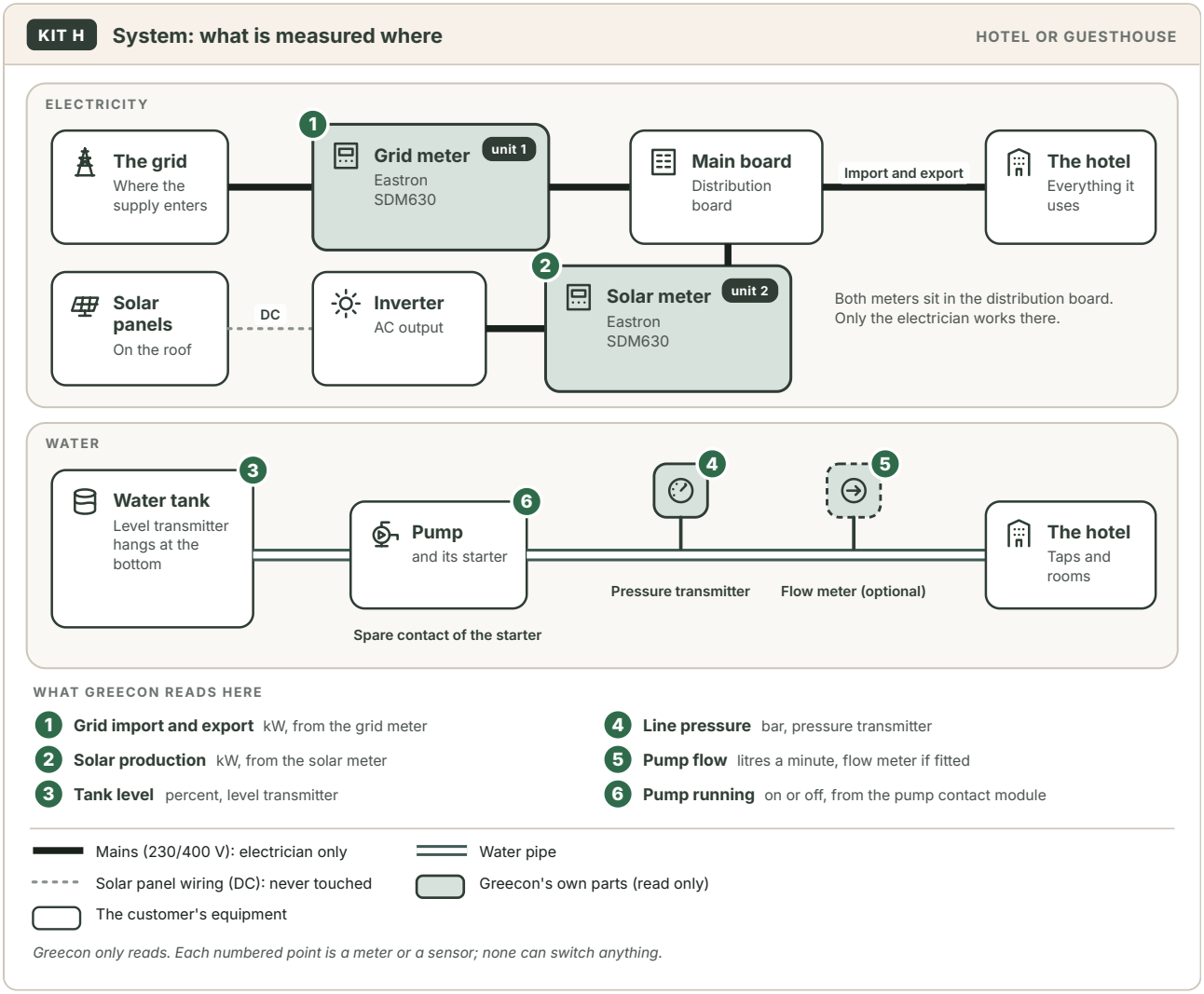
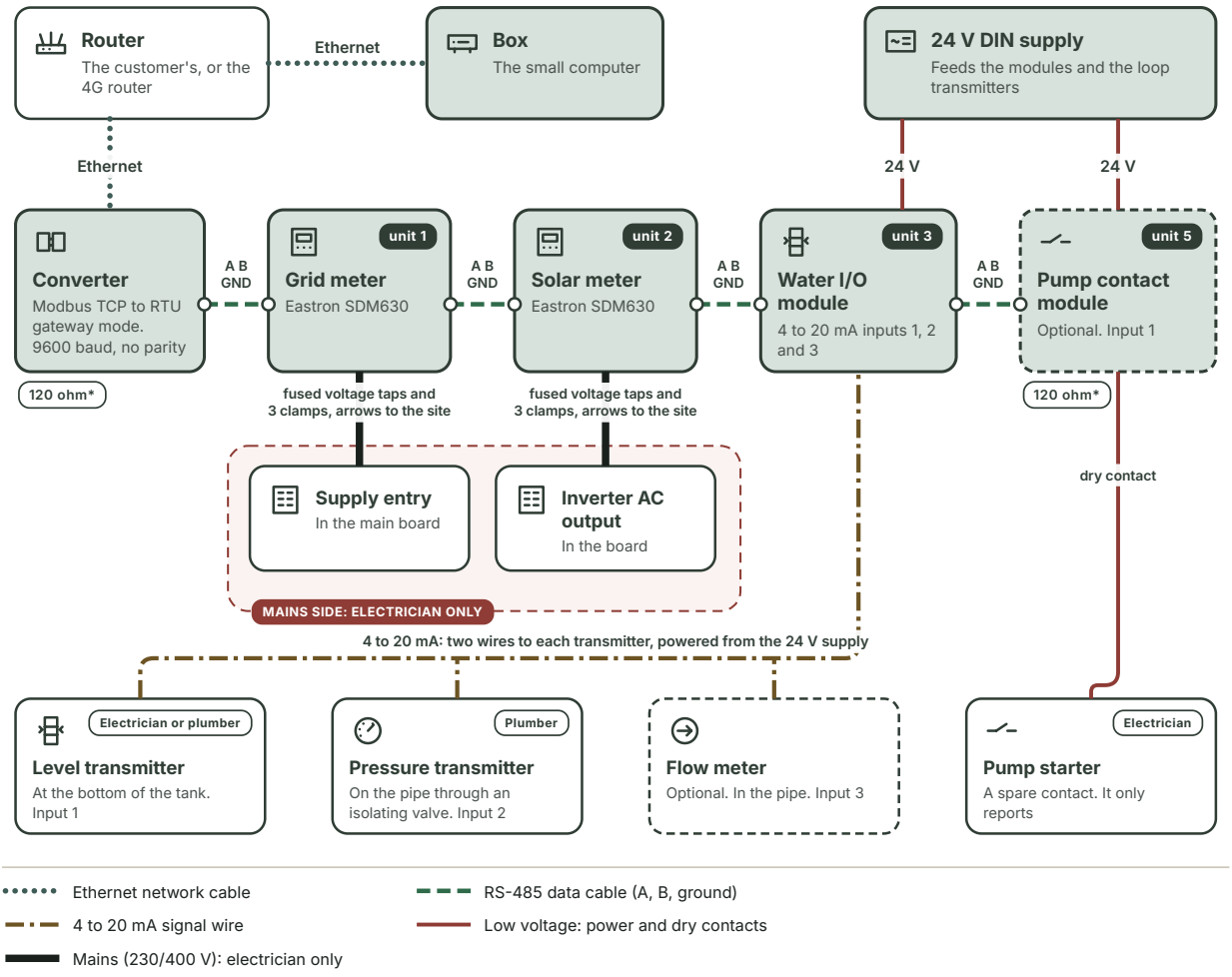


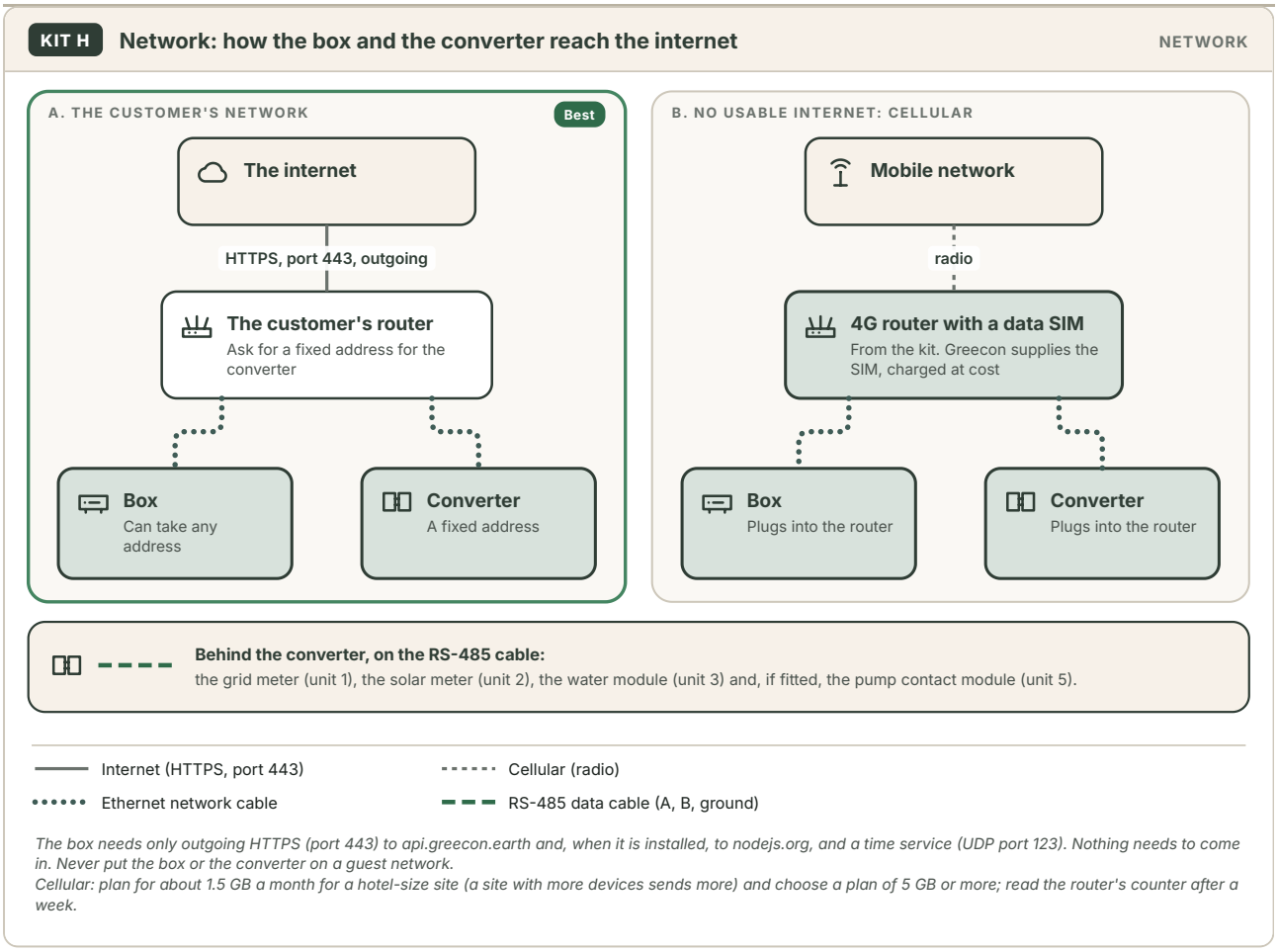
Figure H-1. Kit H, hotel or guesthouse. System: what is measured where. The numbers mark what Greecon reads; they are listed under the drawing.

KIT H **Wiring: the data cable, the signal wires and the mains side** BUS AND SIGNALS



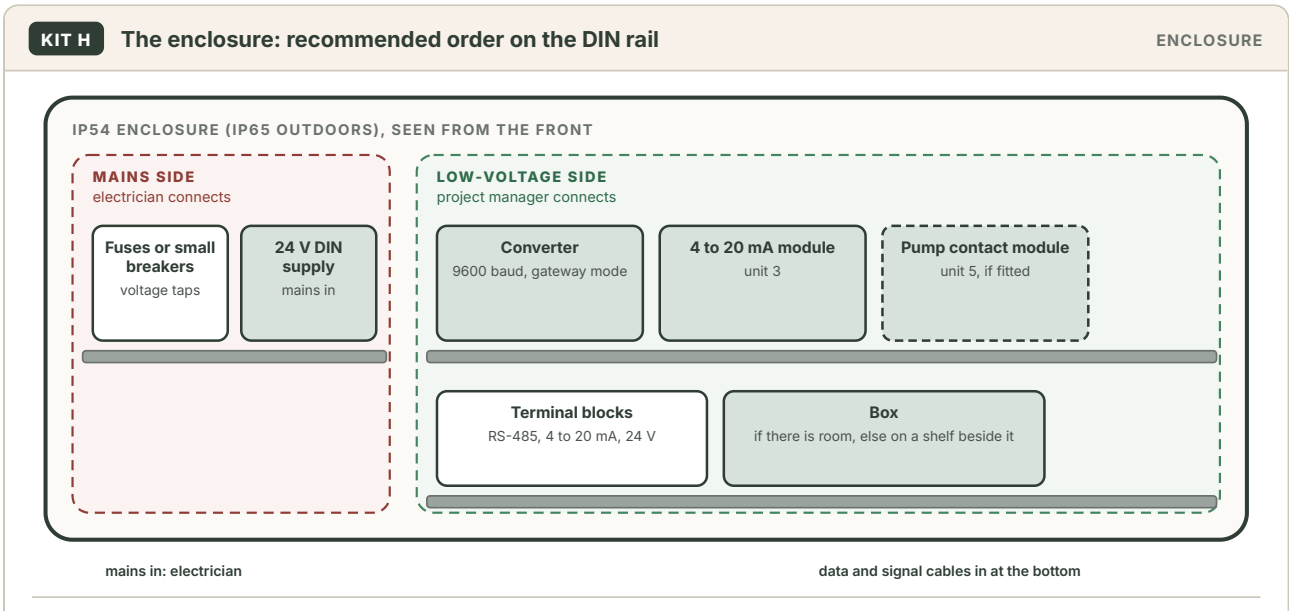
Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The electrician wires the 4 to 20 mA loops and the contact. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure H-2. Kit H. Wiring: one RS-485 cable from the converter through the grid meter, the solar meter and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module. Mains work is inside the dashed box.



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure H-3. Kit H. Network: the box and the converter on the customer's router (best) or on a 4G router.



The meters are not in this enclosure: they sit in the distribution board at the point they measure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure H-4. Kit H. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

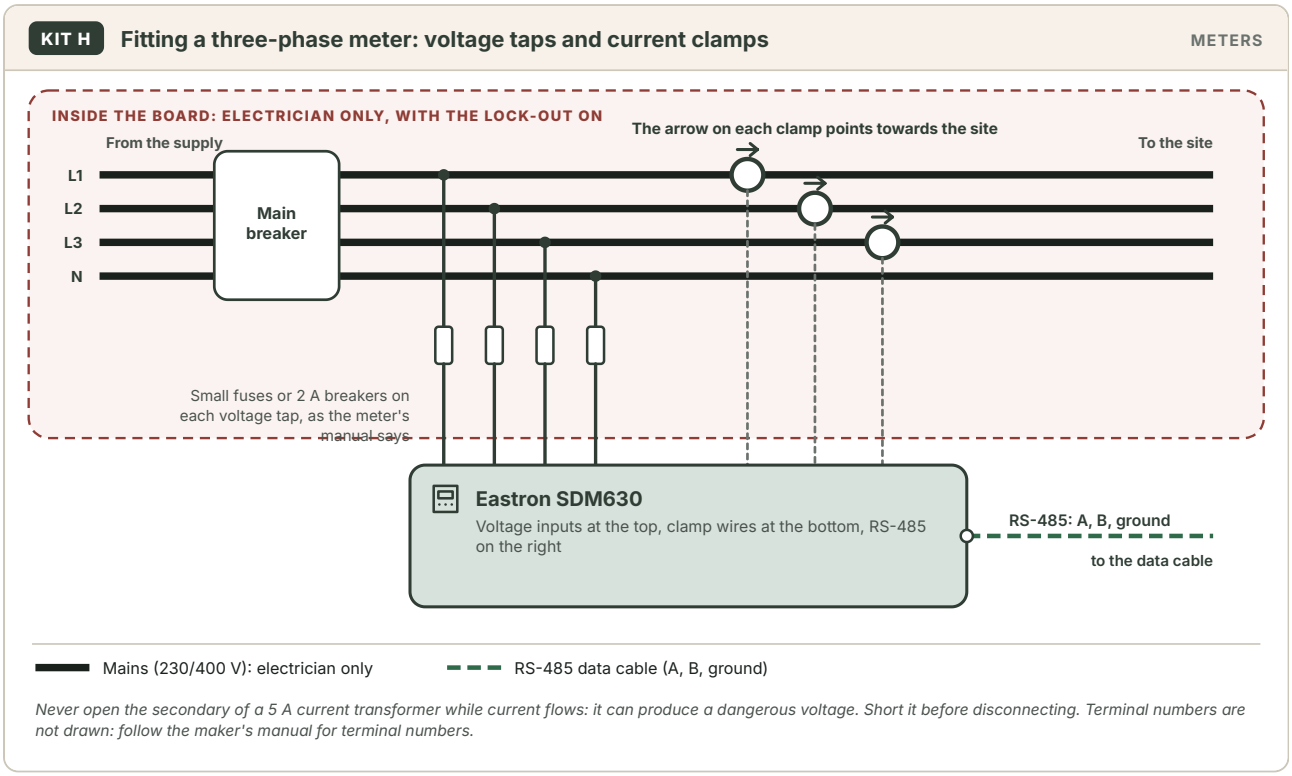


Figure H-5. Kit H. Fitting the grid meter and the solar meter: fused voltage taps and three clamps each, arrows towards the site.

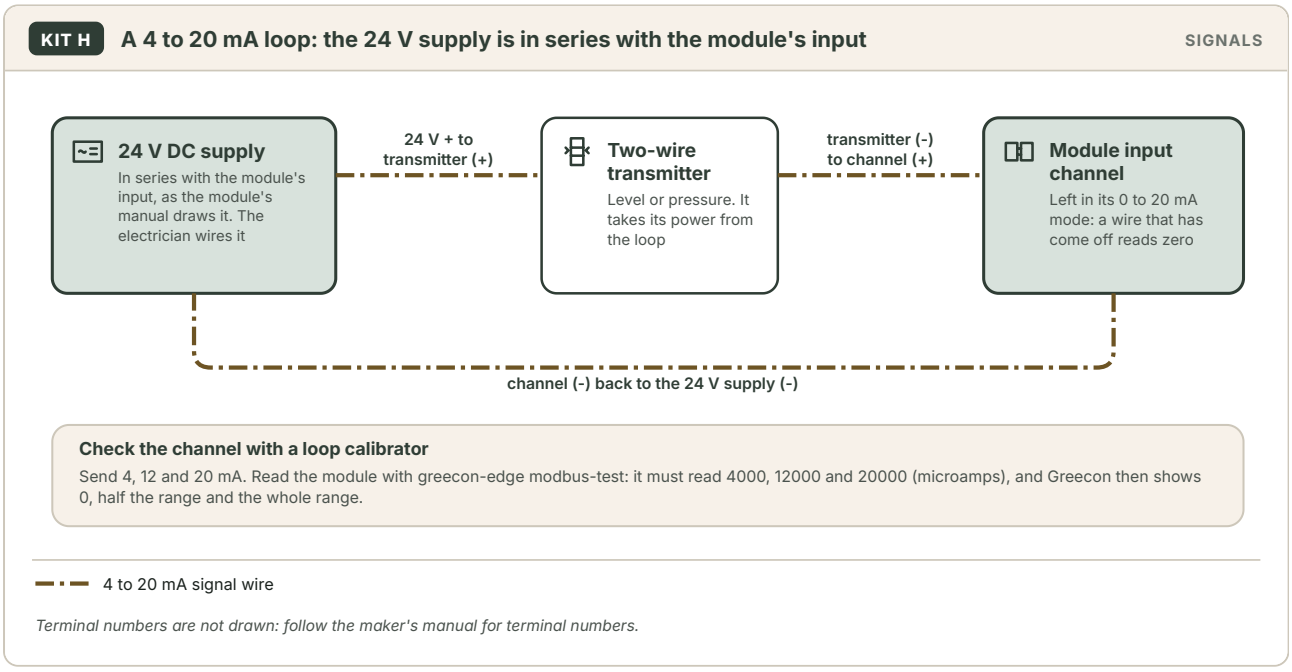


Figure H-6. Kit H. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

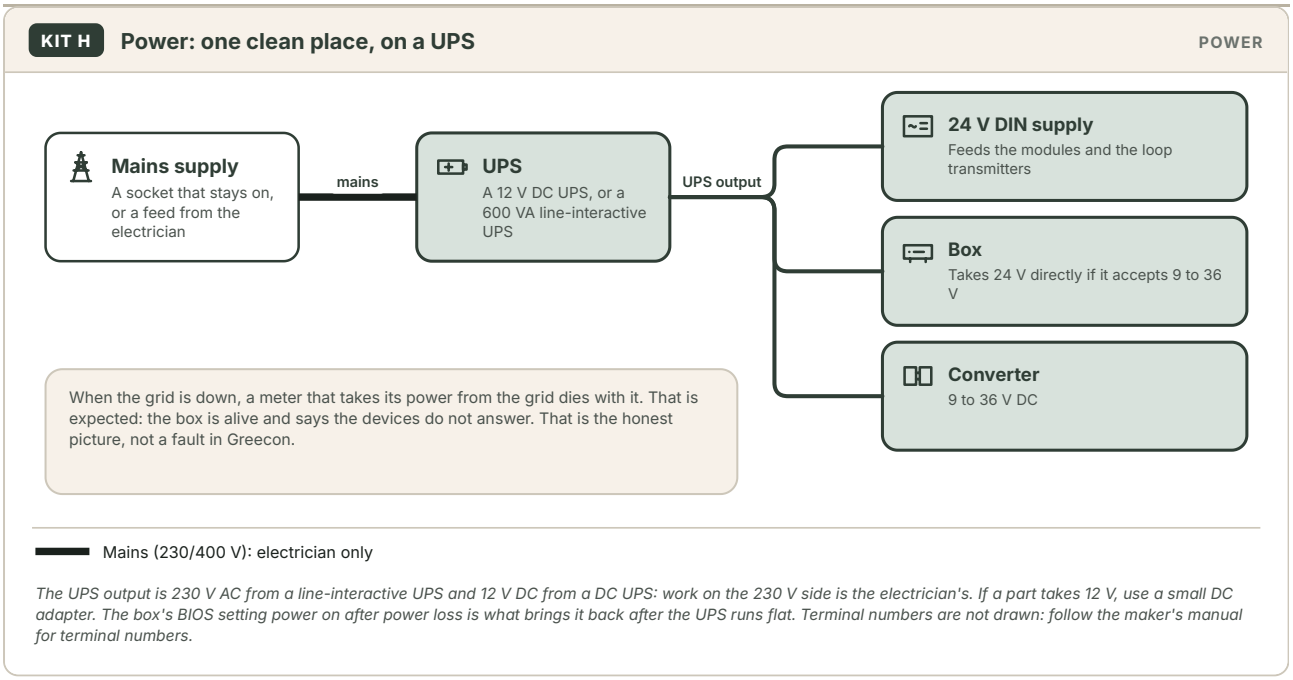


Figure H-7. Kit H. Power: the UPS feeds the box, the converter and the 24 V supply.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box, and more for the first kit's numbers and its desk test: time it, and write it down. PROJECT MANAGER

IN GREECON

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Hotel or guesthouse**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.

- [ ] **Write down that the papers are signed:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`, and its first step is **Order form signed and data terms agreed**. With the signed order form and the agreed data terms in front of you, write in the box who signed for the customer and on what date (it cannot be left empty), and press **The papers are signed**. Nothing goes into a customer's building without both papers.

- [ ] **Print the install card:** the link **The install card** is at the top of the setup page. It is your list for the visit: what to bring, what each device is set to and its number, how it is wired, the converter's settings, and what the customer's network is asked for. At the bottom are three messages to copy: send the first to the customer's contact two or three days before, and the second to whoever looks after their network. Each is in English with an *Albanian draft* folded under it, and someone who writes Albanian every day reads the draft before it is sent.

- [ ] **Get the install line:** on the same page press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> **STOP** **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

THE BOX ON YOUR DESK

The operating system and the pasted install line are still a terminal job; everything after them is done on the setup page.

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in Part 4, section 4.3.

- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).

- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`

- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.

- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.

- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit, the printed pack and the printed install card.

THE NUMBERS AND THE CONVERTER

The kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office; on site you only check the labels and do not change them. The numbers for this kit are in the table under **Numbers, addresses and ranges** and on the install card. Everything on one cable is **9600 baud, 8 data bits, no parity, 1 stop bit**.

☐ **Set each meter's number on its own display,** one meter at a time (delivered as unit 1, two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page. Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains.

☐ **Number the two Waveshare modules** with the maker's setting tool and a USB-to-RS-485 adapter, one at a time (each is delivered as address 1, and there is no switch): **3** on the 4 to 20 mA module (Modbus RTU Analog Input 8CH) and **5** on the pump contact module (Modbus RTU IO 8CH). Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default 0 to 20 mA mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.

☐ **Set the converter** in its web page: mode **Modbus TCP to RTU (gateway)** (not transparent, not raw TCP, not Modbus RTU over TCP), 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site: a fixed one is best, and Greecon finds the converter either way.

☐ **Label every device with its number** and the converter with its settings, and photograph the labels.

☐ **Test the kit on the desk.** The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (Part 3, section 3.11). Anything that differs from this book is reported to Greecon.



If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.



The first time, rehearse the whole job on a desk with a laptop standing in for the kit: Part 3, section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN + PROJECT MANAGER	**Before anyone touches anything:** the signed papers are in your hand, the electrician is there, and the lock-out is agreed. **Walk the site together.** Agree where each meter and sensor goes. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the main board, the inverter, the tank room and the starter. Have the router and a socket that stays on ready, and the person who can give the converter a fixed address.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the board to be worked on.
☐	4	ELECTRICIAN	**Fit the grid meter** at the supply entry: voltage taps with their small fuses or 2 A breakers as the meter's manual says, and the three current clamps around the incoming phases, **arrow towards the site**. Figure H-5.
☐	5	ELECTRICIAN	**Fit the solar meter** on the inverter's AC output, after the inverter and before the distribution board: the same, clamps with the arrow towards the site.
☐	6	ELECTRICIAN	**Connect the pump's run contact** (a spare auxiliary contact of the starter, as a dry contact) to **input 1 of the pump contact module**, as the module's manual draws it (its DI COM left unconnected). Never to a control circuit, and never to the module's relay outputs: the contact only reports. If the starter has no spare contact there is no module: it is taken off the list in setup step 4.
☐	7	ELECTRICIAN + PROJECT MANAGER	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts, and **wire their 4 to 20 mA loops** with the 24 V supply in series, as the module's manual draws it: **level on input 1, pressure on input 2, flow meter on input 3.** The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve. Figure H-6. You check each is on its own input and number the wires at both ends.
☐	8	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the two modules (4 to 20 mA and pump contact), the 24 V supply and the UPS (the box too, if there is room, otherwise on a shelf beside it). Figure H-4.
☐	9	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the grid meter, on to the solar meter, then to the 4 to 20 mA module and the pump contact module: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. The electrician connects its ends at the meters' A and B terminals inside the board. Figure H-2.
☐	10	PROJECT MANAGER	**Check the labels:** the number on each device against the table under *Numbers, addresses and ranges* (grid meter 1, solar meter 2, water module 3, pump contact module 5). They were set at the office: do not change them. Photograph each label.
☐	11	PROJECT MANAGER	**Check the converter's label** (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502) and **connect the network:** plug the box and the converter into the customer's router, or into the sockets of the 4G router if the site has no usable internet. Ask for a fixed address for the converter and write it down. Figure H-3.

	#	Who	Do this
☐	12	ELECTRICIAN / PROJECT MANAGER	**Photograph before closing:** every meter's label (model and serial number), the wiring, the converter's settings. Then close the board.
☐	13	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	14	PROJECT MANAGER	**Power up** the UPS, the 24 V supply, the converter and the box.
☐	15	ELECTRICIAN	**Stay until commissioning is done.**
☐	16	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list: **Find equipment**, the ranges, and the electrician's name and licence number.

Numbers, addresses and ranges

The numbers on the cable were set at the office. The box finds where the converter is. You give the range printed on each transmitter's label. You type no IP address, no port and no register.

THE NUMBERS ON THE CABLE

All four devices are behind one converter: they share its address and port 502, and differ only by number. Check each label against this table on install day. **Do not change a number on site.**

Device in Greecon	Make and model	Number	Label checked	Address found (the converter's)
Grid meter	Eastron SDM630	1	☐	
Solar meter	Eastron SDM630	2	☐	
Water I/O module	Waveshare Modbus RTU Analog Input 8CH	3	☐	
Pump contact module (optional)	Waveshare Modbus RTU IO 8CH	5	☐	

THE EASTRON METERS AND THE PUMP CONTACT MODULE

Nothing to type: the template fills in their registers. For the meters it is total active power, input register 52 (a manual's 30053; counted from zero), a 32-bit floating point number with the high half first, in watts, multiplied by 0.001 to make kW; for the pump contact module it is digital input 1. Check the module by eye: with the pump running, **Pump running** reads 1; stopped, 0. Test from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
```

THE RANGES ON THE LABELS

One form for each transmitter (step 5 of the setup list), in the label's own words. Read the range from the label on the transmitter and measure how tall the tank is inside, from the bottom to the top of the water when it is full.

Reading	Input	The form asks for	Range on the label	Tank height
Tank level	1	Level transmitter: the range on the label (m or bar) and how tall the tank is inside (m)		
Line pressure	2	Pressure transmitter: the top of the range (bar)		
Pump flow	3	Flow meter: the top of the range (lpm, litres a minute)		

Greecon works out the register and the scale: 4 mA is 0 and 20 mA is the range (for the tank, range ÷ tank height × 100: a 0 to 5 m transmitter in a tank 4 m tall gives 125, and a full tank, at 16.8 mA, reads 100). **A transmitter shorter than the tank is refused:** it cannot see the top of the water. **No flow meter?** Tick **Not fitted at this site** under Flow meter: its reading is taken away, and the handover says that the no-flow alert cannot be raised. **No spare contact for the pump?** Press **The site does not have this: remove it** under the pump contact module in step 4.

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to twelve steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself: you press buttons only for what a person must check or say. Near the top are links to the site page, to **The install card** (the card for the visit) and to the **Handover sheet**.

	Step	Title on the setup list	Done by
☐	1	**Order form signed and data terms agreed**	Project manager
☐	2	**Prices from the customer's bills**	Project manager (optional)
☐	3	**Install the box**	Project manager
☐	4	**Where each device is on the network**	Project manager
☐	5	**Which register holds which reading**	Project manager
☐	6	**Every device answers the box**	The platform, by itself
☐	7	**Readings arrive and look right**	The platform, by itself
☐	8	**Someone is told when the site needs attention**	Project manager
☐	9	**Grid meter direction checked**	Project manager, with the electrician
☐	10	**Alert limits agreed with the customer**	Project manager
☐	11	**Electrical work done by a licensed electrician**	Project manager, with the electrician
☐	12	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

STEP 1 ORDER FORM SIGNED AND DATA TERMS AGREED

PROJECT MANAGER

DO	Nothing goes into a customer's building until the order form is signed and the data terms are agreed. Greecon keeps the papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date** (for example *Signed for the hotel by Besa Krasniqi on 19 October*): it cannot be left empty. Press **The papers are signed**. Do this at the office, and bring copies on install day.
YOU SHOULD SEE	The step turns **Done**: *Signed and agreed.*, with your name, the time and what you wrote. They are on the handover sheet.
IF YOU DO NOT	Until it is done, commissioning stays grey. The button will not work while the box is empty. If a paper turns out to be wrong, press **Reopen** (if the site had been commissioned, that is taken back too).

STEP 2 PRICES FROM THE CUSTOMER'S BILLS

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

STEP 3 INSTALL THE BOX

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	• *Waiting for the install command to be run on the box.* The line has not been run yet. • *The box has its credential but has not reported yet.* Wait a minute and check the network cable. • *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. • *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

STEP 4 WHERE EACH DEVICE IS ON THE NETWORK

PROJECT MANAGER

DO	Press **Find equipment**. Nobody types an IP address, a port or a unit number here. - The box looks at the site's own network (local private addresses only, port 502, about a minute) and only reads. The page says *The box is looking at the network now.* and updates by itself. The platform recognises what answered and fills in at once only what is certain: a device with no address, found at the kit's own number. The rest waits for you. - A device **Found** but not filled in (another address, another number, another meter model): read the line and, if it makes sense, press **Use this address** (or *Use these N addresses*). An optional device the site does not have: press **The site does not have this: remove it**. - The soil probe, the weather sensor and a battery are entered by hand: under the device open **Enter the address by hand** (soil probe 6, weather sensor 7). - Box not on your network (a desk rehearsal), or a converter on another port: open **I know the converter's address**, type its address (and its port, if not 502), and press **Look at this address**.
YOU SHOULD SEE	Each device gets a line in plain words: **Set** (found, address filled in), **Found**, **Not found**, **Not sure which** or **Enter by hand**. When it worked the step is **Done** (*Every device has an address.*), and the button reads **Look again**.
IF YOU DO NOT	- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* An optional device the site does not have: press **The site does not have this: remove it**. - **Not sure which**: Greecon will not guess, because a wrong guess could swap the grid meter and the solar meter. Read each device's label, use **Enter the address by hand** under each with the number on its label, then **Look again**. Numbers are set at the office: a part with a wrong number goes back there. - *No Modbus converter answered on this network:* check that the converter is on and plugged into the same router as the box. *A converter answers at 192.168.1.50, but nothing answers behind it:* check each meter's address, the speed and the A and B wires. *All answered with exactly the same words:* the converter is not in gateway mode. *The box did not pick this up:* check that it shows as online in the step above. - Readings that stop after it had worked: the router may have given the converter a new address. Press **Find equipment**, then **Use this address**.

STEP 5 WHICH REGISTER HOLDS WHICH READING

PROJECT MANAGER

DO	The Eastron meters and the pump contact module need nothing typed. For the 4 to 20 mA module the page asks, for each transmitter, only for the range printed on its label, in the label's own words: - **Level transmitter** (input 1): the **Range on the label**, for example 5, its **Unit on the label** (metres, or bar) and **How tall the tank is inside** in metres, for example 4: from the bottom to the top of the water when full. - **Pressure transmitter** (input 2): the **Top of the range** in bar; 10 is usual. - **Flow meter** (input 3): the **Top of the range** in litres a minute. Greecon works out the register and the scale. For a transmitter the site does not have (no flow meter), tick **Not fitted at this site**: Greecon asks first, then takes the reading and its history away. Press **Save the ranges**. Equipment Greecon does not know (the soil probe, the weather sensor, a battery) is mapped by hand, from the maker's register table: under the reading press **Choose the register** and fill in the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero: 30053 is address 52, 40001 is address 0), the kind of number (and which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.* Under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (0 to 5 m in a tank 4 m tall); the tank level is a percentage of the tank, so a full tank reads 100.
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *Needs the range printed on the transmitter's label.* (the 4 to 20 mA module) or *No register chosen.* (equipment Greecon does not know). A transmitter shorter than the tank is refused: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* A reading the site does not have holds the list here: tick **Not fitted at this site**, or press **The site does not have this: remove it** (step 4) for an optional device.

STEP 6 EVERY DEVICE ANSWERS THE BOX

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings).* On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2).* Look in this order: the converter's address may have changed (press **Find equipment** again, then **Use this address**); the number on the device's label against the install card; the speed at the converter; A and B swapped on the data cable; the converter not in gateway mode. Nothing answers at all: the converter is off, or it is on another network than the box. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

STEP 7 READINGS ARRIVE AND LOOK RIGHT

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* **Pump running** reads 1 while the pump runs and 0 while it is stopped.
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter a reading below zero is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank); one far too high is a range or tank height typed wrongly: give the ranges again (step 5).

STEP 8 SOMEONE IS TOLD WHEN THE SITE NEEDS ATTENTION

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	■ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ■ *Nobody is set to be told:* add the people. ■ *Nobody on the list has an email address* (or a phone number): add one for each person. ■ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ■ *Test messages are limited to 6 an hour.* Wait and try again.

STEP 9 GRID METER DIRECTION CHECKED

PROJECT MANAGER | ELECTRICIAN

DO	Greecon cannot see which way the meter's clamps point. With the sun down or the solar switched off, switch on a known load (a kettle or a heater). **Grid import must go up.** If **Grid export** goes up instead, the clamps are the wrong way round: the electrician turns them (with the lock-out on) and you check again. Then press **I checked this** with a note of what you saw.
YOU SHOULD SEE	Before: **To do**, *Greecon cannot see which way the meter's clamps point.* After you confirm: **Done**, *Checked on site,* with your name and the time.
IF YOU DO NOT	Import and export swapped: the clamps are the wrong way round. The electrician turns them; you repeat the check. A reversed clamp swaps what the site buys with what it sells, and every figure with it.

STEP 10 ALERT LIMITS AGREED WITH THE CUSTOMER

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

STEP 11 ELECTRICAL WORK DONE BY A LICENSED ELECTRICIAN

PROJECT MANAGER | ELECTRICIAN

DO	Fitting meters in a board and wiring transmitters and contacts is the **licensed electrician's** work, under the customer's lock-out. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number** (for example *Arben Hoxha, licence E-1234*), or that no electrical work was needed: it cannot be left empty. Press **The electrical work is done**.
YOU SHOULD SEE	The step turns **Done**: *Done by a licensed electrician.*, with your name, the time and what you wrote (they are on the handover sheet). The platform does not check the licence: what you write is the record.
IF YOU DO NOT	Until it is done, commissioning stays grey. Do not press the button while the work is open or the lock-out is still on. The button will not work while the box is empty.

STEP 12 COMMISSIONED

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 12) with a note. It stays grey until every other step is done, the papers (step 1) and the electrician's name and licence number (step 11) included. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Grid meter's display against Greecon. Read the display (kW or W) and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Solar meter's display against Greecon. The same, on the solar meter.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on (a kettle or a heater), the sun down or the solar off.	Import rises; export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's own display.	Within about 5 percent			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running. The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail

Check and how	Pass when	Greecon shows	Real	Result
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); **Energy now**, the flow moving from the solar to the site and the grid; **Value**, what the solar kept on site this month and the prices it uses; the site's own page under **Sites**, with the tank level, the pressure and whether the pump runs; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Send the third message on The install card** to the customer's contact the same day, with the handover sheet attached (its Albanian is a draft: have it read first).
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT H · HOTEL OR GUESTHOUSE

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Alert limits agreed (tank Critical below, Low below; no-flow level)	
Without a flow meter or the pump's run contact the no-flow alert cannot be raised: said to the owner	☐ Said ☐ Not applicable
Prices entered (electricity, export, water) and where they came from	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, Energy now, Value, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
The solar meter reads zero at midday	A clamp closed on the wrong cable, a closed breaker upstream, or the inverter is limited to zero export	Compare with the meter's own display and test the register from the box (`greecon-edge modbus-test`). Ask the electrician to confirm that the clamps are on the inverter's output phases. If the plant is zero-export, midday solar can honestly be low: check the inverter's display
The tank level is wrong (above 100 percent, below zero, or does not move)	The range or the tank's height was given wrongly; a wire has come off (the module reads 0 and the level reads below zero); a blocked vented cable; or the transmitter is not at the bottom of the tank or not on input 1	Give the ranges again (**Save the ranges**). Check that the transmitter hangs at the bottom with its vented cable free and fixed. Send 4, 12 and 20 mA with the loop calibrator: the module must read 4000, 12000 and 20000
The pump shows running when it is stopped (and the no-flow alert, an Act now, opens while the pump is stopped)	The pump's running contact is inverted	The electrician swaps the contact's normally-open and normally-closed wires, or uses the opposite contact. Until it is right the no-flow alert cannot be trusted

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: the converter's address changed (press **Find equipment** again), the number on the device's label not the one on the install card, a wrong speed at the converter, A and B swapped on the data cable, the converter not in gateway mode.

5.2 Site pack S: Solar plant with grid connection

Solar and grid, with an optional battery. Planning estimate: About EUR 600 to 1,200. This pack is also printed on its own as PDF 05 (05-Site-Pack-S-Solar-Plant.pdf).

At a glance

WHO IT SUITS
A solar plant with a grid connection, with or without a battery. It works with any inverter brand, because it measures at the AC cable with its own meter and not through the inverter maker's portal.

WHAT IT MEASURES
How much the solar produces, the grid (import and export) and, if wanted, the site's own consumption (a load meter) and a battery's charge and power.

PLANNING ESTIMATE
About EUR 600 to 1,200.

TIME
Survey about 1 hour. Install day 4 to 6 hours with the electrician. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Solar plant with grid connection

STANDARD SETTINGS
Unit numbers: grid meter 1, solar meter 2, optional load meter 4, set at the office. A battery is addressed by hand (for example 8). 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode.

The order of the work: survey, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the labels, and the 12 or 24 V supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with their 24 V supply, as the module's manual draws it.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

HOW TO READ THIS PACK

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, the pump's run contact, and the 4 to 20 mA loops.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure S-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Solar plant with grid connection** makes up to four devices. Take off what the site does not have, with **The site does not have this: remove it** (setup step 4, below). What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Grid import and **Grid export**	Grid meter: an Eastron SDM630 where the supply enters, unit 1. One meter gives both: positive power is energy coming in, negative is energy sent out	kW	None
Solar production	Solar meter: an Eastron SDM630 on the inverter's AC output, unit 2. It reads the size of the power, so it works with any inverter brand	kW	None
Energy consumption (optional)	Load meter (optional): an Eastron SDM630 on the feed to the loads, unit 4, if the owner wants the site's own consumption measured	kW	None
Battery charge (optional)	Battery system (optional): the battery's own Modbus, its address entered and its readings mapped by hand from the maker's register table	%	Low below 25 %
Battery power (optional)	The same device. Positive while charging	kW	None

If the maker of the battery will not publish its Modbus register table, leave the battery out: do not guess registers. Solar surplus has no standard limit: the platform says nothing about it rather than guess.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *Grid and solar; a third on the site's own consumption if the owner wants it measured.*	2, or 3	90 to 180 each
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet; not in the hardware guide's list for this kit, see the network rules.*	0 or 1	150 to 200

About EUR 600 to 1,200. For a battery add nothing but the maker's Modbus register table. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in Part 4.

THE PROJECT MANAGER BRINGS

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch
- ☐ Spare fuses and terminals
- ☐ A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the electrician if the plan needs one. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

ASK, IN THIS ORDER

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money? Write it in their words. It is the reason they buy.
3. The electricity price per kWh, **what is paid for energy sent to the grid, if anything,** and the last twelve bills. The platform shows what solar kept on site at their price; without a price it shows kWh only.
4. What brand is the inverter, and where is its AC output? **Is the plant allowed to send energy to the grid?** If not, tick *This site cannot send energy to the grid* in the prices.
5. Where does the supply enter the building? Is it three-phase?
6. Is there a battery? Which maker and model, and does the maker publish the Modbus register table? Does it speak Modbus on the site's network, or on its own RS-485 port?
7. Does the owner want the site's own consumption measured (a third meter)?
8. Is there a network socket where the box will sit, or cellular signal? (Look at your phone there.)
9. Is there a power socket for the box, and is it on a circuit that stays on?
10. Who is the customer's electrician, and can they be there on install day?

LOOK AT, AND PHOTOGRAPH

- The supply entry and the main board (closed): where a meter could be fitted with the clamps around the incoming phases.
- The inverter and its AC output cable.
- The battery and its communication port, if there is one. Do not open it.
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable.
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable).



CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT S · SOLAR PLANT WITH GRID CONNECTION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Prices from the bills: electricity per kWh; paid for energy sent to the grid	
Twelve electricity bills collected	☐ Yes ☐ Later
Allowed to send energy to the grid?	☐ Yes ☐ No (tick the box in the prices)
Supply: single or three phase; main breaker size (A)	
Inverter: brand, size (kW), where its AC output cable is	
Battery: maker and model; register table available; connection (network or RS-485)	☐ No battery
Meters needed: grid, solar, load (and where)	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit S and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers. Mains work is inside the dashed red box and is the electrician's.

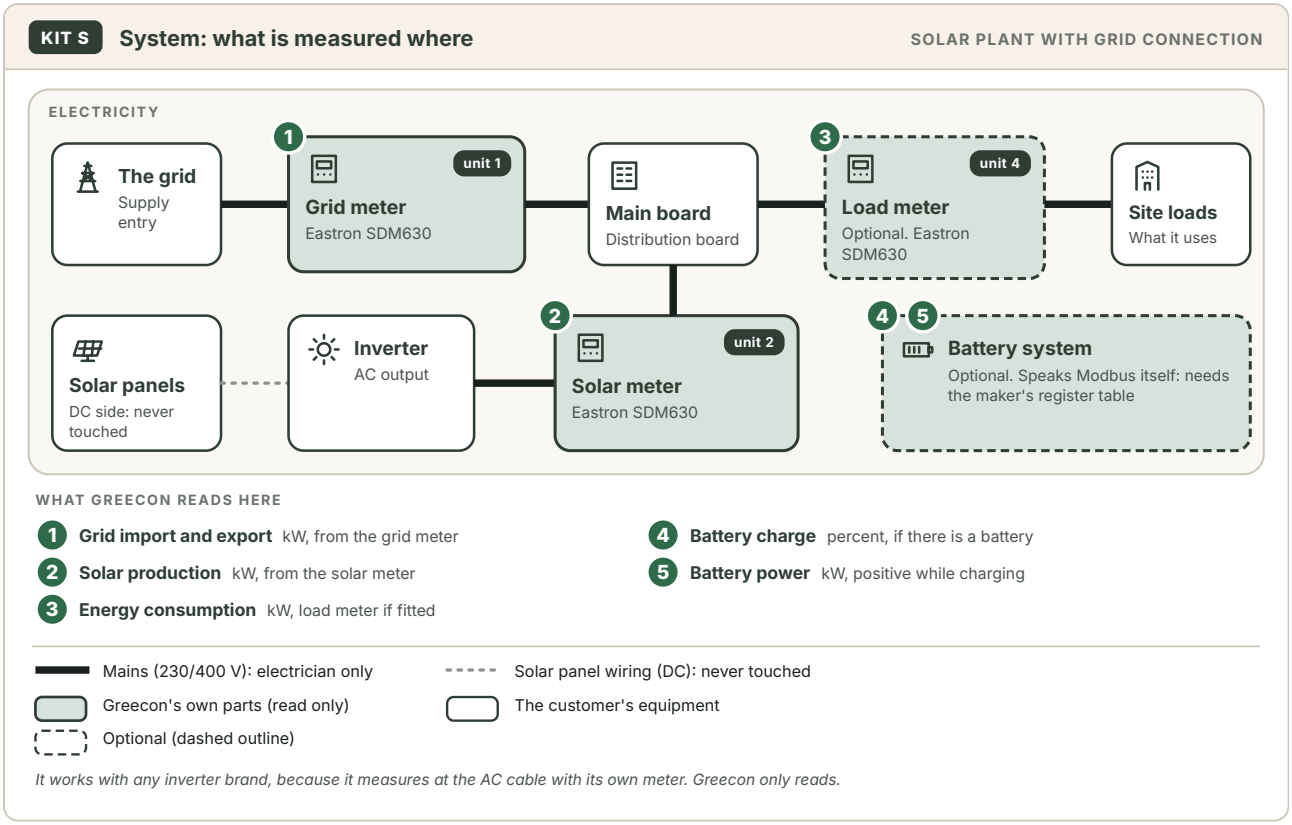
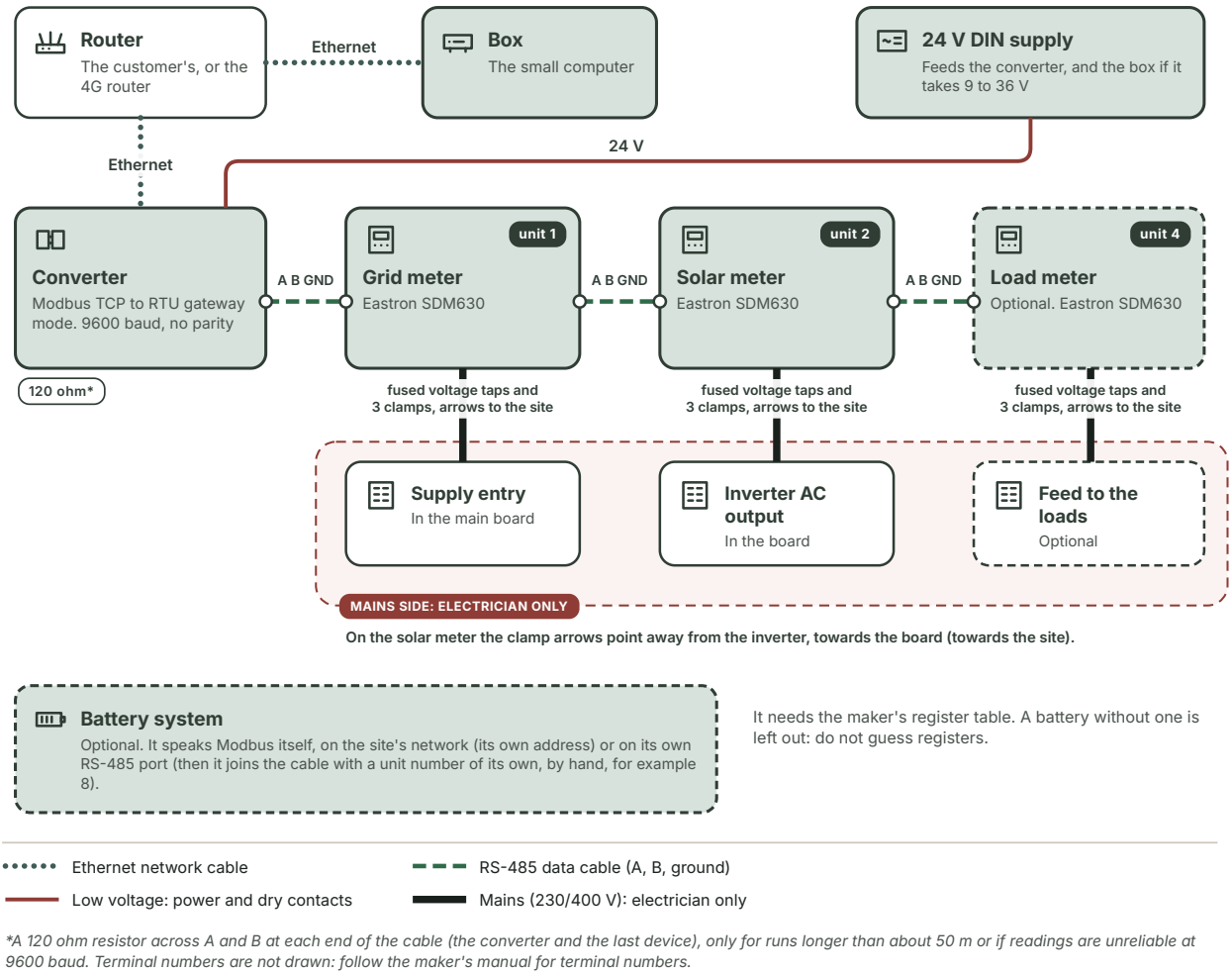


Figure S-1. Kit S, solar plant with grid connection. System: what is measured where. The dashed boxes are optional.

KIT S **Wiring: the data cable and the mains side** BUS AND SIGNALS



A 120 ohm resistor across A and B at each end of the cable (the converter and the last device), only for runs longer than about 50 m or if readings are unreliable at 9600 baud. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure S-2. Kit S. Wiring: one RS-485 cable from the converter through the grid meter, the solar meter and, if fitted, the load meter. Mains work is inside the dashed box.

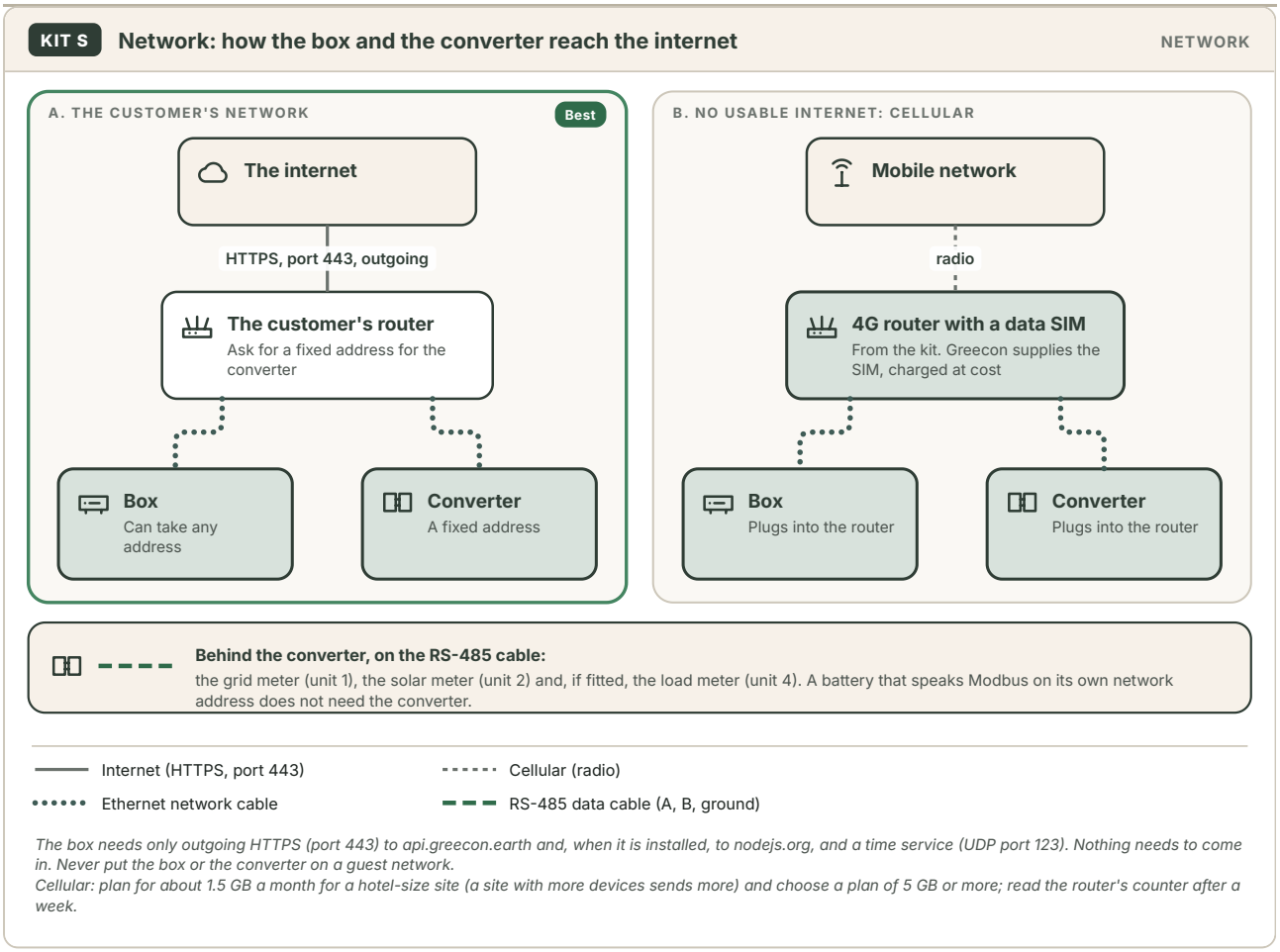


Figure S-3. Kit S. Network: the box and the converter on the customer's router (best) or on a 4G router.

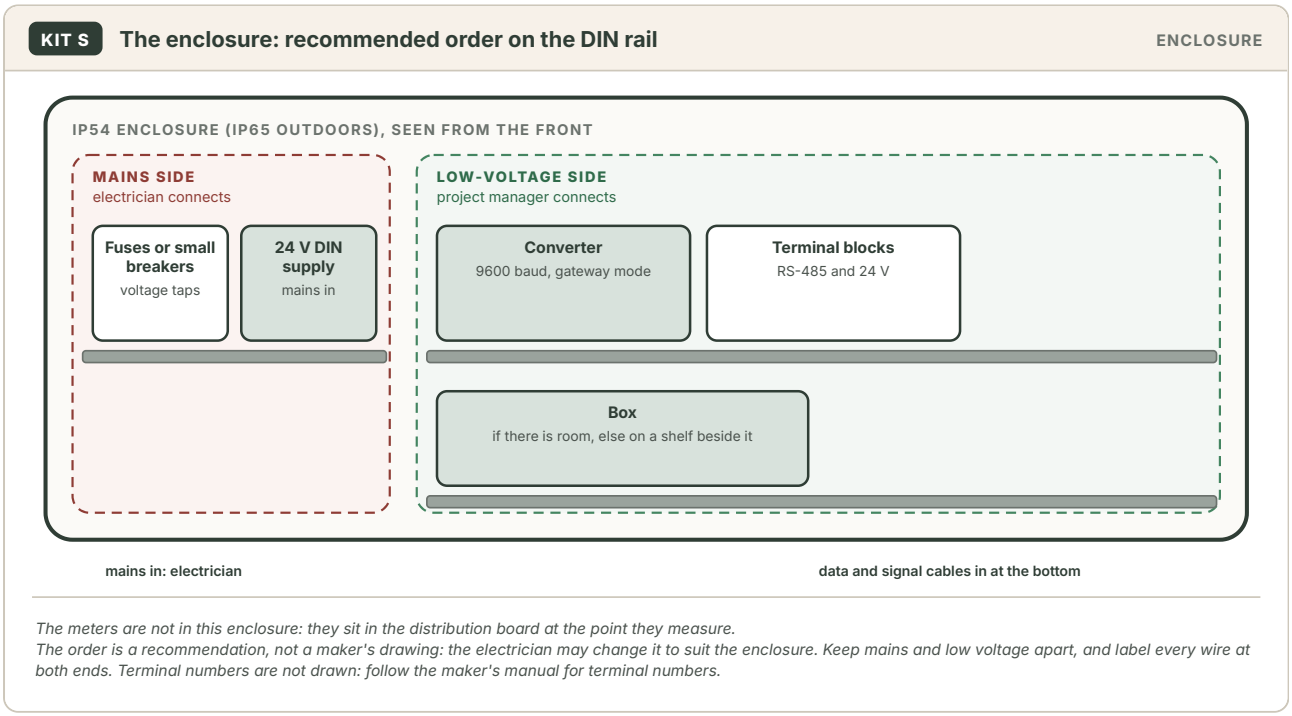


Figure S-4. Kit S. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

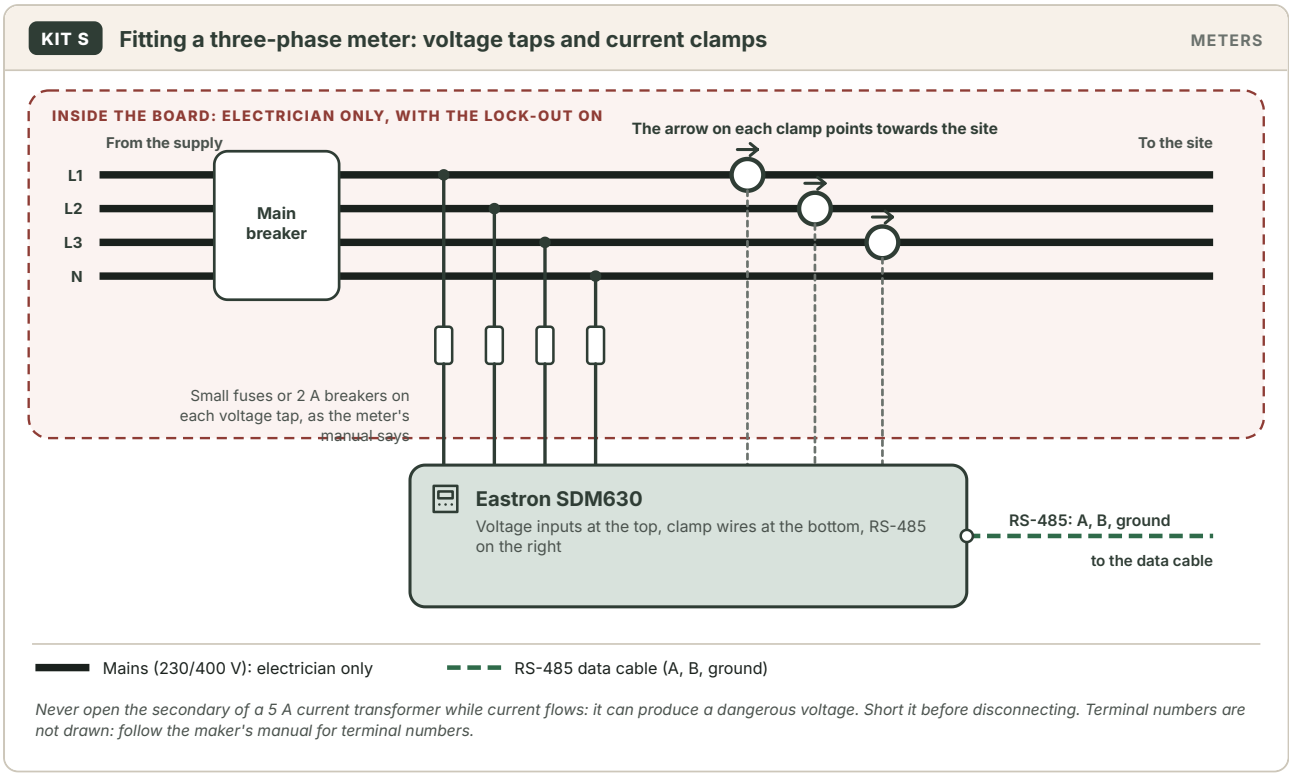


Figure S-5. Kit S. Fitting the grid meter and the solar meter (and the load meter): fused voltage taps and three clamps each, arrows towards the site.

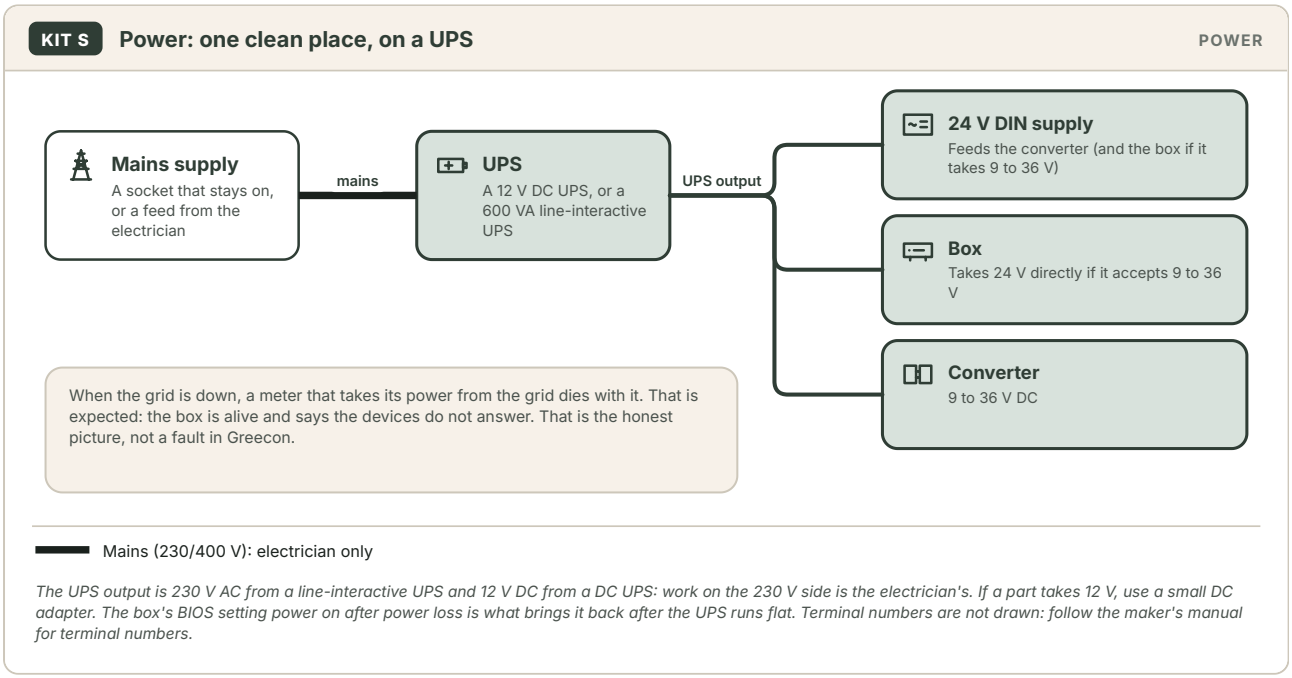


Figure S-6. Kit S. Power: the UPS feeds the box, the converter and the 24 V supply.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box, and more for the first kit's numbers and its desk test: time it, and write it down. PROJECT MANAGER

IN GREECON

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Solar plant with grid connection**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.
- [ ] **Write down that the papers are signed:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`, and its first step is **Order form signed and data terms agreed**. With the signed order form and the agreed data terms in front of you, write in the box who signed for the customer and on what date (it cannot be left empty), and press **The papers are signed**. Nothing goes into a customer's building without both papers.
- [ ] **Print the install card:** the link **The install card** is at the top of the setup page. It is your list for the visit: what to bring, what each device is set to and its number, how it is wired, the converter's settings, and what the customer's network is asked for. At the bottom are three messages to copy: send the first to the customer's contact two or three days before, and the second to whoever looks after their network. Each is in English with an *Albanian draft* folded under it, and someone who writes Albanian every day reads the draft before it is sent.
- [ ] **Get the install line:** on the same page press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> **STOP** **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

THE BOX ON YOUR DESK

The operating system and the pasted install line are still a terminal job; everything after them is done on the setup page.

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in Part 4, section 4.3.
- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`
- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit, the printed pack and the printed install card.

THE NUMBERS AND THE CONVERTER

The kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office; on site you only check the labels and do not change them. The numbers for this kit are in the table under **Numbers, addresses and ranges** and on the install card. Everything on one cable is **9600 baud, 8 data bits, no parity, 1 stop bit**.

- [ ] **Set each meter's number on its own display,** one meter at a time (delivered as unit 1, two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page. Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains.
- [ ] **Number the two Waveshare modules** with the maker's setting tool and a USB-to-RS-485 adapter, one at a time (each is delivered as address 1, and there is no switch): **3** on the 4 to 20 mA module (Modbus RTU Analog Input 8CH) and **5** on the pump contact module (Modbus RTU IO 8CH). Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default 0 to 20 mA mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- [ ] **Set the converter** in its web page: mode **Modbus TCP to RTU (gateway)** (not transparent, not raw TCP, not Modbus RTU over TCP), 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site: a fixed one is best, and Greecon finds the converter either way.
- [ ] **Label every device with its number** and the converter with its settings, and photograph the labels.
- [ ] **Test the kit on the desk.** The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (Part 3, section 3.11). Anything that differs from this book is reported to Greecon.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the kit: Part 3, section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN + PROJECT MANAGER	**Before anyone touches anything:** the signed papers are in your hand, the electrician is there, and the lock-out is agreed. **Walk the site together.** Agree where each meter goes. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the main board and the inverter. Have the router and a socket that stays on ready, and the person who can give the converter a fixed address.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the board to be worked on.
☐	4	ELECTRICIAN	**Fit the grid meter** at the supply entry: voltage taps with their small fuses or 2 A breakers as the meter's manual says, and the three current clamps around the incoming phases, **arrow towards the site**. Figure S-5.
☐	5	ELECTRICIAN	**Fit the solar meter** on the inverter's AC output, after the inverter and before the distribution board: the same, clamps with the arrow towards the site.
☐	6	ELECTRICIAN	**Fit the load meter** (only if the owner wants the site's own consumption measured) on the feed to the loads, as its manual says.
☐	7	CUSTOMER	**Battery (if there is one):** the customer's installer, or the battery's maker, enables the battery's Modbus connection and gives you its address, unit number and register table. Nobody from Greecon opens the battery or touches its DC wiring.
☐	8	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the 24 V supply and the UPS (the box too, if there is room, otherwise on a shelf beside it). Figure S-4.
☐	9	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the grid meter, on to the solar meter, and on to the load meter if there is one: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. The electrician connects its ends at the meters' A and B terminals inside the board. Figure S-2.
☐	10	PROJECT MANAGER	**Check the labels:** the number on each meter against the table under *Numbers, addresses and ranges* (grid meter 1, solar meter 2, load meter 4). They were set at the office: do not change them. Photograph each label and each display.
☐	11	PROJECT MANAGER	**Check the converter's label** (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502) and **connect the network:** plug the box and the converter into the customer's router, or into the sockets of the 4G router if the site has no usable internet. Ask for a fixed address for the converter and write it down. Figure S-3.
☐	12	ELECTRICIAN + PROJECT MANAGER	**Photograph before closing:** every meter's label (model and serial number), the wiring, the converter's settings. Then close the board.
☐	13	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	14	PROJECT MANAGER	**Power up** the UPS, the 24 V supply, the converter and the box.

	#	Who	Do this
☐	15	ELECTRICIAN	**Stay until commissioning is done.**
☐	16	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list: **Find equipment**, the battery by hand if there is one, and the electrician's name and licence number.

Numbers, addresses and ranges

The numbers on the cable were set at the office. The box finds where the converter is. You type no IP address, no port and no register for the meters, and kit S has no transmitters, so there is no range to give. Only a battery is entered by hand.

THE NUMBERS ON THE CABLE

The meters are behind one converter: they share its address and port 502, and differ only by number. Check each label against this table on install day. **Do not change a number on site.**

Device in Greecon	Make and model	Number	Label checked	Address found (the converter's)
Grid meter	Eastron SDM630	1	☐	
Solar meter	Eastron SDM630	2	☐	
Load meter (optional)	Eastron SDM630	4	☐	
Battery system (optional)	The customer's own	By hand: from its manual (for example 8 if it joins the cable)		Its own, or the converter's

THE EASTRON METERS

Their registers are already filled in by the template: total active power, input register 52 (a manual's 30053; counted from zero), a 32-bit floating point number with the high half first, in watts, multiplied by 0.001 to make kW. Check them against the meter's manual only if a reading looks wrong. Test one from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
```

THE BATTERY, IF THERE IS ONE

The box cannot recognise a battery on a network, so the step **Where each device is on the network** says **Enter by hand**: open **Enter the address by hand** under it and fill in its **IP address** (its own, or the converter's), **Port** and **Unit number**. Its two readings are mapped by hand from the maker's register table (step 5 of the setup list): press **Choose the register** under each reading's name. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero.

Reading	Register kind	Address	Kind of number	How it becomes the reading	Notes
Battery charge					percent
Battery power					kW, positive while charging

Take off what the site does not have: in step 4 press **The site does not have this: remove it** under the load meter, and under the battery, if the site has none (it asks you to confirm, because the device's readings go with it). A reading with no register holds the setup list at step 5. If the plant is not allowed to send energy to the grid, tick *This site cannot send energy to the grid* in the prices (step 2).

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to twelve steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself: you press buttons only for what a person must check or say. Near the top are links to the site page, to **The install card** (the card for the visit) and to the **Handover sheet**.

	Step	Title on the setup list	Done by
☐	1	**Order form signed and data terms agreed**	Project manager
☐	2	**Prices from the customer's bills**	Project manager (optional)
☐	3	**Install the box**	Project manager
☐	4	**Where each device is on the network**	Project manager
☐	5	**Which register holds which reading**	Project manager
☐	6	**Every device answers the box**	The platform, by itself
☐	7	**Readings arrive and look right**	The platform, by itself
☐	8	**Someone is told when the site needs attention**	Project manager
☐	9	**Grid meter direction checked**	Project manager, with the electrician
☐	10	**Alert limits agreed with the customer**	Project manager (only if the plant has a battery)
☐	11	**Electrical work done by a licensed electrician**	Project manager, with the electrician
☐	12	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

STEP 1 ORDER FORM SIGNED AND DATA TERMS AGREED

PROJECT MANAGER

DO	Nothing goes into a customer's building until the order form is signed and the data terms are agreed. Greecon keeps the papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date** (for example *Signed for the hotel by Besa Krasniqi on 19 October*): it cannot be left empty. Press **The papers are signed**. Do this at the office, and bring copies on install day.
YOU SHOULD SEE	The step turns **Done**: *Signed and agreed.*, with your name, the time and what you wrote. They are on the handover sheet.
IF YOU DO NOT	Until it is done, commissioning stays grey. The button will not work while the box is empty. If a paper turns out to be wrong, press **Reopen** (if the site had been commissioned, that is taken back too).

STEP 2 PRICES FROM THE CUSTOMER'S BILLS

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m^3; a typical month before (the kWh bought and the m^3 of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m^3* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

STEP 3 INSTALL THE BOX

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	• *Waiting for the install command to be run on the box.* The line has not been run yet. • *The box has its credential but has not reported yet.* Wait a minute and check the network cable. • *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. • *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

STEP 4 WHERE EACH DEVICE IS ON THE NETWORK

PROJECT MANAGER

DO	Press **Find equipment**. Nobody types an IP address, a port or a unit number here. - The box looks at the site's own network (local private addresses only, port 502, about a minute) and only reads. The page says *The box is looking at the network now.* and updates by itself. The platform recognises what answered and fills in at once only what is certain: a device with no address, found at the kit's own number. The rest waits for you. - A device **Found** but not filled in (another address, another number, another meter model): read the line and, if it makes sense, press **Use this address** (or *Use these N addresses*). An optional device the site does not have: press **The site does not have this: remove it**. - The soil probe, the weather sensor and a battery are entered by hand: under the device open **Enter the address by hand** (soil probe 6, weather sensor 7). - Box not on your network (a desk rehearsal), or a converter on another port: open **I know the converter's address**, type its address (and its port, if not 502), and press **Look at this address**.
YOU SHOULD SEE	Each device gets a line in plain words: **Set** (found, address filled in), **Found**, **Not found**, **Not sure which** or **Enter by hand**. When it worked the step is **Done** (*Every device has an address.*), and the button reads **Look again**.
IF YOU DO NOT	- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* An optional device the site does not have: press **The site does not have this: remove it**. - **Not sure which**: Greecon will not guess, because a wrong guess could swap the grid meter and the solar meter. Read each device's label, use **Enter the address by hand** under each with the number on its label, then **Look again**. Numbers are set at the office: a part with a wrong number goes back there. - *No Modbus converter answered on this network:* check that the converter is on and plugged into the same router as the box. *A converter answers at 192.168.1.50, but nothing answers behind it:* check each meter's address, the speed and the A and B wires. *All answered with exactly the same words:* the converter is not in gateway mode. *The box did not pick this up:* check that it shows as online in the step above. - Readings that stop after it had worked: the router may have given the converter a new address. Press **Find equipment**, then **Use this address**.

STEP 5 WHICH REGISTER HOLDS WHICH READING

PROJECT MANAGER

DO	The Eastron meters and the pump contact module need nothing typed. For the 4 to 20 mA module the page asks, for each transmitter, only for the range printed on its label, in the label's own words: - **Level transmitter** (input 1): the **Range on the label**, for example 5, its **Unit on the label** (metres, or bar) and **How tall the tank is inside** in metres, for example 4: from the bottom to the top of the water when full. - **Pressure transmitter** (input 2): the **Top of the range** in bar; 10 is usual. - **Flow meter** (input 3): the **Top of the range** in litres a minute. Greecon works out the register and the scale. For a transmitter the site does not have (no flow meter), tick **Not fitted at this site**: Greecon asks first, then takes the reading and its history away. Press **Save the ranges**. Equipment Greecon does not know (the soil probe, the weather sensor, a battery) is mapped by hand, from the maker's register table: under the reading press **Choose the register** and fill in the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero: 30053 is address 52, 40001 is address 0), the kind of number (and which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.* Under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (0 to 5 m in a tank 4 m tall); the tank level is a percentage of the tank, so a full tank reads 100.
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *Needs the range printed on the transmitter's label.* (the 4 to 20 mA module) or *No register chosen.* (equipment Greecon does not know). A transmitter shorter than the tank is refused: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* A reading the site does not have holds the list here: tick **Not fitted at this site**, or press **The site does not have this: remove it** (step 4) for an optional device.

STEP 6 EVERY DEVICE ANSWERS THE BOX

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings).* On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2).* Look in this order: the converter's address may have changed (press **Find equipment** again, then **Use this address**); the number on the device's label against the install card; the speed at the converter; A and B swapped on the data cable; the converter not in gateway mode. Nothing answers at all: the converter is off, or it is on another network than the box. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

STEP 7 READINGS ARRIVE AND LOOK RIGHT

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* **Pump running** reads 1 while the pump runs and 0 while it is stopped.
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter a reading below zero is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank); one far too high is a range or tank height typed wrongly: give the ranges again (step 5).

STEP 8 SOMEONE IS TOLD WHEN THE SITE NEEDS ATTENTION

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	■ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ■ *Nobody is set to be told:* add the people. ■ *Nobody on the list has an email address* (or a phone number): add one for each person. ■ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ■ *Test messages are limited to 6 an hour.* Wait and try again.

STEP 9 GRID METER DIRECTION CHECKED

PROJECT MANAGER | **ELECTRICIAN**

DO	Greecon cannot see which way the meter's clamps point. With the sun down or the solar switched off, switch on a known load (a kettle or a heater). **Grid import must go up.** If **Grid export** goes up instead, the clamps are the wrong way round: the electrician turns them (with the lock-out on) and you check again. Then press **I checked this** with a note of what you saw.
YOU SHOULD SEE	Before: **To do**, *Greecon cannot see which way the meter's clamps point.* After you confirm: **Done**, *Checked on site,* with your name and the time.
IF YOU DO NOT	Import and export swapped: the clamps are the wrong way round. The electrician turns them; you repeat the check. A reversed clamp swaps what the site buys with what it sells, and every figure with it.

STEP 10 ALERT LIMITS AGREED WITH THE CUSTOMER

PROJECT MANAGER *only if the plant has a battery*

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

STEP 11 ELECTRICAL WORK DONE BY A LICENSED ELECTRICIAN

PROJECT MANAGER | **ELECTRICIAN**

DO	Fitting meters in a board and wiring transmitters and contacts is the **licensed electrician's** work, under the customer's lock-out. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number** (for example *Arben Hoxha, licence E-1234*), or that no electrical work was needed: it cannot be left empty. Press **The electrical work is done**.
YOU SHOULD SEE	The step turns **Done**: *Done by a licensed electrician.*, with your name, the time and what you wrote (they are on the handover sheet). The platform does not check the licence: what you write is the record.
IF YOU DO NOT	Until it is done, commissioning stays grey. Do not press the button while the work is open or the lock-out is still on. The button will not work while the box is empty.

STEP 12 COMMISSIONED

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 12) with a note. It stays grey until every other step is done, the papers (step 1) and the electrician's name and licence number (step 11) included. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Grid meter's display against Greecon. Read the display (kW or W) and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Solar meter's display against Greecon. The same, on the solar meter.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Load meter's display against Greecon (if fitted).	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on (a kettle or a heater), the sun down or the solar off. On a plant that exports at noon, do it at night or with the inverter off.	Import rises; export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's own display.	Within about 5 percent; grid export is the difference between production and what the site is using			☐ Pass ☐ Fail
Battery charge (if fitted). Against the battery's own display.	The two agree (write both numbers)			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail

Check and how	Pass when	Greecon shows	Real	Result
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); **Energy now**, the flow moving from the solar to the site and the grid; **Value**, what the solar kept on site this month and the prices it uses; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Send the third message on The install card** to the customer's contact the same day, with the handover sheet attached (its Albanian is a draft: have it read first).
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT S · SOLAR PLANT WITH GRID CONNECTION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Prices entered (electricity, export) and where they came from; zero-export box ticked if needed	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Battery: limits agreed (Low below), or no battery	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, Energy now, Value, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
The solar meter reads zero at midday	A clamp closed on the wrong cable, a closed breaker upstream, or the inverter is limited to zero export	Compare with the meter's own display and test the register from the box (`greecon-edge modbus-test`). Ask the electrician to confirm that the clamps are on the inverter's output phases. If the plant is zero-export, midday solar can honestly be low: check the inverter's display
Import and export are swapped (export rises when a load is switched on)	The grid meter's clamps are the wrong way round	The electrician turns them, with the lock-out on; you repeat the direction check (step 9 of the setup list). A reversed clamp swaps what the site buys with what it sells, and every figure with it. Left reversed, Greecon notices at night or in cloud and opens a Check today alert, *The grid meter may be fitted the wrong way round*; at midday it cannot
A reading is a thousand times too big	Watts shown as kilowatts, typically on a hand-mapped battery reading	In the register form multiply by 0.001 (the Eastron meters already do)

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: the converter's address changed (press **Find equipment** again), the number on the device's label not the one on the install card, a wrong speed at the converter, A and B swapped on the data cable, the converter not in gateway mode.

5.3 Site pack F: Farm or greenhouse with irrigation

Soil and weather, with the water supply behind it. Planning estimate: About EUR 700 to 1,500. This pack is also printed on its own as PDF 06 (06-Site-Pack-F-Farm-or-Greenhouse.pdf).

At a glance

WHO IT SUITS
A farm or a greenhouse with irrigation: soil moisture and weather at the field, with the water supply behind it (a tank, a pump, the pressure in the pipe).

WHAT IT MEASURES
The moisture of the soil (one probe for each zone), the air temperature and humidity, the tank, the line pressure and, if there is a pump to watch, the pump's flow and whether it runs.

PLANNING ESTIMATE
About EUR 700 to 1,500.

TIME
Survey about 1 hour, with the grower. Install day 4 to 6 hours. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Farm or greenhouse with irrigation

STANDARD SETTINGS
Soil probe 6, weather sensor 7, water module 3, pump contact module 5 (if there is a pump to watch), all set before the kit leaves the office. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode. A cellular router is the likelier case in a field.

STOP

Greecon gives no irrigation advice until the grower's own "start irrigating below" value is entered. That number depends on the soil, the crop and the probe's depth, and nobody can set it for the grower. After the first week of readings, ask the grower what reading means "time to water", and enter it as the limit (step 10 of the setup list). **Frost works the same way:** ask at what air temperature frost damages the crop, and in which weeks; enter it as **Frost alert at or below** on the air temperature (step 10). It opens an Act now alert, so agree who is woken. Map the air temperature as a signed number (int16), or a frosty morning reads as a hot afternoon.

The order of the work: survey with the grower, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the labels, and the 12 or 24 V supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with their 24 V supply, as the module's manual draws it.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

HOW TO READ THIS PACK

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, the pump's run contact, and the 4 to 20 mA loops.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure F-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Farm or greenhouse with irrigation** makes four devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Soil moisture	Soil probe: a soil moisture probe that speaks RS-485 Modbus, one for each zone, unit 6 (not an approved model: address and registers entered by hand)	%	None. **Start irrigating below** has no standard value: the grower's own is entered after the first week
Air temperature and **Humidity**	Weather sensor: a temperature and humidity sensor that speaks RS-485 Modbus, unit 7 (not an approved model: address and registers entered by hand)	degrees Celsius, %	None
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow (if there is a pump to watch)	Water I/O module, unit 3: the flow meter's 4 to 20 mA	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running (if there is a pump to watch)	Pump contact module (optional), unit 5: the pump starter's spare contact	on or off	None

A sensor whose maker does not document its registers cannot be mapped, so it cannot be used. Soil readings are only as good as the contact between the probe and the soil: the handover records the probe's depth and where it is.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router**	4G, DIN rail, data SIM. *The likelier case in a field.*	0 or 1	150 to 200
☐	**Soil moisture probe**	RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each
☐	**Temperature and humidity sensor**	RS-485 Modbus, with a published register table	1	20 to 80
☐	**Pump contact module** *(optional)*	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *If there is a pump to watch.*	0 or 1	30 to 60
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *If there is a pump to watch.*	0 or 1	150 to 500

About EUR 700 to 1,500. One soil probe for each zone. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in Part 4.

THE PROJECT MANAGER BRINGS

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter

- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The USB-to-RS-485 adapter is how you set a probe's or a sensor's unit number before it is buried or mounted. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the grower and, if the plan needs one, the electrician. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

ASK, IN THIS ORDER

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or a crop? Write it in their words. It is the reason they buy.
3. **Which crop, which soil, how deep are the roots, and where in the field is typical** (not the wettest or the driest spot)? The irrigation limit depends on them.
4. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Is there a pump to watch, and has its starter a spare "running" contact?
5. Is there a network socket in the pump house, or cellular signal? (Look at your phone there.)
6. Is there a power socket for the box, on a circuit that stays on?
7. How far is the probe's place from the box? The data cable is planned at under 100 metres. If it is further, a probe on a radio gateway that speaks Modbus is the alternative; the gateway's maker must publish its register table. Ask the founder before promising it.
8. Who is the customer's electrician, and can they be there on install day?

LOOK AT, AND PHOTOGRAPH

- The field: where each probe goes, and the weather sensor's place.
- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.
- The pump house: where the box and the cellular router would sit, dry, 30 cm free, no heat, within reach of a socket.
- The route for the data cable from the field to the box (note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Crop; soil; root depth (cm)	
Zones: how many probes; where each goes (typical, not wettest or driest)	
Distance from the probe places to the box (m); wired or radio gateway	
Tank: height (m), water source; pump (kW); starter location; spare running contact on the starter; pipe pressure (bar)	☐ No pump to watch
Sensors needed: level, pressure, flow, pump running (and where)	
Network: customer's cable or cellular; signal in the pump house	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit F and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers.

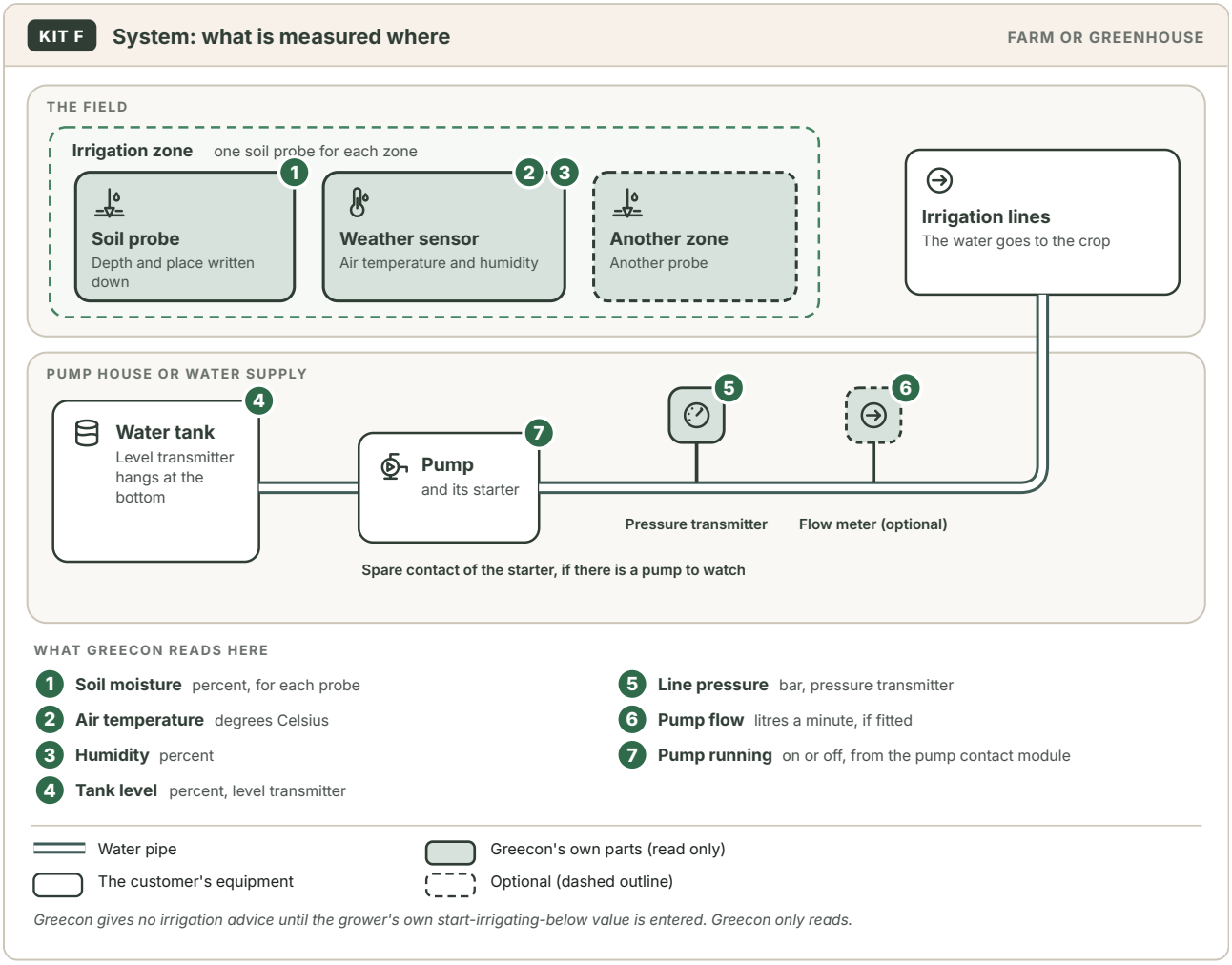
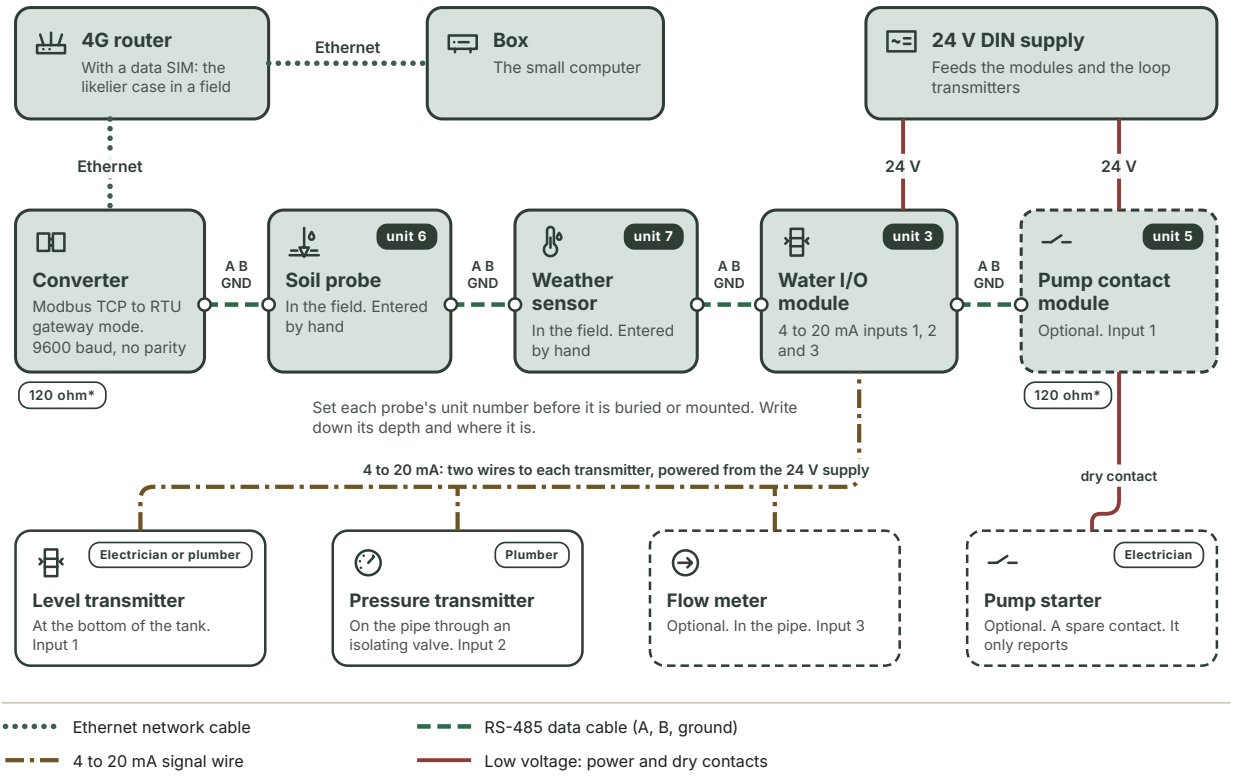


Figure F-1. Kit F, farm or greenhouse with irrigation. System: what is measured where. The dashed parts are optional.

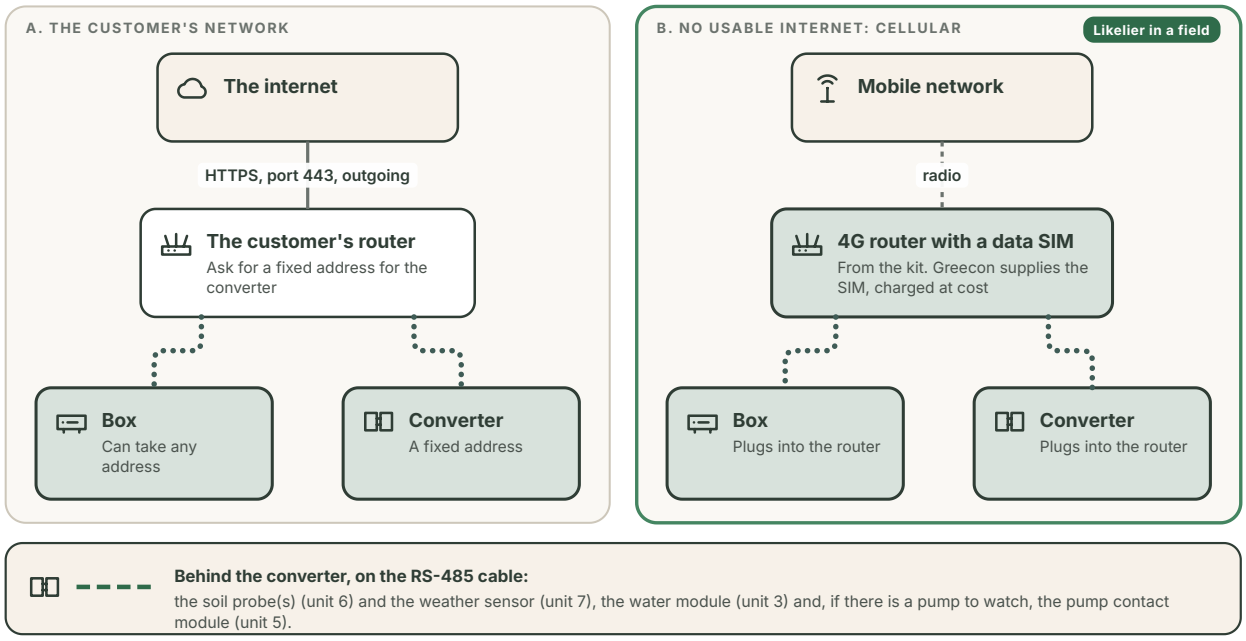
KIT F **Wiring: the data cable and the signal wires** BUS AND SIGNALS



Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The soil probe and the weather sensor need a published register table, and their addresses are entered by hand. The electrician wires the 4 to 20 mA loops and the contact. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure F-2. Kit F. Wiring: one RS-485 cable from the converter through the soil probe, the weather sensor and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module.

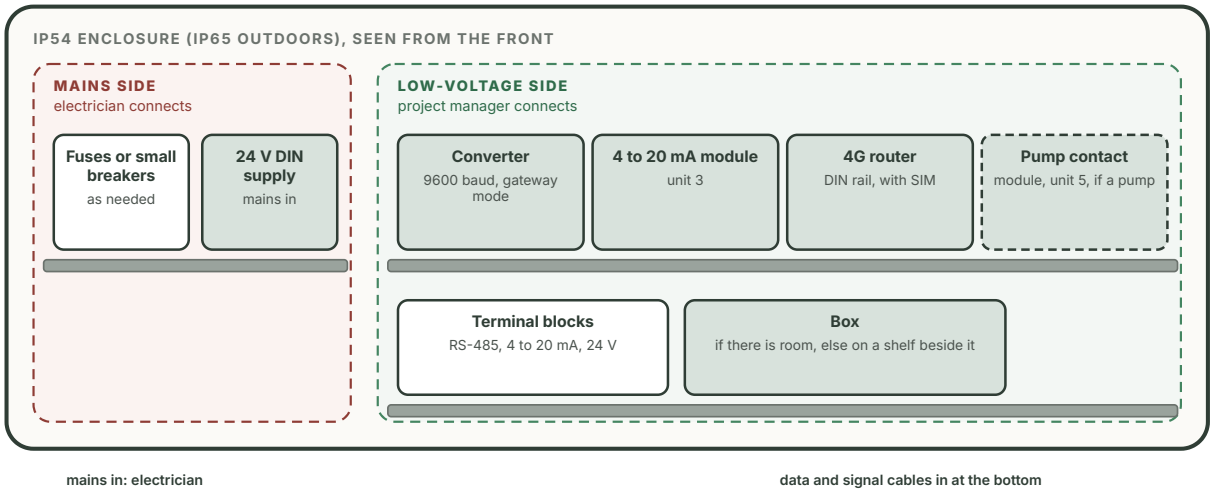
KIT F **Network: how the box and the converter reach the internet** NETWORK



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure F-3. Kit F. Network: a 4G router with a data SIM is the likelier case in a field; the customer's router is better where there is one.

KIT F **The enclosure: recommended order on the DIN rail** ENCLOSURE



The probes and the weather sensor are in the field, on the data cable. Check the signal where the router sits.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure F-4. Kit F. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

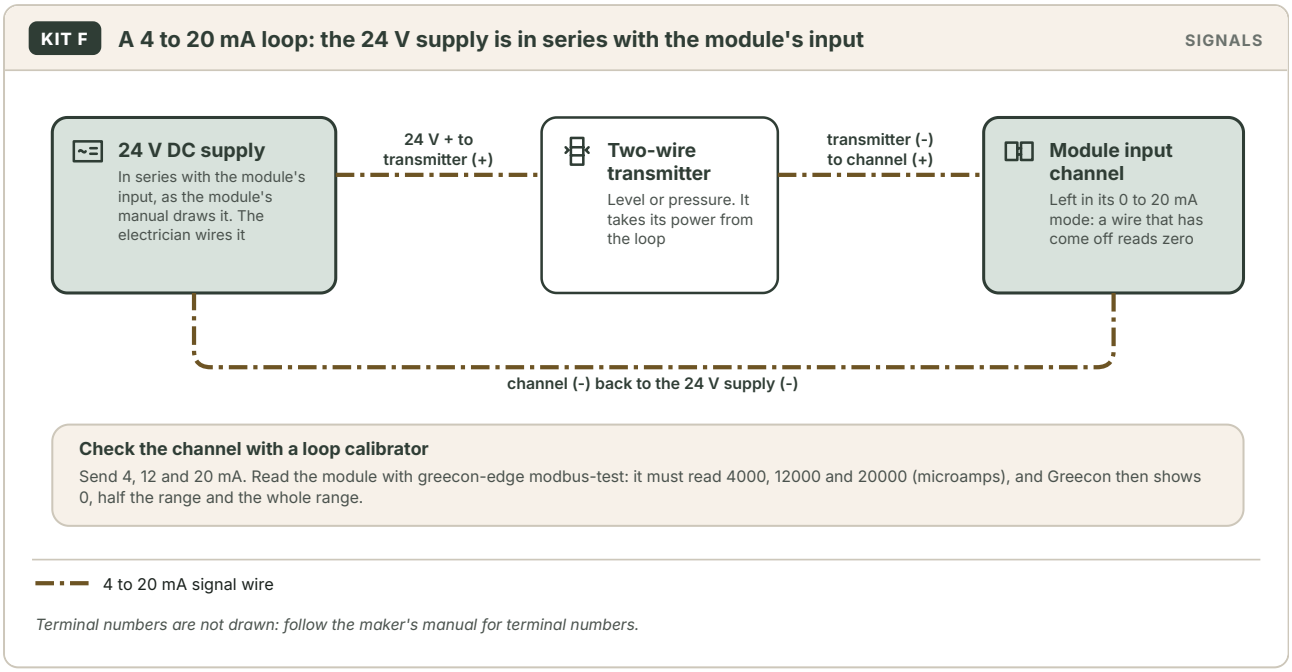


Figure F-5. Kit F. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

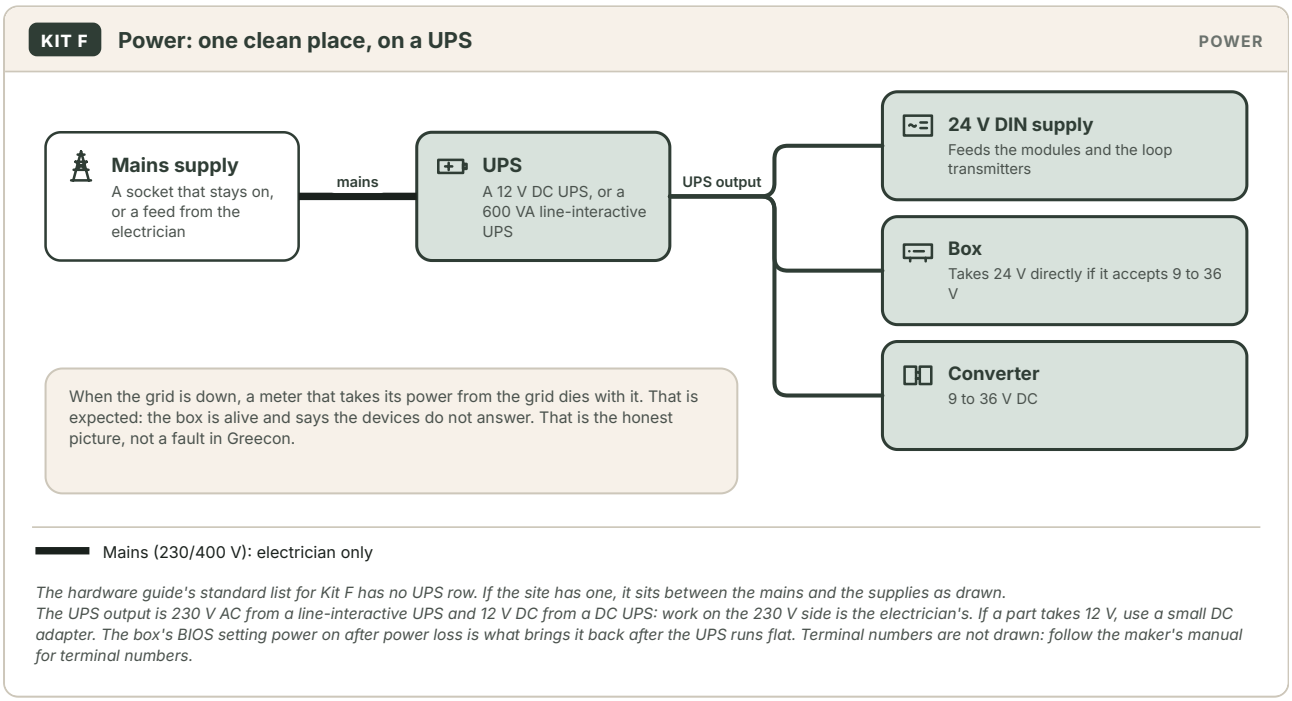


Figure F-6. Kit F. Power. The hardware guide's standard list for this kit has no UPS: decide at the survey and write it down.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box, and more for the first kit's numbers and its desk test: time it, and write it down. PROJECT MANAGER

IN GREECON

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Farm or greenhouse with irrigation**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.
- [ ] **Write down that the papers are signed:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`, and its first step is **Order form signed and data terms agreed**. With the signed order form and the agreed data terms in front of you, write in the box who signed for the customer and on what date (it cannot be left empty), and press **The papers are signed**. Nothing goes into a customer's building without both papers.
- [ ] **Print the install card:** the link **The install card** is at the top of the setup page. It is your list for the visit: what to bring, what each device is set to and its number, how it is wired, the converter's settings, and what the customer's network is asked for. At the bottom are three messages to copy: send the first to the customer's contact two or three days before, and the second to whoever looks after their network. Each is in English with an *Albanian draft* folded under it, and someone who writes Albanian every day reads the draft before it is sent.
- [ ] **Get the install line:** on the same page press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> STOP
>
> **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

THE BOX ON YOUR DESK

The operating system and the pasted install line are still a terminal job; everything after them is done on the setup page.

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in Part 4, section 4.3.
- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`
- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit, the printed pack and the printed install card.

THE NUMBERS AND THE CONVERTER

The kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office; on site you only check the labels and do not change them. The numbers for this kit are in the table under **Numbers, addresses and ranges** and on the install card. Everything on one cable is **9600 baud, 8 data bits, no parity, 1 stop bit**.

- [ ] **Set each meter's number on its own display,** one meter at a time (delivered as unit 1, two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page. Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains.
- [ ] **Number the two Waveshare modules** with the maker's setting tool and a USB-to-RS-485 adapter, one at a time (each is delivered as address 1, and there is no switch): **3** on the 4 to 20 mA module (Modbus RTU Analog Input 8CH) and **5** on the pump contact module (Modbus RTU IO 8CH). Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default 0 to 20 mA mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- [ ] **Set the converter** in its web page: mode **Modbus TCP to RTU (gateway)** (not transparent, not raw TCP, not Modbus RTU over TCP), 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site: a fixed one is best, and Greecon finds the converter either way.
- [ ] **Label every device with its number** and the converter with its settings, and photograph the labels.
- [ ] **Test the kit on the desk.** The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (Part 3, section 3.11). Anything that differs from this book is reported to Greecon.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the kit: Part 3, section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Before the kit leaves the office, set the unit number of each soil probe (**6**) and of the weather sensor (**7**), with the USB-to-RS-485 adapter and the maker's instructions, and write them down: **set them before the probe is buried or the sensor is mounted.** Make every one different from the others and from the water module (3) and the pump contact module (5): a second probe, for a second zone, takes the next free number (8, then 9), written down.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN · PROJECT MANAGER · CUSTOMER	**Before anyone touches anything:** the signed order form and the data terms are in your hand, the electrician is there, and the customer's lock-out is agreed. **Walk the site together,** with the grower. Agree where each probe and the weather sensor go (typical of the zone, not the wettest or driest spot), and where the enclosure sits. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the pump house, the tank and the starter. Have a socket that stays on ready.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the circuit that feeds the pump house.
☐	4	ELECTRICIAN	**Connect the mains to the 24 V DIN supply** in the enclosure, as its manual shows, with a fuse or small breaker as needed.
☐	5	ELECTRICIAN	**If there is a pump to watch:** connect the pump's run contact (a spare auxiliary contact of the starter, as a dry contact) to **input 1 of the pump contact module**, as the module's manual draws it (its DI COM left unconnected). Never to a control circuit, and never to the module's relay outputs: the contact only reports.
☐	6	ELECTRICIAN · PROJECT MANAGER	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts, and **wire their 4 to 20 mA loops** with the 24 V supply in series, as the module's manual draws it: **level on input 1, pressure on input 2, flow meter on input 3.** The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve. Figure F-5. You check each is on its own input and number the wires at both ends.
☐	7	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the water module, the pump contact module (if there is a pump to watch), the 24 V supply, the 4G router (and the box, if there is room, otherwise on a shelf beside it). Figure F-4.
☐	8	PROJECT MANAGER · CUSTOMER	**Place the soil probe(s)** at the depth and in the place agreed with the grower, in good contact with the soil, as the maker says; mount the weather sensor. **Write down each probe's depth and where it is, and photograph it.**
☐	9	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the soil probe(s) and the weather sensor, and on to the water module and the pump contact module: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. Figure F-2.
☐	10	PROJECT MANAGER	**Check the labels:** the number on each device against the table under *Numbers, addresses and ranges* (soil probe 6, weather sensor 7, water module 3, pump contact module 5). They were set at the office: do not change them. Photograph each label.
☐	11	PROJECT MANAGER	**Check the converter's label** (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502) and **connect the network:** put the SIM in the 4G router, check the signal where it sits, and plug the box and the converter into its sockets (or into the customer's router if the pump house has one). Ask for a fixed address for the converter if it is on the customer's router, and write it down. Figure F-3.

	#	Who	Do this
☐	12	ELECTRICIAN / PROJECT MANAGER	**Photograph before closing:** the wiring, the converter's settings, each device's label. Then close the enclosure.
☐	13	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	14	PROJECT MANAGER	**Power up** the 24 V supply, the converter, the router and the box.
☐	15	ELECTRICIAN	**Stay until commissioning is done.**
☐	16	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list: **Find equipment**, the probe and the sensor by hand, the ranges, and the electrician's name and licence number.

Numbers, addresses and ranges

The numbers on the cable were set at the office. The box finds the water module and the pump contact module on the network. It cannot recognise a soil probe or a weather sensor, so those two are entered by hand, with their registers, from the makers' manuals. For the transmitters you give the range printed on each label.

THE NUMBERS ON THE CABLE

All the devices are behind one converter: they share its address and port 502, and differ only by number. Check each label against this table on install day. **Do not change a number on site.**

Device in Greecon	Make and model	Number	Label checked	Address (the converter's)
Soil probe (zone 1)		6	☐	
Soil probe (zone 2, if any)		8 (the next free number)	☐	
Weather sensor		7	☐	
Water I/O module	Waveshare Modbus RTU Analog Input 8CH	3	☐	
Pump contact module (optional)	Waveshare Modbus RTU IO 8CH	5	☐	

For the soil probes and the weather sensor, under each device open **Enter the address by hand** and fill in its **IP address** (the converter's: the water module's line in the same step shows it, as *Reached at 192.168.1.50:502, unit 3*), its **Port** (502) and its **Unit number**.

REGISTERS FOR THE PROBE AND THE SENSOR: ALL OF THEM FROM THE MAKERS' MANUALS

There is no ready-made profile for a probe or a sensor: each reading is mapped by hand from the maker's register table (step 5 of the setup list). Under each reading's name press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero (30053 is address 52). Write what you enter:

Reading	Register kind	Address	Kind of number	How it becomes the reading
Soil moisture (zone 1)				
Air temperature				
Humidity				

THE PUMP CONTACT MODULE AND THE RANGES

If there is a pump to watch, the pump contact module needs nothing typed: the template fills in its reading (digital input 1). Check it by eye: with the pump running, **Pump running** reads 1; stopped, 0. Test it from the box: `greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0`. With no pump to watch, press **The site does not have this: remove it** under it in step 4.

For the water module, one form for each transmitter (step 5 of the setup list), in the label's own words. Read the range from the label on the transmitter and measure how tall the tank is inside, from the bottom to the top of the water when it is full.

Reading	Input	The form asks for	Range on the label	Tank height
Tank level	1	Level transmitter: the range on the label (m or bar) and how tall the tank is inside (m)		
Line pressure	2	Pressure transmitter: the top of the range (bar)		
Pump flow	3	Flow meter: the top of the range (lpm, litres a minute)		

Greecon works out the register and the scale: 4 mA is 0 and 20 mA is the range (for the tank, range ÷ tank height × 100: a 0 to 5 m transmitter in a tank 4 m tall gives 125, and a full tank, at 16.8 mA, reads 100). **A transmitter shorter than the tank is refused:** it cannot see the top of the water. Tick **Not fitted at this site** for the readings the site does not have (no flow meter): the reading is taken away and nothing waits for it.

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to twelve steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself: you press buttons only for what a person must check or say. Near the top are links to the site page, to **The install card** (the card for the visit) and to the **Handover sheet**.

	Step	Title on the setup list	Done by
☐	1	**Order form signed and data terms agreed**	Project manager
☐	2	**Prices from the customer's bills**	Project manager (optional)

	Step	Title on the setup list	Done by
☐	3	**Install the box**	Project manager
☐	4	**Where each device is on the network**	Project manager
☐	5	**Which register holds which reading**	Project manager
☐	6	**Every device answers the box**	The platform, by itself
☐	7	**Readings arrive and look right**	The platform, by itself
☐	8	**Someone is told when the site needs attention**	Project manager
☐	10	**Alert limits agreed with the customer**	Project manager
☐	11	**Electrical work done by a licensed electrician**	Project manager, with the electrician
☐	12	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

Not on this kit. The platform leaves out a step that does not apply to the site, so the list for kit F has no step 9 (Grid meter direction checked). The steps keep their order; the count on the setup page is out of the steps that do apply, not out of 12.

STEP 1 ORDER FORM SIGNED AND DATA TERMS AGREED

PROJECT MANAGER

DO	Nothing goes into a customer's building until the order form is signed and the data terms are agreed. Greecon keeps the papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date** (for example *Signed for the hotel by Besa Krasniqi on 19 October*): it cannot be left empty. Press **The papers are signed**. Do this at the office, and bring copies on install day.
YOU SHOULD SEE	The step turns **Done**: *Signed and agreed.*, with your name, the time and what you wrote. They are on the handover sheet.
IF YOU DO NOT	Until it is done, commissioning stays grey. The button will not work while the box is empty. If a paper turns out to be wrong, press **Reopen** (if the site had been commissioned, that is taken back too).

STEP 2 PRICES FROM THE CUSTOMER'S BILLS

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

STEP 3 INSTALL THE BOX

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	• *Waiting for the install command to be run on the box.* The line has not been run yet. • *The box has its credential but has not reported yet.* Wait a minute and check the network cable. • *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. • *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

STEP 4 WHERE EACH DEVICE IS ON THE NETWORK

PROJECT MANAGER

DO	Press **Find equipment**. Nobody types an IP address, a port or a unit number here. - The box looks at the site's own network (local private addresses only, port 502, about a minute) and only reads. The page says *The box is looking at the network now.* and updates by itself. The platform recognises what answered and fills in at once only what is certain: a device with no address, found at the kit's own number. The rest waits for you. - A device **Found** but not filled in (another address, another number, another meter model): read the line and, if it makes sense, press **Use this address** (or *Use these N addresses*). An optional device the site does not have: press **The site does not have this: remove it**. - The soil probe, the weather sensor and a battery are entered by hand: under the device open **Enter the address by hand** (soil probe 6, weather sensor 7). - Box not on your network (a desk rehearsal), or a converter on another port: open **I know the converter's address**, type its address (and its port, if not 502), and press **Look at this address**.
YOU SHOULD SEE	Each device gets a line in plain words: **Set** (found, address filled in), **Found**, **Not found**, **Not sure which** or **Enter by hand**. When it worked the step is **Done** (*Every device has an address.*), and the button reads **Look again**.
IF YOU DO NOT	- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* An optional device the site does not have: press **The site does not have this: remove it**. - **Not sure which**: Greecon will not guess, because a wrong guess could swap the grid meter and the solar meter. Read each device's label, use **Enter the address by hand** under each with the number on its label, then **Look again**. Numbers are set at the office: a part with a wrong number goes back there. - *No Modbus converter answered on this network:* check that the converter is on and plugged into the same router as the box. *A converter answers at 192.168.1.50, but nothing answers behind it:* check each meter's address, the speed and the A and B wires. *All answered with exactly the same words:* the converter is not in gateway mode. *The box did not pick this up:* check that it shows as online in the step above. - Readings that stop after it had worked: the router may have given the converter a new address. Press **Find equipment**, then **Use this address**.

STEP 5 WHICH REGISTER HOLDS WHICH READING

PROJECT MANAGER

DO	The Eastron meters and the pump contact module need nothing typed. For the 4 to 20 mA module the page asks, for each transmitter, only for the range printed on its label, in the label's own words: - **Level transmitter** (input 1): the **Range on the label**, for example 5, its **Unit on the label** (metres, or bar) and **How tall the tank is inside** in metres, for example 4: from the bottom to the top of the water when full. - **Pressure transmitter** (input 2): the **Top of the range** in bar; 10 is usual. - **Flow meter** (input 3): the **Top of the range** in litres a minute. Greecon works out the register and the scale. For a transmitter the site does not have (no flow meter), tick **Not fitted at this site**: Greecon asks first, then takes the reading and its history away. Press **Save the ranges**. Equipment Greecon does not know (the soil probe, the weather sensor, a battery) is mapped by hand, from the maker's register table: under the reading press **Choose the register** and fill in the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero: 30053 is address 52, 40001 is address 0), the kind of number (and which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.* Under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (0 to 5 m in a tank 4 m tall); the tank level is a percentage of the tank, so a full tank reads 100.
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *Needs the range printed on the transmitter's label.* (the 4 to 20 mA module) or *No register chosen.* (equipment Greecon does not know). A transmitter shorter than the tank is refused: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* A reading the site does not have holds the list here: tick **Not fitted at this site**, or press **The site does not have this: remove it** (step 4) for an optional device.

STEP 6 EVERY DEVICE ANSWERS THE BOX

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings).* On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2).* Look in this order: the converter's address may have changed (press **Find equipment** again, then **Use this address**); the number on the device's label against the install card; the speed at the converter; A and B swapped on the data cable; the converter not in gateway mode. Nothing answers at all: the converter is off, or it is on another network than the box. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

STEP 7 READINGS ARRIVE AND LOOK RIGHT

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* **Pump running** reads 1 while the pump runs and 0 while it is stopped.
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter a reading below zero is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank); one far too high is a range or tank height typed wrongly: give the ranges again (step 5).

STEP 8 SOMEONE IS TOLD WHEN THE SITE NEEDS ATTENTION

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	■ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ■ *Nobody is set to be told:* add the people. ■ *Nobody on the list has an email address* (or a phone number): add one for each person. ■ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ■ *Test messages are limited to 6 an hour.* Wait and try again.

STEP 10 ALERT LIMITS AGREED WITH THE CUSTOMER

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

STEP 11 ELECTRICAL WORK DONE BY A LICENSED ELECTRICIAN

PROJECT MANAGER ELECTRICIAN

DO	Fitting meters in a board and wiring transmitters and contacts is the **licensed electrician's** work, under the customer's lock-out. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number** (for example *Arben Hoxha, licence E-1234*), or that no electrical work was needed: it cannot be left empty. Press **The electrical work is done**.
YOU SHOULD SEE	The step turns **Done**: *Done by a licensed electrician.*, with your name, the time and what you wrote (they are on the handover sheet). The platform does not check the licence: what you write is the record.
IF YOU DO NOT	Until it is done, commissioning stays grey. Do not press the button while the work is open or the lock-out is still on. The button will not work while the box is empty.

STEP 12 COMMISSIONED

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 12) with a note. It stays grey until every other step is done, the papers (step 1) and the electrician's name and licence number (step 11) included. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Soil probe. In the air, then in a glass of water, then in the soil.	Three different readings: low in the air, high in the water, plausible in the soil			☐ Pass ☐ Fail
Weather sensor. Against a hand-held thermometer at the sensor. (This check is added by the pack: the launch documents give no tolerance, so write both numbers and judge.)	Plausible for the time of day			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running (if there is a pump to watch). The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail

Check and how	Pass when	Greecon shows	Real	Result
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); the site's own page under **Sites**, with the soil moisture, the air temperature and humidity, the tank level and the pressure; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Write the probe's depth and place** on the handover record, and say that soil readings are only as good as the contact between the probe and the soil.
- ☐ **Say that Greecon gives no irrigation advice yet,** and agree when you will ask what reading means "time to water" (after the first week).
- ☐ **Ask at what air temperature frost damages the crop, and in which weeks.** Enter it as the frost limit; say that the alert is an Act now message, possibly in the night, and agree who is woken.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Send the third message on The install card** to the customer's contact the same day, with the handover sheet attached (its Albanian is a draft: have it read first).
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Each probe: depth, and where it is	
Alert limits agreed (tank Critical below, Low below; no-flow level if a pump is watched)	
“Start irrigating below”: not set yet; to be asked after the first week, on	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, the site’s page, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer’s folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
A probe or the weather sensor does not answer	A wrong unit number (many are delivered as unit 1, the same as another device: that is why every number is set at the office, before the probe is buried), a number or an address entered by hand wrongly, a speed or parity that does not match the converter, A and B swapped, no power	Check each in that order. Test the register from the box (`greecon-edge modbus-test`) against the maker's table. A device whose number was set before burying is easy to check; one that was not must be dug up, so every number is set before the probe goes in the ground
The soil reading looks wrong (constant, or the same in air and in water)	A wrong register, the wrong register kind (Input or Holding), a wrong scale, or poor contact between the probe and the soil	Re-test the register against the maker's table; repeat the air, water and soil check; check the probe's depth and contact
A frosty morning reads as a hot afternoon	The air temperature was mapped as an unsigned number	Map it as int16 (signed), with the scale of the sensor's table
No irrigation advice is shown	"Start irrigating below" has no standard value	After the first week of readings, ask the grower what reading means "time to water" and enter it as the limit (setup step 10)

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: the converter's address changed (press **Find equipment** again), the number on the device's label not the one on the install card, a wrong speed at the converter, A and B swapped on the data cable, the converter not in gateway mode.

5.4 Site pack W: Water tank and pump station

Tank, pump and the pipe it feeds. Planning estimate: About EUR 650 to 1,300. This pack is also printed on its own as PDF 07 (07-Site-Pack-W-Water-Tank-and-Pump-Station.pdf).

At a glance

WHO IT SUITS
A water tank and pump station: a tank, a pump and the pipe it feeds, with the pump's electricity use if wanted. The simplest site, and a good first rehearsal.

WHAT IT MEASURES
The level in the tank, the line pressure, the pump's flow (with a flow meter), whether it runs and, optionally, the pump's electricity use.

PLANNING ESTIMATE
About EUR 650 to 1,300.

TIME
Survey about 1 hour. Install day 4 to 6 hours with the electrician. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Water tank and pump station

STANDARD SETTINGS
Unit numbers: water module 3, optional pump meter 4, pump contact module 5 (if the starter has a spare contact), all set at the office. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode.

The order of the work: survey, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the labels, and the 12 or 24 V supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with their 24 V supply, as the module's manual draws it.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

HOW TO READ THIS PACK

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, the pump's run contact, and the 4 to 20 mA loops.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure W-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Water tank and pump station** makes three devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow	Water I/O module, unit 3: the flow meter's 4 to 20 mA (optional)	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running	Pump contact module (optional), unit 5: the pump starter's spare contact	on or off	None
Energy consumption (optional)	Pump meter (optional): an Eastron SDM230 on the pump's feed, unit 4	kW	None

Without the flow meter, or without the pump's running contact, "the pump is running but no water is flowing" cannot be raised, and the handover should say so.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)

	Item	Specification	Qty	Planning estimate (EUR)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200
☐	**Single-phase meter** *(optional)*	Eastron SDM230, on the pump's feed, to measure its electricity. *To measure the pump's electricity.*	0 or 1	35 to 60

About EUR 650 to 1,300. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in Part 4.

THE PROJECT MANAGER BRINGS

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch

- [ ] Spare fuses and terminals
- [ ] A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the electrician if the plan needs one. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

ASK, IN THIS ORDER

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or water? Write it in their words. It is the reason they buy.
3. Where are the water tank, the pump and its starter? How tall is the tank? Has the starter a spare "running" contact?
4. **What is the pipe's pressure range?** It decides the pressure transmitter's range.
5. **What does the pump do when the tank is low, and when it is full?** And who is told today when something goes wrong?
6. Is there a network socket in the plant room, or cellular signal? (Look at your phone there.)
7. Is there a power socket for the box, and is it on a circuit that stays on?
8. Does the owner want the pump's own electricity measured (an optional single-phase meter on the pump's feed)?
9. Who is the customer's electrician, and can they be there on install day?

LOOK AT, AND PHOTOGRAPH

- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.
- Where the level and pressure transmitters can be fitted: the pressure transmitter on the pipe through an isolating valve; the level transmitter at the bottom of the tank.
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable.
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT W · WATER TANK AND PUMP STATION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Tank: height (m), water source; drinking water?	
Pump (kW); starter location; spare running contact on the starter; what it does when the tank is low and when it is full	
Pipe pressure range (bar), so the pressure transmitter's range	
Sensors needed: level, pressure, flow, pump running (and where)	
Pump meter wanted?	☐ Yes ☐ No
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit W and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers. Mains work is inside the dashed red box and is the electrician's.

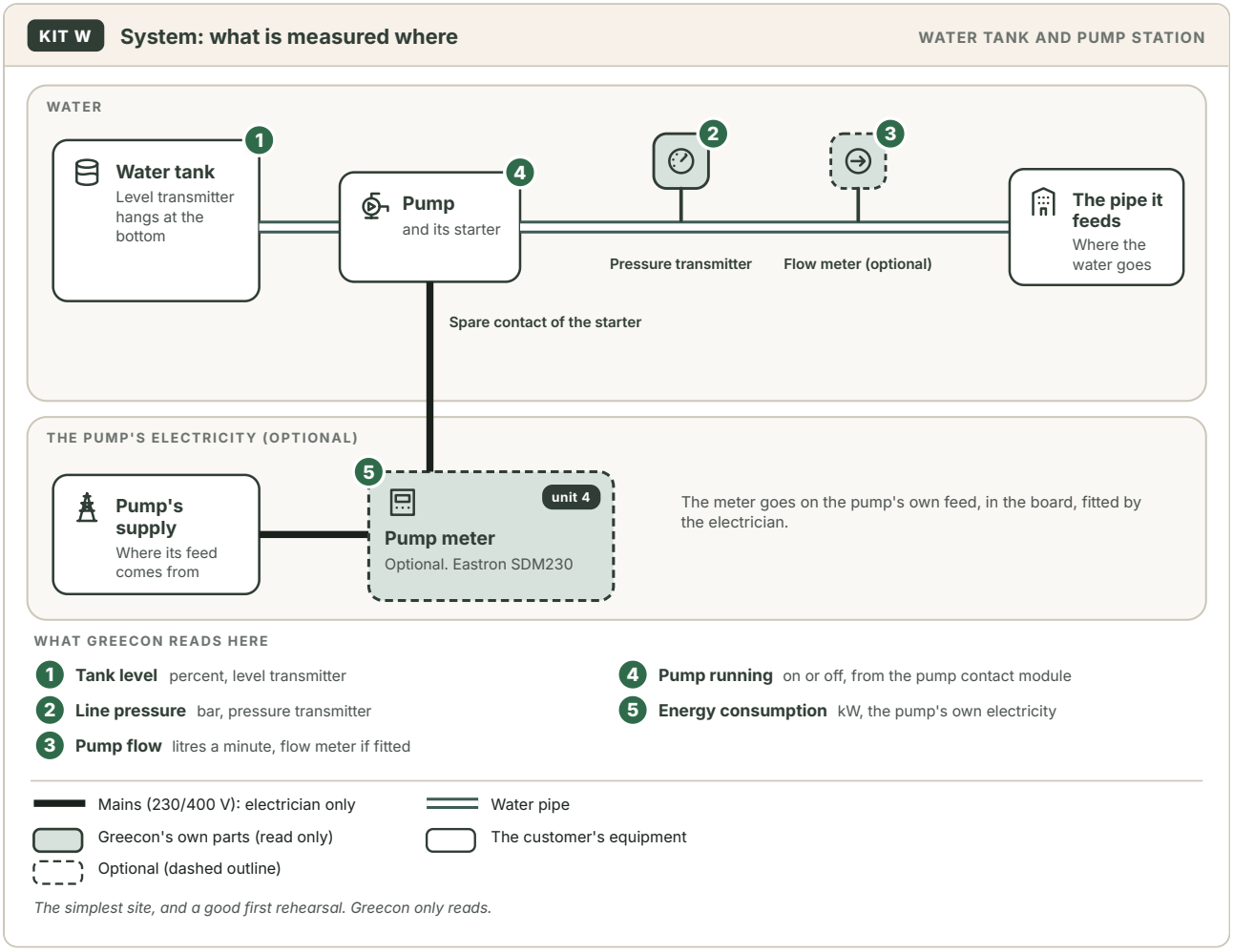
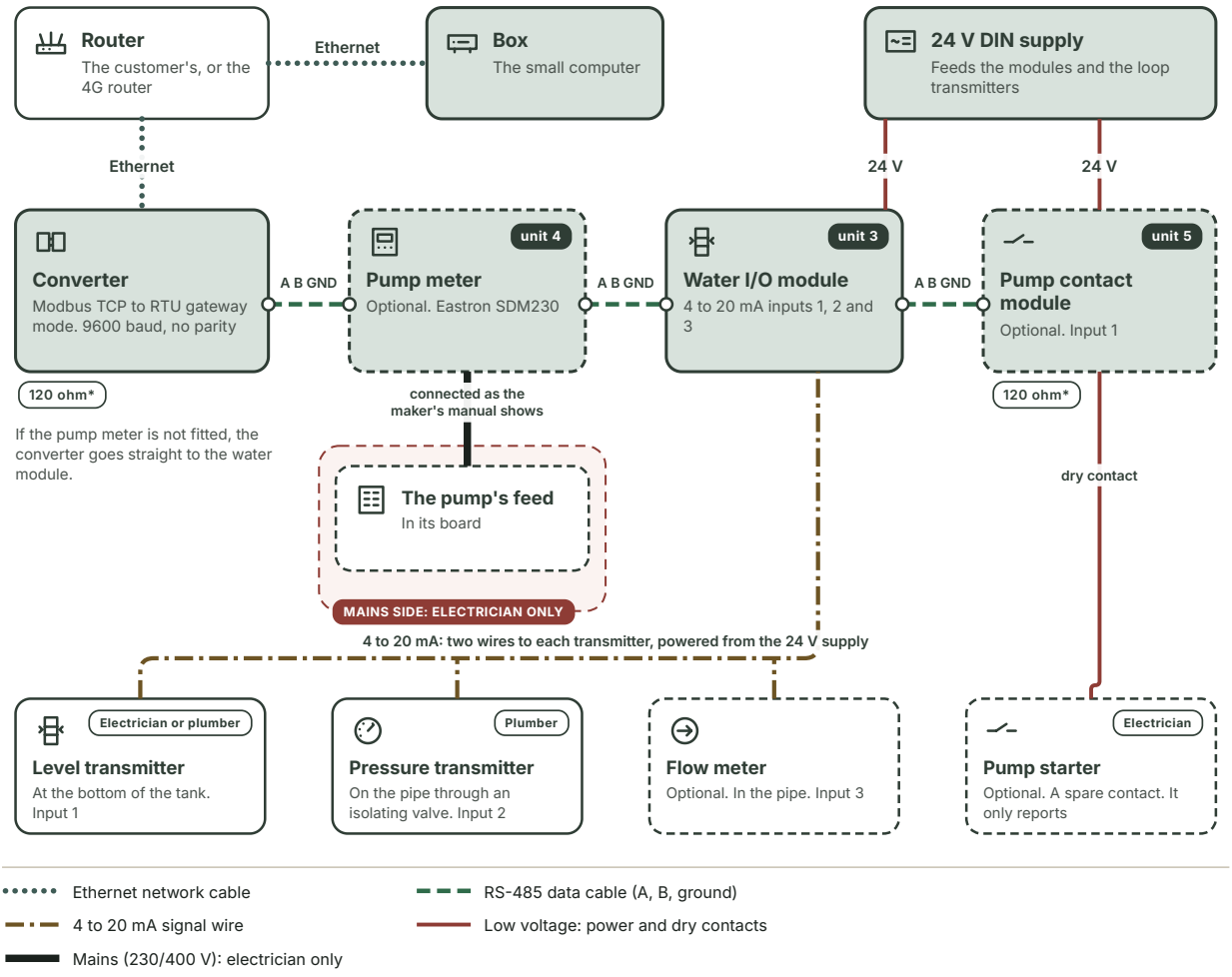


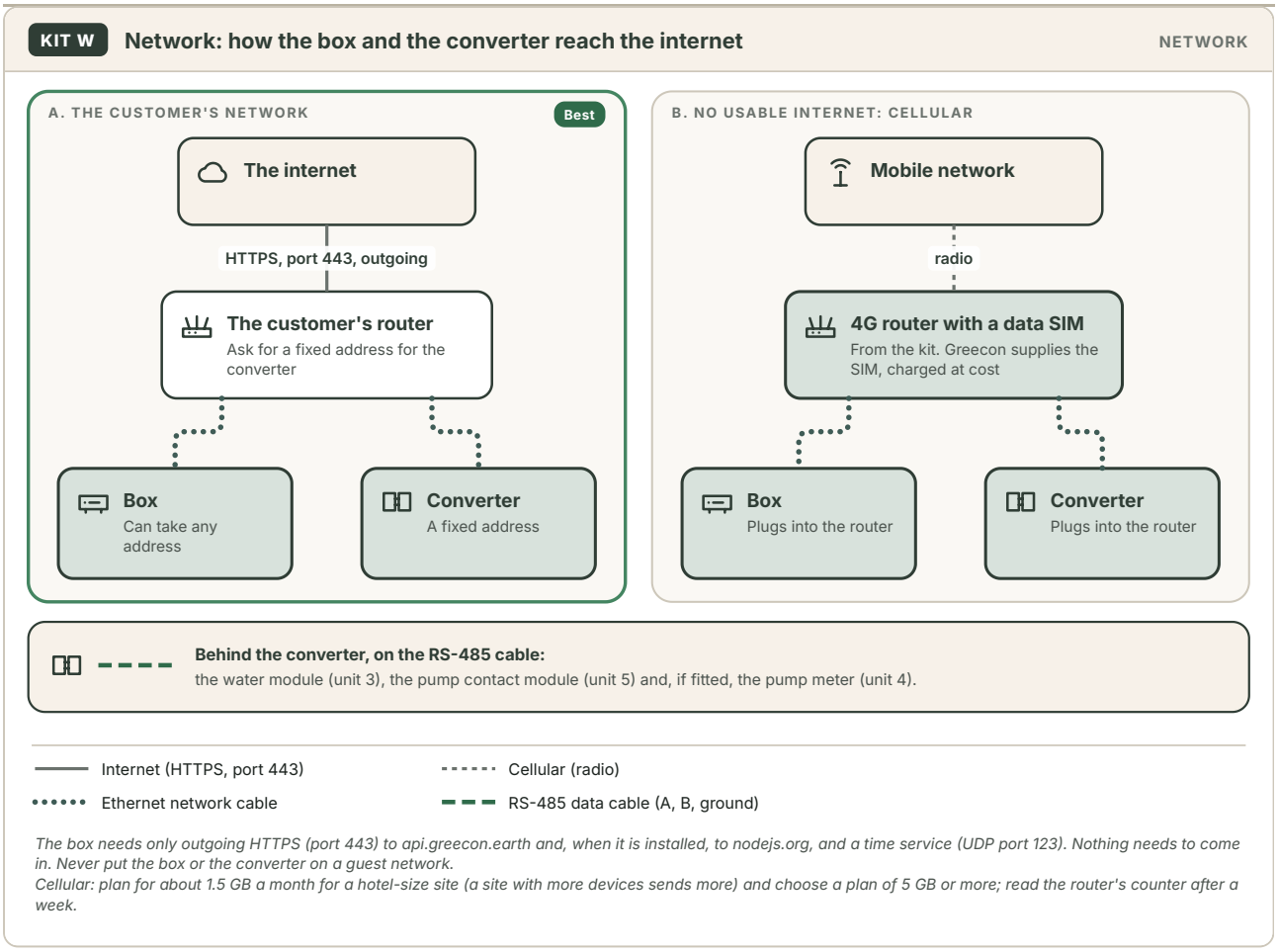
Figure W-1. Kit W, water tank and pump station. System: what is measured where. The dashed parts are optional.

KIT W **Wiring: the data cable and the signal wires** BUS AND SIGNALS



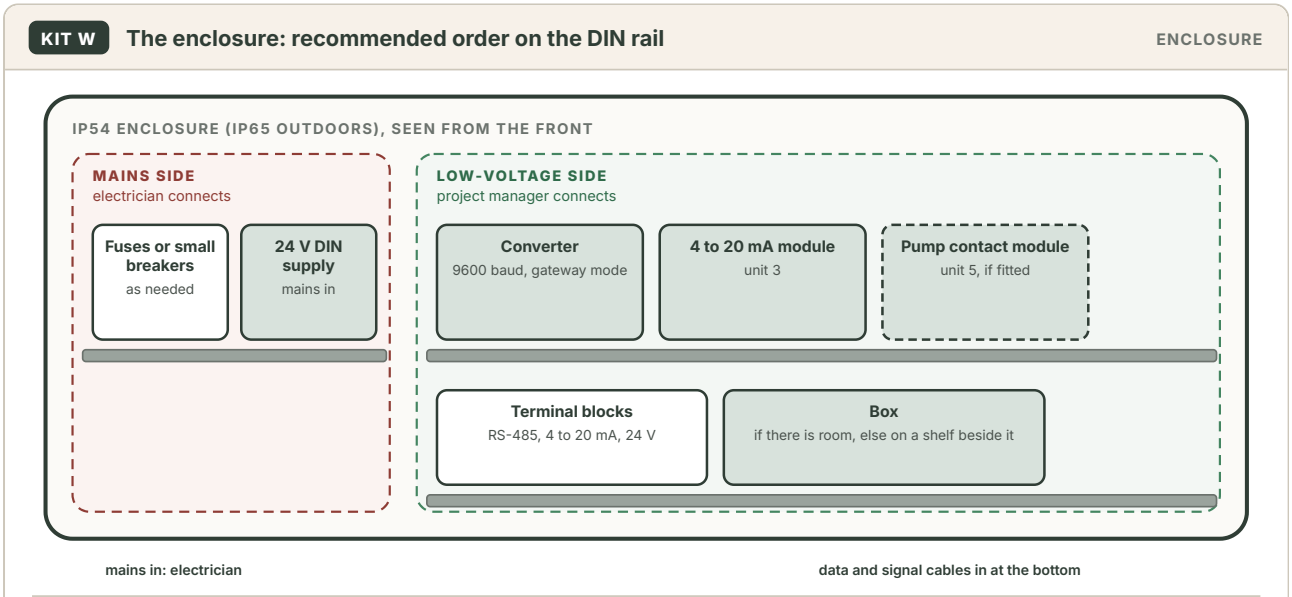
Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The pump meter is connected as its maker's manual shows. The electrician wires the 4 to 20 mA loops and the contact. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-2. Kit W. Wiring: one RS-485 cable from the converter through the pump meter (if fitted) and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module.



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure W-3. Kit W. Network: the box and the converter on the customer's router (best) or on a 4G router.



The optional pump meter sits in the board that feeds the pump, not in this enclosure.
The order is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure W-4. Kit W. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

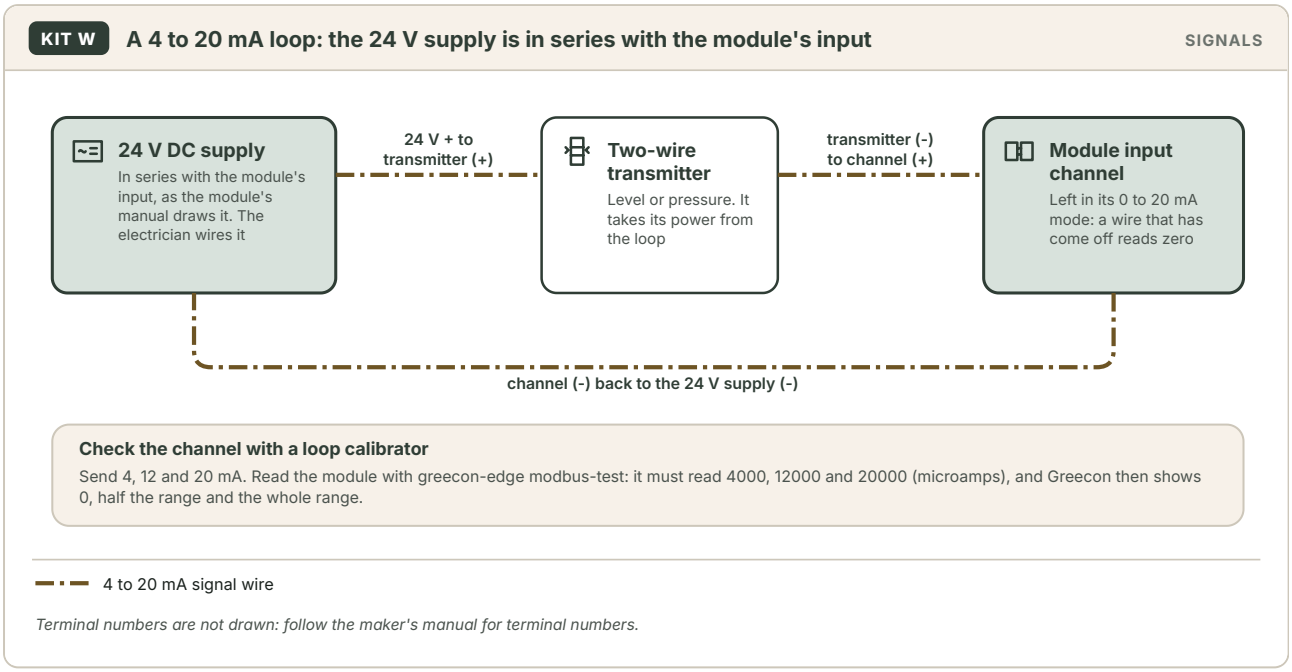


Figure W-5. Kit W. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

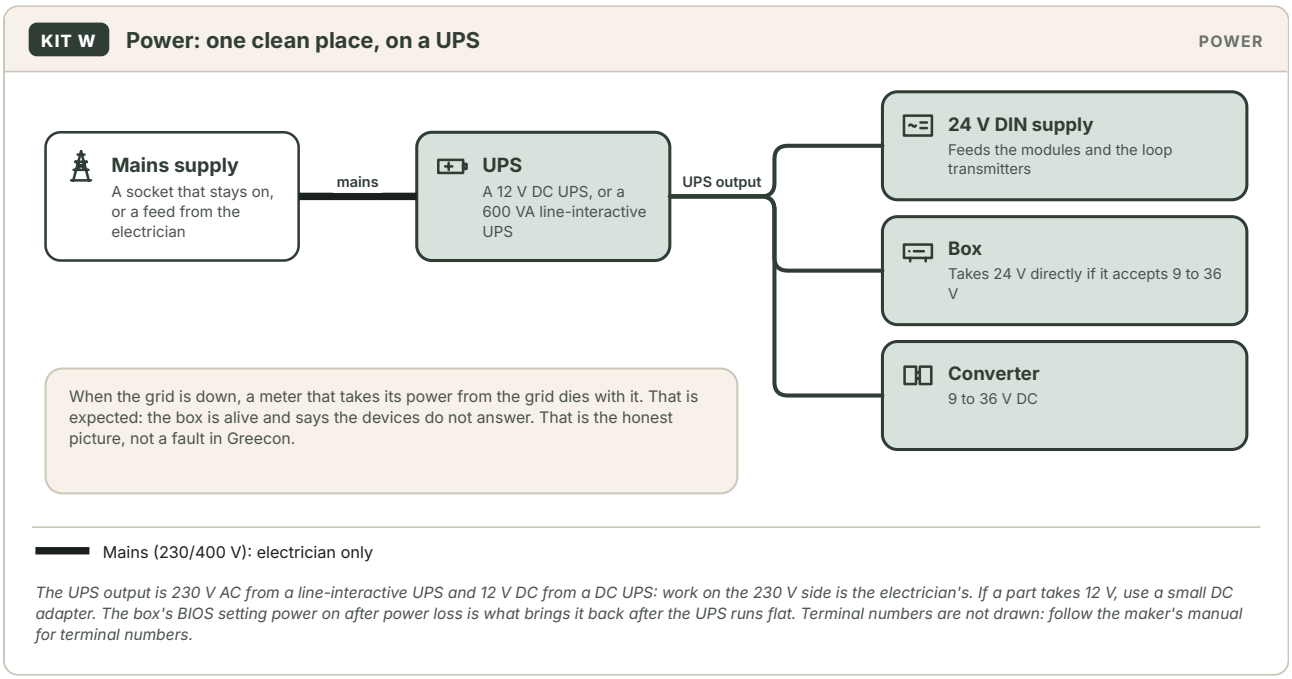


Figure W-6. Kit W. Power: the UPS feeds the box, the converter and the 24 V supply.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box, and more for the first kit's numbers and its desk test: time it, and write it down. PROJECT MANAGER

IN GREECON

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.

- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.

- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.

- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Water tank and pump station**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.

- [ ] **Write down that the papers are signed:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`, and its first step is **Order form signed and data terms agreed**. With the signed order form and the agreed data terms in front of you, write in the box who signed for the customer and on what date (it cannot be left empty), and press **The papers are signed**. Nothing goes into a customer's building without both papers.

- [ ] **Print the install card:** the link **The install card** is at the top of the setup page. It is your list for the visit: what to bring, what each device is set to and its number, how it is wired, the converter's settings, and what the customer's network is asked for. At the bottom are three messages to copy: send the first to the customer's contact two or three days before, and the second to whoever looks after their network. Each is in English with an *Albanian draft* folded under it, and someone who writes Albanian every day reads the draft before it is sent.

- [ ] **Get the install line:** on the same page press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> **STOP** **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

THE BOX ON YOUR DESK

The operating system and the pasted install line are still a terminal job; everything after them is done on the setup page.

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in Part 4, section 4.3.

- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).

- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`
- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit, the printed pack and the printed install card.

THE NUMBERS AND THE CONVERTER

The kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office; on site you only check the labels and do not change them. The numbers for this kit are in the table under **Numbers, addresses and ranges** and on the install card. Everything on one cable is **9600 baud, 8 data bits, no parity, 1 stop bit**.

- [ ] **Set each meter's number on its own display,** one meter at a time (delivered as unit 1, two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page. Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains.
- [ ] **Number the two Waveshare modules** with the maker's setting tool and a USB-to-RS-485 adapter, one at a time (each is delivered as address 1, and there is no switch): **3** on the 4 to 20 mA module (Modbus RTU Analog Input 8CH) and **5** on the pump contact module (Modbus RTU IO 8CH). Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default 0 to 20 mA mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- [ ] **Set the converter** in its web page: mode **Modbus TCP to RTU (gateway)** (not transparent, not raw TCP, not Modbus RTU over TCP), 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site: a fixed one is best, and Greecon finds the converter either way.
- [ ] **Label every device with its number** and the converter with its settings, and photograph the labels.
- [ ] **Test the kit on the desk.** The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (Part 3, section 3.11). Anything that differs from this book is reported to Greecon.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the kit: Part 3, section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN / PROJECT MANAGER	**Before anyone touches anything:** the signed papers are in your hand, the electrician is there, and the lock-out is agreed. **Walk the site together.** Agree where each sensor goes. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the tank, the pump and the starter. Have the router and a socket that stays on ready, and the person who can give the converter a fixed address.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the board to be worked on.
☐	4	ELECTRICIAN	**Connect the pump's run contact** (a spare auxiliary contact of the starter, as a dry contact) to **input 1 of the pump contact module**, as the module's manual draws it (its DI COM left unconnected). Never to a control circuit, and never to the module's relay outputs: the contact only reports. If the starter has no spare contact there is no module: it is taken off the list in setup step 4.
☐	5	ELECTRICIAN / PROJECT MANAGER	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts, and **wire their 4 to 20 mA loops** with the 24 V supply in series, as the module's manual draws it: **level on input 1, pressure on input 2, flow meter on input 3.** The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve. Figure W-5. You check each is on its own input and number the wires at both ends.
☐	6	ELECTRICIAN	**Fit the pump meter** (only if wanted) on the pump's feed, as its maker's manual shows.
☐	7	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the water module, the pump contact module, the 24 V supply and the UPS (the box too, if there is room, otherwise on a shelf beside it). Figure W-4.
☐	8	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the pump meter if there is one, and on to the water module and the pump contact module: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. The electrician connects its end at the pump meter's A and B terminals. Figure W-2.
☐	9	PROJECT MANAGER	**Check the labels:** the number on each device against the table under *Numbers, addresses and ranges* (water module 3, pump contact module 5, pump meter 4). They were set at the office: do not change them. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400); Greecon set it to 9600 on its display at the office. Photograph each label and the pump meter's display.
☐	10	PROJECT MANAGER	**Check the converter's label** (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502) and **connect the network:** plug the box and the converter into the customer's router, or into the sockets of the 4G router if the site has no usable internet. Ask for a fixed address for the converter and write it down. Figure W-3.
☐	11	ELECTRICIAN / PROJECT MANAGER	**Photograph before closing:** the wiring, the converter's settings, each device's label. Then close the board.

	#	Who	Do this
☐	12	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	13	PROJECT MANAGER	**Power up** the UPS, the 24 V supply, the converter and the box.
☐	14	ELECTRICIAN	**Stay until commissioning is done.**
☐	15	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list: **Find equipment**, the ranges, and the electrician's name and licence number.

Numbers, addresses and ranges

The numbers on the cable were set at the office. The box finds where the converter is. You give the range printed on each transmitter's label. You type no IP address, no port and no register.

THE NUMBERS ON THE CABLE

All the devices are behind one converter: they share its address and port 502, and differ only by number. Check each label against this table on install day. **Do not change a number on site.**

Device in Greecon	Make and model	Number	Label checked	Address found (the converter's)
Water I/O module	Waveshare Modbus RTU Analog Input 8CH	3	☐	
Pump contact module (optional)	Waveshare Modbus RTU IO 8CH	5	☐	
Pump meter (optional)	Eastron SDM230	4	☐	

THE METERS AND THE PUMP CONTACT MODULE

Nothing to type: the template fills in their registers. For the pump meter, if there is one, it is total active power, input register 12 on the SDM230 (a manual's 30013; counted from zero), a 32-bit floating point number with the high half first, in watts, multiplied by 0.001 to make kW; for the pump contact module it is digital input 1. Check the module by eye: with the pump running, **Pump running** reads 1; stopped, 0. Test from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 4 --type input --address 12 --data-type
float32
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
```

THE RANGES ON THE LABELS

One form for each transmitter (step 5 of the setup list), in the label's own words. Read the range from the label on the transmitter and measure how tall the tank is inside, from the bottom to the top of the water when it is full.

Reading	Input	The form asks for	Range on the label	Tank height
Tank level	1	Level transmitter: the range on the label (m or bar) and how tall the tank is inside (m)		
Line pressure	2	Pressure transmitter: the top of the range (bar)		
Pump flow	3	Flow meter: the top of the range (lpm, litres a minute)		

Greecon works out the register and the scale: 4 mA is 0 and 20 mA is the range (for the tank, range ÷ tank height × 100: a 0 to 5 m transmitter in a tank 4 m tall gives 125, and a full tank, at 16.8 mA, reads 100). **A transmitter shorter than the tank is refused:** it cannot see the top of the water. **No flow meter?** Tick **Not fitted at this site** under Flow meter: its reading is taken away, and the handover says that the no-flow alert cannot be raised. **No pump meter, or no spare contact for the pump?** Press **The site does not have this: remove it** under the device in step 4 (it asks you to confirm, because the device's readings go with it).

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to twelve steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself: you press buttons only for what a person must check or say. Near the top are links to the site page, to **The install card** (the card for the visit) and to the **Handover sheet**.

	Step	Title on the setup list	Done by
☐	1	**Order form signed and data terms agreed**	Project manager
☐	2	**Prices from the customer's bills**	Project manager (optional)
☐	3	**Install the box**	Project manager
☐	4	**Where each device is on the network**	Project manager
☐	5	**Which register holds which reading**	Project manager
☐	6	**Every device answers the box**	The platform, by itself
☐	7	**Readings arrive and look right**	The platform, by itself
☐	8	**Someone is told when the site needs attention**	Project manager
☐	10	**Alert limits agreed with the customer**	Project manager
☐	11	**Electrical work done by a licensed electrician**	Project manager, with the electrician
☐	12	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

Not on this kit. The platform leaves out a step that does not apply to the site, so the list for kit W has no step 9 (Grid meter direction checked). The steps keep their order; the count on the setup page is out of the steps that do apply, not out of 12.

STEP 1 ORDER FORM SIGNED AND DATA TERMS AGREED

PROJECT MANAGER

DO	Nothing goes into a customer's building until the order form is signed and the data terms are agreed. Greecon keeps the papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date** (for example *Signed for the hotel by Besa Krasniqi on 19 October*): it cannot be left empty. Press **The papers are signed**. Do this at the office, and bring copies on install day.
YOU SHOULD SEE	The step turns **Done**: *Signed and agreed.*, with your name, the time and what you wrote. They are on the handover sheet.
IF YOU DO NOT	Until it is done, commissioning stays grey. The button will not work while the box is empty. If a paper turns out to be wrong, press **Reopen** (if the site had been commissioned, that is taken back too).

STEP 2 PRICES FROM THE CUSTOMER'S BILLS

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

STEP 3 INSTALL THE BOX

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	• *Waiting for the install command to be run on the box.* The line has not been run yet. • *The box has its credential but has not reported yet.* Wait a minute and check the network cable. • *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. • *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

STEP 4 WHERE EACH DEVICE IS ON THE NETWORK

PROJECT MANAGER

DO	Press **Find equipment**. Nobody types an IP address, a port or a unit number here. - The box looks at the site's own network (local private addresses only, port 502, about a minute) and only reads. The page says *The box is looking at the network now.* and updates by itself. The platform recognises what answered and fills in at once only what is certain: a device with no address, found at the kit's own number. The rest waits for you. - A device **Found** but not filled in (another address, another number, another meter model): read the line and, if it makes sense, press **Use this address** (or *Use these N addresses*). An optional device the site does not have: press **The site does not have this: remove it**. - The soil probe, the weather sensor and a battery are entered by hand: under the device open **Enter the address by hand** (soil probe 6, weather sensor 7). - Box not on your network (a desk rehearsal), or a converter on another port: open **I know the converter's address**, type its address (and its port, if not 502), and press **Look at this address**.
YOU SHOULD SEE	Each device gets a line in plain words: **Set** (found, address filled in), **Found**, **Not found**, **Not sure which** or **Enter by hand**. When it worked the step is **Done** (*Every device has an address.*), and the button reads **Look again**.
IF YOU DO NOT	- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* An optional device the site does not have: press **The site does not have this: remove it**. - **Not sure which**: Greecon will not guess, because a wrong guess could swap the grid meter and the solar meter. Read each device's label, use **Enter the address by hand** under each with the number on its label, then **Look again**. Numbers are set at the office: a part with a wrong number goes back there. - *No Modbus converter answered on this network:* check that the converter is on and plugged into the same router as the box. *A converter answers at 192.168.1.50, but nothing answers behind it:* check each meter's address, the speed and the A and B wires. *All answered with exactly the same words:* the converter is not in gateway mode. *The box did not pick this up:* check that it shows as online in the step above. - Readings that stop after it had worked: the router may have given the converter a new address. Press **Find equipment**, then **Use this address**.

STEP 5 WHICH REGISTER HOLDS WHICH READING

PROJECT MANAGER

DO	The Eastron meters and the pump contact module need nothing typed. For the 4 to 20 mA module the page asks, for each transmitter, only for the range printed on its label, in the label's own words: - **Level transmitter** (input 1): the **Range on the label**, for example 5, its **Unit on the label** (metres, or bar) and **How tall the tank is inside** in metres, for example 4: from the bottom to the top of the water when full. - **Pressure transmitter** (input 2): the **Top of the range** in bar; 10 is usual. - **Flow meter** (input 3): the **Top of the range** in litres a minute. Greecon works out the register and the scale. For a transmitter the site does not have (no flow meter), tick **Not fitted at this site**: Greecon asks first, then takes the reading and its history away. Press **Save the ranges**. Equipment Greecon does not know (the soil probe, the weather sensor, a battery) is mapped by hand, from the maker's register table: under the reading press **Choose the register** and fill in the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero: 30053 is address 52, 40001 is address 0), the kind of number (and which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.* Under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (0 to 5 m in a tank 4 m tall); the tank level is a percentage of the tank, so a full tank reads 100.
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *Needs the range printed on the transmitter's label.* (the 4 to 20 mA module) or *No register chosen.* (equipment Greecon does not know). A transmitter shorter than the tank is refused: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* A reading the site does not have holds the list here: tick **Not fitted at this site**, or press **The site does not have this: remove it** (step 4) for an optional device.

STEP 6 EVERY DEVICE ANSWERS THE BOX

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings).* On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2).* Look in this order: the converter's address may have changed (press **Find equipment** again, then **Use this address**); the number on the device's label against the install card; the speed at the converter; A and B swapped on the data cable; the converter not in gateway mode. Nothing answers at all: the converter is off, or it is on another network than the box. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

STEP 7 READINGS ARRIVE AND LOOK RIGHT

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* **Pump running** reads 1 while the pump runs and 0 while it is stopped.
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter a reading below zero is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank); one far too high is a range or tank height typed wrongly: give the ranges again (step 5).

STEP 8 SOMEONE IS TOLD WHEN THE SITE NEEDS ATTENTION

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	■ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ■ *Nobody is set to be told:* add the people. ■ *Nobody on the list has an email address* (or a phone number): add one for each person. ■ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ■ *Test messages are limited to 6 an hour.* Wait and try again.

STEP 10 ALERT LIMITS AGREED WITH THE CUSTOMER

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard)*. After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

STEP 11 ELECTRICAL WORK DONE BY A LICENSED ELECTRICIAN

PROJECT MANAGER ELECTRICIAN

DO	Fitting meters in a board and wiring transmitters and contacts is the **licensed electrician's** work, under the customer's lock-out. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number** (for example *Arben Hoxha, licence E-1234*), or that no electrical work was needed: it cannot be left empty. Press **The electrical work is done**.
YOU SHOULD SEE	The step turns **Done**: *Done by a licensed electrician.*, with your name, the time and what you wrote (they are on the handover sheet). The platform does not check the licence: what you write is the record.
IF YOU DO NOT	Until it is done, commissioning stays grey. Do not press the button while the work is open or the lock-out is still on. The button will not work while the box is empty.

STEP 12 COMMISSIONED

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 12) with a note. It stays grey until every other step is done, the papers (step 1) and the electrician's name and licence number (step 11) included. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running. The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail
Pump meter's display against Greecon (if fitted). Read the display and Greecon at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- [ ] **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
- [ ] **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); the site's own page under **Sites**, with the tank level, the pressure and whether the pump runs; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- [ ] **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- [ ] **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- [ ] **Send the third message on The install card** to the customer's contact the same day, with the handover sheet attached (its Albanian is a draft: have it read first).
- [ ] **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT W · WATER TANK AND PUMP STATION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Alert limits agreed (tank Critical below, Low below; no-flow level)	
Without a flow meter or the pump's run contact the no-flow alert cannot be raised: said to the owner	☐ Said ☐ Not applicable
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, the site's page, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
The pump shows running when it is stopped (and the no-flow alert, an Act now, opens while the pump is stopped)	The pump's running contact is inverted	The electrician swaps the contact's normally-open and normally-closed wires, or uses the opposite contact. Until it is right the no-flow alert cannot be trusted
The tank level is wrong (above 100 percent, below zero, or does not move)	The range or the tank's height was given wrongly; a wire has come off (the module reads 0 and the level reads below zero); a blocked vented cable; or the transmitter is not at the bottom of the tank or not on input 1	Give the ranges again (**Save the ranges**). Check that the transmitter hangs at the bottom with its vented cable free and fixed. Send 4, 12 and 20 mA with the loop calibrator: the module must read 4000, 12000 and 20000
A device does not answer (the pump meter, the water module or the pump contact module)	The converter's address changed; a number or a speed on the device that is not the one on the install card (an SDM230 can be delivered at 2400 baud, the V1.6 manual says, not the converter's 9600; the meters and the modules are delivered as number 1); A and B swapped; the converter not in gateway mode	Press **Find equipment** again. Read the number on the device's label (and a meter's speed on its display) against the install card: if they differ, the part was not set at the office, so take it back there or fit the spare that was numbered there, and do not change it on site. Then check A and B and the converter's gateway mode

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: the converter's address changed (press **Find equipment** again), the number on the device's label not the one on the install card, a wrong speed at the converter, A and B swapped on the data cable, the converter not in gateway mode.

PART 6

Commissioning, handover and what to leave

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. Handover is the thirty minutes after it: the owner sees the product working on their own equipment, and is left with what they need. Do them in this order, on install day or the day after, with the electrician still there for the checks that need them. The checks are the ones for the site's kind; the four site packs (Part 5) print them as a record with room for the two numbers, and Appendix A has a blank record for any site.

PROJECT MANAGER ELECTRICIAN CUSTOMER

STOP

Nobody starts, stops or opens equipment for a check except the owner's own person. They know what else it affects. The project manager reads and compares; the electrician works on the mains side under the lock-out; Greecon only reads.

6.1 Commissioning checks by kind of site

Compare, in this order, and write the pair of numbers on the handover.

Check	How	Pass
Every meter's display against Greecon	Read the meter's display (kW or W) and Greecon at the same time	Within 5 percent, or 1 kW on a small load
Grid direction	A known load on, the sun down: grid import rises	Import up, export unchanged
Solar at night	Solar production	Zero, or a few watts
Solar at midday	Against the inverter's display	Within about 5 percent
Tank level	A dipstick or the sight gauge at two levels	Within 5 percentage points
Pressure	Against the pressure gauge on the same line	Within 0.2 bar
Flow	A bucket test (fill a measured volume, time it), or the flow meter's own display	Within 10 percent
Pump running	The owner starts the pump; Greecon shows running within 30 seconds	Changes both ways
Soil probe	In the air (low), in a glass of water (high), in the soil (plausible)	Three different readings

Check	How	Pass
4 to 20 mA channels	With a loop calibrator in place of the transmitter, send 4, 12 and 20 mA	The reading at the bottom, the middle and the top of the transmitter's range (for a 0 to 10 bar transmitter: 0, 5 and 10 bar)
A test message	Each person on the list	Arrived
Power cut	Unplug the box for 30 seconds, plug it in	Back reporting within 3 minutes, and the box says nothing about readings lost
Internet cut *(the first install, and any time the box or its software is new)*	Cut **only the site's internet** for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router**: if the box cannot reach the converter it reads nothing, and there is nothing to test. Then restore it	`greecon-edge status` shows a queue while it is out, and a few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down: it is the evidence for the statement that the box keeps its readings.**

On the tank's channel the top of the range reads more than 100 when the transmitter is longer than the tank (a 0 to 5 m transmitter in a tank 4 m tall reads 125 at 20 mA), and Greecon may call that impossible while you test: that is the test, not a fault. A full tank reads 100 (16.8 mA in that example). The module itself must read 4000, 12000 and 20000 at 4, 12 and 20 mA:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

(The address is 0 for input 1, 1 for input 2, 2 for input 3.) Put the transmitter back when you finish.

Do not **start, stop or open** any equipment yourself. The owner's own person does, because they know what else it affects.

6.2 Handover

1. On the setup page open **Open the handover sheet**. Check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and site, says where each device is (address, unit number, registers), who is told and how to reach them, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
2. Show the owner, in ten minutes, in this order: the **Overview** (is everything all right, in one sentence); **Energy now**, the flow moving from the solar to the site and the grid; **Value**, what the solar kept on site this month (and the prices it uses); **Who is told** (under **Settings**) and a live **test message** arriving on their phone (and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert); **Reports**, under **Downloads**, where **their data is theirs**: every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want (and if they stop, they can ask Greecon for every reading it holds, as a file).
3. Say: *Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.*

4. Leave: the printed sheet (signed), the box's label, a card with `info@greecon.earth` and your phone.
5. Photographs and the survey form go in the shared drive, in a folder named for the customer.
6. The same day, send the customer's contact the third message on **The install card** (its subject is *Greecon is now watching* and the site's name), with the handover sheet attached. It is in English, with an *Albanian draft* folded under it: someone who writes Albanian every day reads the draft before it is sent.
7. Press **Back to the Greecon console** (the banner's button) to leave the customer's account: it returns you to **Today**.

6.3 What to leave with the customer

Leave the customer with these, and nothing more:

- ☐ **The printed handover sheet, signed.** It says where each device is, who is told, what was commissioned and by whom, and how to reach Greecon.
- ☐ **The box's label** on the box: the site name and the date. Never a code or a password written on the box.
- ☐ **A card** with `info@greecon.earth` and your phone number.
- ☐ **Their own logins.** The owner chose a password of their own at their first sign-in; the people who should also look at the screens have a login of their own (the Operator role is enough), and everyone on the **Who is told** list knows about the *I have seen this* link in each alert message, which needs no login. The temporary passwords were told by phone, never by email.
- ☐ **The words** in section 6.2: what Greecon is and is not, how an alert reaches people, and that their data is theirs.

Do not leave: the install line or its code (it is the box's key, and it has been used), any password written down, or a promise of a saving.

What you keep, for Greecon: the survey form and every photograph (every meter's label, the wiring before the board was closed, the converter's settings, the box's label) in the shared drive, in a folder named for the customer; the commissioning record with its pairs of numbers, and the internet-cut result, which is the evidence for the statement that the box keeps its readings; and a line in the hardware register for each part: the make and model, the supplier, the date, the serial number or the batch, and the invoice.

6.4 When a check fails

A check that fails is not commissioned. Do not press **Commission this site** until it passes. Look, in this order, at what the table says; the longer answers are in Appendix C and in the runbook for a reading that looks wrong.

Check that fails	Look at first
A meter's display and Greecon differ by more than 5 percent	The model fitted against the one on the install card (an SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only); on a meter with 5 A current transformers, the CT ratio set in the meter's menu; a clamp on the wrong phase or cable; the two meters' numbers swapped (the meter at the supply entry must be number 1). Test the register from the box with `greecon-edge modbus-test` against the meter's display

Check that fails	Look at first
Grid direction: export rises when the load is switched on	The clamps are the wrong way round: the electrician turns them, with the lock-out on, and you repeat the check
Solar reads more than a few watts at night, or nothing at midday	The solar meter's clamps are on the wrong cable; or the inverter is limited to zero export
The tank reading is off by more than 5 percentage points, or above 100	The range and the tank's height as they were given (give the ranges again: a wrong height reads too high or too low); the transmitter at the bottom of the tank with its vented cable free; the transmitter on input 1
Pressure is off by more than 0.2 bar	The top of the range as it was given; the transmitter on input 2; check the channel with the loop calibrator
Flow is off by more than 10 percent	The top of the range as it was given (in litres a minute); the flow meter on input 3; its straight run in the pipe
The pump shows running when it is stopped (and the no-flow alert opens)	The pump's running contact is inverted: the electrician swaps its normally-open and normally-closed wires, or uses the opposite contact
The reading at 4, 12 and 20 mA is not the bottom, the middle and the top	Read the module from the box with `greecon-edge modbus-test` (unit 3, input registers 0 to 2; the command is in the hardware reference): 4, 12 and 20 mA must read 4000, 12000 and 20000. If they do, give the range again. If they do not, the loop is wrong (a wire, the 24 V supply, the wrong input): the electrician checks it
A test message did not arrive	The address, the number's country code, spam; look at **What was sent** under **Who is told**
After a power cut the box is not back within 3 minutes	The BIOS "power on after power loss" setting; the clock; `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`
After the internet cut the queue does not empty	`greecon-edge status` and the box's log say what it is waiting for; then the box is offline: see the runbook for an offline box

Write down what was wrong and what you changed, on the commissioning record. A reading that was wrong stays as it was, with a note: do not edit history.

PART 7

Quick runbooks

For: the founder and the project manager. Each runbook says what you see, what to check, what to do, what to tell the customer, and what to write down afterwards. Commands are in `code`; places in the app are in **bold**.

The product exists so that people hear about problems before their guests do. These pages exist so that **Greecon hears about its own problems before its customers do.**

R0 The routine

When	What	How long
Every morning (the project manager; the founder when away)	Read the morning email from Greecon ("Greecon, Mon 19 Oct: …"). Open **Today**: the sites under **Needs a look**, worst first. For anything not **Stable**, follow the runbook below. If everything is **Stable**, do nothing.	5 min
Same day	A customer who is **Attention needed** or **Unknown**: phone them *before* they phone you ("we saw your box was offline since 03:10; we are looking into it").	
Every Friday (founder)	The pipeline sheet (the Sales kit (document 04, not part of these guides)); the hardware register; take the weekly backup (R13); look at Railway **Usage**	30 min
Every month (founder)	Restore drill (R13); read the last month's alerts at each live site with the owner: *useful? too many?* (R5); look at the audit log for anything unexpected; check Resend and Twilio balances; update packages (R14 lists who can do what)	2 hours
Every six months	Re-check the legal list (document 22); read the claims register (document 21) against what is being said; renew `security.txt` before 30 September 2027	half a day

Who owns what. The **founder** owns every incident about the platform or about security, and is the person the outside uptime check calls. The **project manager** is the first person for a customer's site and deputises for the founder. **Nobody promises a 24-hour response yet** (the customer promise is: acknowledged within four business hours, Monday to Saturday). The platform does the night shift: Act now reaches the *customer's own* people at once.

Severity.

	What	Target
S1	The platform is down; sign-in does not work for anyone; a possible leak of data	Acknowledge to affected customers within 30 minutes of finding out (a target, not a promise)
S2	One customer's site is silent or wrong (box offline, readings stale)	Same working day
S3	A wrong value, a message not received, a question	Within four business hours

While a clock runs. Safety events, security incidents and personal-data incidents have legal clocks (4 hours, 24 hours, 72 hours) that depend on the law that applies to the *customer*. R12 says what to do, and the clocks are in document 21, section 4. Write down the **time you found out** first.

R1 A customer's box is offline

You see: Today, under *Needs a look*, says *The box has not reported for N: it is offline*; or the customer's morning message says readings are out of date. **Or you are told:** Greecon's own team gets a message when a customer's box has been silent for a quarter of an hour (*A customer's box went silent*, also under **Who is told → What was sent**): the client, the site, since when, its last reading, and whether any message has gone to the customer (it has not, by this one). **By day it goes at once; in the team's quiet hours (22:00 to 07:00 unless changed under Who is told) it waits, shown there as *Waiting* with the time it is due, and goes when they end, unless the box has reported again by then, in which case it is dropped and the log says why:** a box that goes quiet at midnight is on **Today** at once and its message arrives at 07:00; nobody is woken for it. If three or more boxes go silent within ten minutes it is **one** message that says so: that is Greecon's platform or a provider they share, not that many customers losing power at once: go to R7. If the email service cannot take the message (it is down, or the day's allowance is used up) it is tried again every five minutes for six hours; after that, the morning message and **Today** are what say it. A box that keeps coming back and going again is told of three times a day at most, so a poor connection at one farm cannot fill the inboxes (it stays on **Today** while it is silent).

Check, in this order:

1. **Is the whole site offline?** Phone the owner or manager. *"Is the power on in the plant room? Is the router's light on?"* A power cut at the site is the commonest cause, and ends by itself.
2. **Is it the site's internet?** If they have other internet things down, it is the router or the line. Ask them to restart the router. (The box keeps about three days of readings, for a site of up to about thirteen devices, and sends them when the internet returns. **Nothing is lost** if it returns within that time.)

3. **Is it the box?** If power and internet are fine: on the box (keyboard and screen, or SSH on the site's network): `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. Read the last lines: they say what is wrong (cannot reach Greecon, a clock far out, the credential refused).
4. **Is it Greecon?** If every customer's box is offline at once, the platform is down: R7.

Fix: restart the service `sudo systemctl restart greecon-edge`; if the clock is out `sudo timedatectl set-ntp true`; if the box's log says *the platform does not recognise this box's credential*, make a new code: on the site's setup page, **Install a box, or replace this one**, then run the new line on the box. A box that has been off for hours reports in again by itself.

Tell the customer: what you saw and when, what you did, what they need to do. If nothing was lost, say so.

Afterwards: a line in the incident log (date, site, cause, minutes offline). If it is the *same site twice*, find the cause: a loose plug, a router that restarts nightly.

R2 The box is online but a site has no readings, or a device "does not answer"

You see: the box is **Online** and the site is **Attention needed** or **Unknown**; the setup page shows the device as **Needs attention** with *no answer from 192.168.1.50:502 (unit 2)*, and on the box `greecon-edge status` lists it as *NOT OK*.

Check:

1. **Is the converter powered and on the network?** From the box: `ping 192.168.1.50` (its address on the setup page).
2. **Has the converter's address changed?** Press **Find equipment**. If the converter answers at a new address, the devices say **Found** there: R17.
3. **Does one device fail or all of them?** All: the converter or the cable. One: that device (its number, its power, its wires).
4. **From the box:** `greecon-edge modbus-test --host <address> --unit <n> --type input --address 52 --data-type float32`. *Could not read: no answer* means the device or the cable; *has no register at that address* means the register is wrong (the device answers); a value means the path works.
5. **A meter powered from the grid** goes dark in a power cut: expected, and it returns with the power.

Fix: reseat the converter's power; check **A** and **B** are not swapped; check the number on the device's label against the install card; check speed and parity match the converter's. The numbers and speeds are set at the office and are not changed on site. If a device was replaced, the new one is **numbered and set at the office first**, like the one it replaces (a new meter or module is delivered as number 1): fit the spare that was numbered there, then press **Find equipment**.

Tell the customer: which device, since when, and whether their alerts for it can be trusted meanwhile (they cannot while it does not answer: the platform shows *Unknown*, never *fine*).

R3 A reading looks wrong

You see: a value that is zero, never changes, is ten or a thousand times too big or small, is negative where it cannot be, or has import and export swapped.

Symptom	Likely cause	Fix
Zero all the time	A clamp on the wrong cable, a closed breaker upstream, the wrong register, or the inverter is limited to zero export	`modbus-test` against the meter's display; confirm the clamp is on the right phase
Constant	A frozen device, a wrong register kind (input vs holding)	`modbus-test` twice a minute apart
x1000 too big	Watts shown as kilowatts, on a reading mapped by hand (the Eastron meters already scale it)	In the register form: multiply by **0.001**
Tank above 100 percent, or a value that "cannot be a tank level"	The range or the tank's height was given wrongly, a wire has come off (the reading is below zero), or the transmitter is on the wrong input	Give the ranges again (Part 3, the ranges step) and see R18
Import and export swapped	Clamps the wrong way round	The electrician turns them; repeat the direction check. Greecon notices on its own at night or in cloud (a Check today alert, *The grid meter may be fitted the wrong way round*, after five minutes of energy going out of a plant that nothing is producing for); at midday it cannot tell
The pump shows running when stopped	The contact is inverted	Swap the contact's normally-open and normally-closed wires, or use the opposite contact. Until it is right, and if a flow meter is fitted, the platform sees a running pump with no flow and opens the no-flow alert (an Act now) whenever the pump is stopped; when the pump really runs it shows stopped

Tell the customer when a number they relied on was wrong, and for how long. **Do not edit history:** a wrong reading stays as it was, with a note. The platform keeps what was measured.

R4 A customer did not get a message, or got one that was wrong

Check:

1. **Who is told** at that customer (sign in, **Open for setup**): is the person on the list, with the right email and number *with the country code*?

2. **What was sent**, on the same page: each message, to whom, when, and its result: **Sent** (the email or text provider took it), **Not sent** (the **Why** column says why), **Waiting**, or **Not needed** (for example, someone acknowledged the alert before it went). **Sent** does not mean the person saw it. After **Send a test message** the page says *Accepted for delivery*, which means the same, so ask the person whether it arrived.
3. **Email in spam?** Ask them to look, and to mark it not spam. **Text:** which network? (some operators replace a sender name; see Part 2, step 8).
4. **Quiet hours** hold Check today until morning; only Act now is sent at night.
5. **Hourly limits:** everything that is not Act now (Check today, and the notes that an alert was acknowledged or has cleared) is limited to 60 emails and 20 texts an hour per account; past that a message is dropped and the **Why** column says so. Act now has a limit of its own, 120 emails and 60 texts an hour, and past it a message is **held**, not dropped, and sent as soon as the hour allows. A condition flapping all day should not reach either; if it did, find out why.
6. **Is the platform's messaging off?** The API's log at start-up says so ("Alert messages are OFF").
7. **A message that could not be delivered** (a key refused, an empty text balance, the provider down): an alert message is tried again every five minutes for the first hour and every half hour after that, for as long as the alert is open and nobody has acknowledged it, and **Who is told** says "N alert messages could not be delivered, the last because …". Each failed try stays in the log as a line of its own. A Check today email that falls due in the quiet hours, for any reason, waits for the morning; Act now never waits. A way of sending that is not set up on the platform at all (no text provider yet, say) is shown in the log as **Not sent**, is tried once an hour rather than every five minutes, and is **not** counted as "could not be delivered": **Who is told** says once that text messages are off, and Greecon's morning message does not name the customer for it. Mend the cause (the key, the balance, the address); the next try sends it. Greecon's own morning message names the customers whose alert messages are refused or not arriving.
8. **A customer who is Suspended** is sent nothing at all, and what had been planned for it is dropped with the reason in the log. An alert that opened while the customer was suspended is messaged normally, from the next round after it is made active again. **An alert that was already part way through its messages when the customer was suspended is not told again from where it stood**: look at **Alerts** for that customer when you turn it back on, and tell the people yourself if something is still open.
9. **The link in an alert message** (*I have seen this*): each alert message holds a link that works for 72 hours and acknowledges that one alert, for the person it was sent to, without signing in. Opening it changes nothing; pressing its button does. It shows in the customer's **Audit log** as *Alert acknowledged with the link in an alert message sent to [name]*. A page that says *This link no longer works* means the 72 hours are over, or the alert has gone, or the account is suspended, or `JWT_SECRET` was changed (changing it ends every link, as it ends every session). Nothing is lost: the person acknowledges in Greecon instead.

Fix: correct the list and send a test message to the person, and confirm it arrived.

An Act now alert that nobody acknowledged (the messages went on down the list until it ended): Act now is passed down the list, the first person at once, the second 30 minutes after the alert began, the third 60 minutes after, and so on, until the list ends or someone says they have seen it. Two ways: the *I have seen this* link in the message, then its button (no sign-in; it works for 72 hours; see point 9), or **Acknowledge** on the **Alerts** page, which only a person who can sign in as an **Owner**, an **Administrator** or an **Operator** can press (a **Viewer** or an **Auditor** cannot). Replying to the email or the text, or phoning someone, does not stop it: the platform cannot hear that. So **tell everyone on the Who is told list about the link**, and give the people who should also look at the screens a login of their own with the Operator role: the customer's owner adds it under **Settings → People**, **Add a person** (a Greecon support session cannot add people). **Settings → People** shows who has a login and with which role.

R5 Too many alerts

An alert nobody acts on trains people to ignore the next. The aim in steady state is **one alert or fewer per site per day**, every one with an action.

1. List the alerts of the last two weeks with the owner. For each: *what did you do?* An alert with no action is a limit that is wrong or a reading that is noisy.
2. Change the **limits** (the tank's *Low below*, *Critical below*) to numbers the owner would act on. Do **not** mute an alert type. Fix its cause.
3. A reading that flaps around a limit: raise the gap between *Low* and *Critical*; the platform already raises one alert per condition and re-opens it within 15 minutes rather than raising a new one.
4. Write down what changed, so the next person knows why.

R6 A person cannot sign in, or someone has left

- **Forgot the password:** the account's **owner** opens **Settings** → **People** and chooses **Reset password** for that person (a temporary password is shown once; they choose their own at first sign-in).
- **A customer's owner is locked out and has no other administrator:** they phone Greecon. **Confirm by phone that it is them** (ask something only the owner would know: the site's name and address, the last alert, who else is on the list). The founder opens **Clients**, finds the client, opens **People** in the client's row, and presses **Reset password** for that person. A temporary password is shown once; give it to them by phone, not by email. It ends every session the person had open, and the reset is written in the customer's own audit trail under the staff member's name. Only the owner of Greecon's account can do it; a project manager can see the people and is told who to ask.
- **Wrong password too often:** sign-in is rate limited per account. Wait a few minutes.
- **A team member leaves:** the founder opens **Team** → **Turn off**. Their session ends on their next click, and so does any customer account they had open. Then **change** `JWT_SECRET` if they knew anything secret (Part 2).
- **A customer's person leaves:** the customer's owner opens **Settings** → **People** and chooses **Turn off** for the person; it ends their session on their next click.
- **The founder is locked out:** R14.

R7 The platform is down

You see: the outside check calls your phone; sign-in does not load; every customer's box shows offline at once.

1. **Railway status:** open railway.com's status page and the project. Is the **api** service *Crashed* or *Deploying*? Is **Postgres** *Crashed*?
2. **Open the api service's Deployments → View logs.** The last red line says why: a setting wrong, the database unreachable, out of memory.
3. **If a deploy just went out:** go back to the previous one (Deployments, the earlier one, **Redeploy**). The database is not rolled back, and does not need to be: new versions only add to it.

4. **If the database is down:** look at Railway's Postgres service; it restarts by itself; if it is full, R8.
5. **While it is down:** boxes keep their readings (about three days, for a site of up to about thirteen devices) and send them when the platform returns. **Customers' own equipment is not affected: Greecon only reads.** Say both, in these words.
6. **When it is back:** for ten minutes after the platform starts listening (`RESTART_GRACE_MINUTES`), Greecon raises nothing about *missing* readings and sends no *box silent* message, and the screens say *Greecon was just restarted; boxes are reconnecting*: the boxes' last words are old because the platform was away, not because they stopped. They send what they kept. After the ten minutes anything still silent is real: R1. A reading that is measured and dangerous (a tank nearly empty) is never held back by this wait.
7. **Tell customers** within 30 minutes of finding out (the template in document 21: "[time, zone] Investigating. Since [time], Greecon's screens and messages are unavailable. Effect on your site: your equipment is not affected; the box keeps its readings and sends them when we are back. Next update by [time]."). Update at the promised time **even if nothing changed.**
8. **Afterwards:** a short written summary within five working days: what happened, how long, what was lost (nothing, if the boxes had it), what changes.

R8 The database is full, or slow, or the bill jumped

- **Railway Usage** shows which service costs what. The database grows with readings: raw readings are kept by the platform; if it nears its volume size, increase the volume in Railway (Postgres service, **Settings** → **Volume**), then plan how long raw readings are kept (a decision for the founder: a retention rule is not built).
- **A box sending too much:** a box that cannot reach the platform keeps one reading a minute; a site with many devices sends more. `greecon-edge status` shows each device. A runaway is unusual: look at the audit log for the site.
- **Slow pages:** a chart is averaged into a few hundred points however long the window, and the longest window is 90 days (**Quarter**, under **Time range** on **Reports** → **Trends**; the platform cuts a longer request to 90 days).

R9 A bad software release

The platform and the box's software are built together and **boxes update themselves** from the platform. A box that fails to start a new version three times **goes back to the one before it** and remembers not to try it again.

1. **Stop the spread:** on the **api** service set `EDGE_AUTO_UPDATE` to `false`. Boxes keep the version they have.
2. **Roll the platform back:** Railway, **Deployments**, the earlier one, **Redeploy**.
3. **Find the cause** in the log, fix it, release again, and turn `EDGE_AUTO_UPDATE` back on (remove the variable).
4. Look at **Today**: each site says its box's version. Look for one that differs from the others.

R10 A box is replaced

A failed box, a stolen one, or a customer who moved the equipment.

1. On the site's setup page press **Install a box, or replace this one**. **Confirm.** A new line is shown, once.
2. Install the operating system on the new box, paste the line. The new box takes the old one's place; **the old one's credential stops working at once.**
3. Nothing else is entered again: addresses and registers are the site's, not the box's.
4. Write the change in the hardware register.

R11 A customer asks for their data, or to leave, or to be deleted

- **A copy of their data:** they can take it themselves, whenever they like, from **Reports**: every reading for the last 31 days, hourly averages for up to a year, their alerts, their audit log and how their account is set up. Every download is recorded in their audit trail. If they cannot, do it for them from inside their account (**Open for setup**; the download is then recorded under your name).
- **Every reading Greecon holds** (when they stop, or when they ask for more than **Reports** gives): **Reports** stops at 31 days of every reading and a year of hourly averages, so make the file by hand. On your laptop, as in R13, open `psql "$DATABASE_PUBLIC_URL"` and paste this as **one line** (`<client id>` is on **Clients**; `<client>` is a short name for the file):

  ```
  \copy (SELECT timestamp_utc, site_id, device_id, point_id, canonical_name, value_numeric,
  value_bool, value_text, unit, quality FROM telemetry_readings WHERE tenant_id = '<client id>'
  ORDER BY timestamp_utc, canonical_name, point_id) TO 'readings-<client>.csv' WITH (FORMAT
  csv, HEADER)
  ```

  The times are UTC. In that file the site, the device and the reading are only ids, so make the key to them as well and send both (also one line):

  ```
  \copy (SELECT s.name AS site, s.id AS site_id, d.name AS device, d.id AS device_id, p.label
  AS reading, p.canonical_name, p.id AS point_id FROM points p JOIN devices d ON d.id =
  p.device_id JOIN sites s ON s.id = p.site_id WHERE p.tenant_id = '<client id>' ORDER BY
  s.name, d.name, p.label) TO 'names-<client>.csv' WITH (FORMAT csv, HEADER)
  ```

  Greecon keeps one reading a minute for each reading point, so a hotel's ten points make about a million rows in 90 days: a large file, which a spreadsheet may not open whole.
- **The messages that were sent** (**What was sent**, under **Who is told**) are in none of the downloads. If the customer asks for them as a file, in the same way:

  ```
  \copy (SELECT created_at, due_at, sent_at, contact_name, channel, destination, kind,
  priority, subject, body, status, error FROM notifications WHERE tenant_id = '<client id>'
  ORDER BY created_at) TO 'messages-<client>.csv' WITH (FORMAT csv, HEADER)
  ```

- **Sending a file you made by hand:** it holds readings of a named site, and the message file holds the names, email addresses and phone numbers of the people who were told, so **not by plain email**. Put it in a shared-drive folder that only the customer's owner can open (by their email address, not "anyone with the link"), tell them by phone that it is there, and delete it when they confirm in writing that they have it. A download from **Reports** is written in the customer's audit trail; **a file made by hand is written in no audit trail**, so write it in the same sheet as the data-deletion log below: who asked, when, which files, how they were sent, when it was confirmed.
- **They stop at day 60 or day 90:** take the equipment away within 14 days; give them the files (the downloads from **Reports**, and every reading Greecon holds, made by hand as above); set the client **Suspended** (**Clients → More → Suspend this client**, which asks once and says what it will do) (it can be turned back on; if you do, read R4, point 8, about alerts that are still open); keep the account until they confirm in writing that they have their data.
- **Deletion (they ask in writing, after they have their data):** this is the only time anything is deleted. The founder does it, in a transaction, after the files are with the customer and the request is in writing:

```
-- in the Railway Postgres service, Data tab or psql with the public connection string
BEGIN;
DELETE FROM tenants WHERE id = '<client id, from Clients>';
DELETE FROM access_requests WHERE lower(email) = lower('<owner email>');
-- the first must say DELETE 1 (if not, ROLLBACK instead); the second says DELETE 1 if the
owner
-- asked through the request-access page from that address, and DELETE 0 if not. Then:
COMMIT;
```

  The first statement removes the account and everything under it: sites, devices, readings, alerts, people, messages sent, audit trail. The database does this by itself: every table that holds a client's data is tied to the client's row, so it goes when the row does (the automatic test checks the people, the settings and the messages sent; for the other tables it follows from how they are built). The one table that is not tied to a client is the list of requests from the request-access page, which holds the name, company, email and phone the owner gave: that is what the second statement removes (if they asked from another address, run it again with that address). **Then restart the API service** (a redeploy) so that the platform's memory forgets them. **Write in the data-deletion log** (a sheet, outside the platform): who asked, when, what was deleted, when it was confirmed to them. **What stays, and what to tell the customer:** copies in **backups** disappear when the backups expire (Railway's daily ones within a week; the weekly dumps you keep: delete those that hold them as you rotate); Greecon's own morning messages to its team, already sent, still name the client and its sites in their text; Resend and Twilio keep their own records of the emails and texts they delivered, on their own schedule; and files already downloaded (by the customer, or by hand) stay where they are.

R12 A security incident, or a vulnerability report

A vulnerability report arrives at `info@greecon.earth` (it is in `security.txt`): reply within two working days with thanks; fix it; do not threaten the reporter; say when it is fixed.

A suspected incident (an account used by someone who should not have it; a key leaked; the platform behaving oddly):

1. **Write down the time you found out.** That starts the clocks.

2. **Contain:** turn off what was misused. A team member's sign-in: **Team** → **Turn off**. A whole customer: **Clients** → **More** → **Suspend this client** (it asks once; everyone there is turned away at once, even with a sign-in that is still valid, and is told the account is suspended and to write to Greecon, not that their password is wrong; the box is turned away too and keeps what it reads; **Reactivate this client** undoes it, asks nothing, and sessions that were open work again without signing in). One person at a customer: ask that customer's owner to press **Turn off** beside them under **Settings** → **People** (Greecon's own people see that list but do not change it). Then change `JWT_SECRET` (signs everyone out); rotate any key that may have leaked (Resend, Twilio, the database password).
3. **Find out what** was seen or changed: the audit trail of the affected customer shows who did what, under whose name.
4. **Decide who must be told and by when**, with a lawyer where you can: the clocks depend on the customer. A customer classed as critical under Albania's Law 25/2024 needs your facts within **4 hours**; GDPR and Albania's Law 124/2024 want a processor to tell the controller **immediately** and the controller the authority within **72 hours**; the Cyber Resilience Act has 24 hours and 72 hours for actively exploited vulnerabilities. The full table is in document 21, section 4, marked *to be confirmed with counsel*.
5. **Tell customers** with the template in document 21 ("Security notice"): what is known, what is not, what to do, when the next update is. **Never** say "no evidence data was accessed" while the investigation is open, and never say "fully secure".
6. **Afterwards:** a written report within one month; what changed.

What is built, and what is not (so that nobody over-promises): a temporary password for each new account, shown once; passwords checked for length and against published ones; sign-in rate limited; sessions that end when a person is turned off or changes their password; every staff action in a customer's account recorded under their name; all customer data separated by account and tested for it; a download recorded in the audit trail. **Not built yet:** two-factor sign-in (the first security work after the fifth site, and before any control of equipment); single sign-on; automated penetration testing. Say so when asked.

R13 Backups and restoring

The rule: a backup that only lives next to the database does not survive losing it.

Every day (automatic): the Postgres service's **Backups** tab, a daily schedule (what your plan allows).

Every Friday (by hand, the founder, ten minutes):

```
# on your laptop, with the PostgreSQL tools installed; DATABASE_PUBLIC_URL is on the Postgres
service's Variables tab
pg_dump "$DATABASE_PUBLIC_URL" --no-owner --format=custom --file="greecon-$(date +%F).dump"
```

Put the file in the company's shared drive (not the laptop), in a folder named `backups`. Keep eight weeks and delete older ones.

Every month, the restore drill (so that you know it works before you need it):

1. In Railway, add a **new** Postgres service named `restore-test` (never restore over the live one).
2. `pg_restore --no-owner --dbname="<its public URL>" greecon-<date>.dump`

3. Look at it: `psql "<its public URL>" -c "select count(*) from users" -c "select count(*) from sites" -c "select count(*) from telemetry_readings"` (one `-c` for each, so that every version of psql shows all three numbers). They are about what you expect.
4. Delete the `restore-test` service.

A real loss: restore into a **new** Postgres service, point the **api** service's `DATABASE_URL` at it (a variable change, then a redeploy), check sign-in and a customer's site, then tell customers what the gap is. Boxes resend what the platform could not take while it was down (about three days, for a site of up to about thirteen devices). They do not keep what the platform had already accepted, so the readings between the last backup and the loss are gone: tell customers which period that is.

R14 When the founder is not there

- **Two people can sign in as Greecon staff**; the project manager can do R1 to R6, R10, R11 and R15 to R18. **Only the founder can**: change settings in Railway; add or remove team members; delete a customer; restore the database.
- **Put in the password manager's emergency access** the person the founder trusts most, and a sealed note: where the password manager is, who owns the Railway and GitHub accounts, who owns the domain.
- **Add a second person as a collaborator on Railway and as an owner on GitHub** as soon as one is hired. The platform never depends on one person's laptop.
- **No one edits the database by hand to reset a password.** A customer's locked-out person is reset from **Clients** (R6). If the founder himself is locked out, set `BOOTSTRAP_ADMIN_EMAIL` and `BOOTSTRAP_ADMIN_PASSWORD` on the API to his address and a new password: it works only for an account that has **no** password, so first clear his password in Railway's Postgres **Data** tab (`UPDATE users SET password_hash = NULL WHERE lower(email) = 'eridon.manuka@greecon.earth';`), then set the two variables and redeploy, sign in, choose a new password in Settings, and remove the variables. Keep the password manager's emergency access so that this never happens.

R15 Find equipment cannot find a device

You see: on the setup page, in the step **Where each device is on the network**, after **Find equipment**: a device marked **Not found** (*Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.*), or *No Modbus converter answered on this network*, or *A converter answers at 192.168.1.50, but nothing answers behind it*, or a device marked **Not sure which**.

Check, in this order:

1. **Is the box online?** Until it is, the button is grey, or the page says *The box did not pick this up*. Do that first (R1).

2. **Which sentence is it?**
   - *No Modbus converter answered on this network.* The converter is off, or not on the same network as the box (a guest network, a second router, a cable in the wrong socket). Look at its power and network lights, and check that the box and the converter are on the same router. At a desk, if the converter is on a different part of the network, or if it uses another port than 502, open **I know the converter's address**, type its address (each part without a leading zero: 192.168.1.50) and, if it is not 502, its port, and press **Look at this address**. The box does not look at itself (127.0.0.1).
   - *A converter answers …, but nothing answers behind it.* The network side works and the cable side does not: **A** and **B** swapped (swap the two wires at the converter); the converter not in **Modbus TCP to RTU (gateway)** mode or not at 9600 baud, 8 data bits, no parity, 1 stop bit; devices without power; a cable that is broken or branched.
   - **Not found** for one device while the others are found: that device's power (a meter powered from the grid is dark in a power cut), the number on its label against the install card, its **A** and **B** wires and the cable to it.
   - **Not sure which**: Greecon refuses to guess, because a wrong guess could swap the grid meter and the solar meter, and every figure with it. *Something answers at 192.168.1.60, unit 1, but not clearly enough to tell whether it is the three-phase SDM630 planned.* Press **Look again**, then check the speed and the A and B wires. *Units 1, 2 at 192.168.1.50 all answered with exactly the same words*: the converter is not in **Modbus TCP to RTU (gateway)** mode, so check its settings page against its label. Two of one kind at the same number on different converters, or at numbers that are not the kit's: read the label on each device, then use **Enter the address by hand** under each, with the number on its label. Two devices on one converter with the same number answer over each other: one was not numbered at the office, and numbers and speeds are not changed on site, so that part goes back there. Then press **Look again**.
3. **From the box:** `greecon-edge find` looks at the network as **Find equipment** does and prints what answers. Then read one device, with the unit number of the device you are looking for:

   ```
   greecon-edge find --host 192.168.1.50 --port 502
   greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
   greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
   greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
   float32
   ```

   The first looks at one address; the others read the 4 to 20 mA module, the pump contact module and a meter. *Could not read: no answer* means the device or the cable; a value means the path works.
4. **A device the site does not have** is not a fault: press **The site does not have this: remove it** under it. **A real Waveshare module that is found but reads something odd:** its registers have not been tried on real hardware: write down what `find` and `modbus-test` print, and tell Greecon.

Fix: what the check found; then **Look again**. The step turns **Done** when every device has the address it was found at.

Tell the customer: only if a visit or a part is needed. Greecon only reads, so nothing of theirs is affected.

Afterwards: write down what was wrong and what you changed (the commissioning record, or the incident log if the site is live).

R16 A device was found at another address

You see: a device marked **Found**, and a button **Use this address** (or *Use these N addresses*): *Found at 192.168.1.60, unit 1: an Eastron meter, reading 1.80 kW at 231 V.* Nothing changes until you press it. Greecon fills in without you only a device that has no address yet, found at the kit's own number, as the model planned.

Check, in this order:

1. **Did the converter move?** Several devices **Found** at one new address, and the old address was the converter's: R17.
2. **Is the number the kit's?** *It is at unit 6, not 3: if it is this site's, use this address, and the install card gives 3 so that the next site looks like this one.* If it is this site's part, it works as it is: write it down, and do not change it on site. If it could be the customer's own device on the same cable, do not press the button: read its label and ask.
3. **Is it another model?** *It is a single-phase SDM230, not the three-phase SDM630 planned. If you use this address, Greecon reads it as a single-phase SDM230.* That is right where the supply is single-phase and the install card lists the meter fitted; if a three-phase meter was meant, do not press the button: the electrician changes the meter. The other way round (*It also answers the three-phase meter's total-power register, so it may be a three-phase SDM630*, on a **Found** or a **Set** line) the meter is read as an SDM230 and shows one phase only: check its label and, if it is an SDM630, remove this device and add the three-phase meter from **Add equipment Greecon already knows**.
4. **Is it something nobody planned?** It is listed under *Also answering, and not part of this site's equipment*: not yours, leave it alone.

Fix: if the line makes sense, press **Use this address**: Greecon sets the address (and reads a meter of the other model as that model), the box picks it up within a minute, and the device shows **Set**. The change is written in the audit trail, old and new. If the line does not make sense, do not press it: find out why first (R15).

Tell the customer: nothing, unless the cause is on their network (a router that changes the converter's address): ask for a fixed address (R17).

Afterwards: a line on the commissioning record: which device, which address, and why.

R17 The converter's address changed

You see: the readings of every device behind one converter stop together (*Last reading N ago. It has stopped arriving*), and each shows **Needs attention**: *no answer from 192.168.1.50:502 (unit 1)*. The box is **Online** and the converter's lights are on. The customer may say that the router was restarted: a router hands out addresses, and it may give the converter a new one.

Check: the box is online, the converter is powered, and more than one device behind it fails at the same moment. From the box, `ping 192.168.1.50` (the address on the setup page) gets no answer, while `greecon-edge find` shows a converter at another address.

Fix: press **Find equipment**. The box finds the converter at its new address and each device behind it says **Found** there. Press **Use these N addresses**: the devices turn **Set** and the box picks them up within a minute. The readings return; the gap stays a gap, because the box could not read the devices while it did not know where they were. If nothing is found, the converter is off or on another network: R15.

To stop it happening again: ask the customer's network administrator for a fixed address for the converter (a "DHCP reservation" on its MAC address in the router): the second message on **The install card** asks for it. Write the address on the converter's label and the handover record.

Tell the customer: what you saw and when, that nothing of theirs was affected (Greecon only reads), and the request for a fixed address.

Afterwards: a line in the incident log (date, site, minutes without readings).

R18 A reading is impossible after the ranges were given

You see: in the step **Readings arrive and look right**, a transmitter's reading cannot be real: *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.*, or *133 cannot be a tank level (0 to 100)*, and likewise for the line pressure and the flow. Or it is believable, but disagrees with a dipstick or a gauge.

Check, in this order:

1. **Below zero: a wire off.** The 4 to 20 mA module is left in its 0 to 20 mA mode on purpose, so that a wire that has come off, or a loop that has lost its 24 V supply, reads **0**: below the 4 mA of an empty tank, and so below zero once the range is applied (about -31 for a 0 to 5 m level transmitter in a tank 4 m tall). Read the module from the box:

   ```
   greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
   ```

   Address 0 is input 1, 1 is input 2, 2 is input 3. **0 means no current.** A working transmitter reads at least 4000 (4 mA, counted in microamps). Is it on the input the install card says (level 1, pressure 2, flow 3)? The electrician checks the loop and its 24 V supply, with the lock-out where needed: you do not work on it.
2. **Far too high or too low: the range or the tank's height was given wrongly.** Read the label again. Too high: the range given is bigger, or the height smaller, than the real one (a tank 4 m tall given as 3 m, with a 0 to 5 m transmitter, reads 133 when it is full). Too low: the other way (a label in bar typed as metres reads about ten times too low: choose bar under **Unit on the label**). Measure the tank inside again, from the bottom, where the transmitter hangs, to the top of the water when it is full. Under **Which register holds which reading** give the ranges again and press **Save the ranges**; the form does not show the level's answer back, so type both numbers. It then says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.*
3. **A transmitter shorter than the tank** is refused when you save (*it cannot see the top of the water*): fit one with a longer range.
4. **A loop calibrator** in place of the transmitter settles it: send 4, 12 and 20 mA, and the module must read 4000, 12000 and 20000. If it does and Greecon is still wrong, it is the range or the height; if it does not, the wiring, the module's input or the loop.
5. **A transmitter the site does not have** (no flow meter) reads 0 and looks like a wire off: tick **Not fitted at this site** under it.

Fix: what the check found. A reading that was wrong stays as it was, with a note: do not edit history.

Tell the customer when a number they relied on was wrong, and for how long.

Afterwards: write down what was wrong (a wire, a range, a height) and what you changed.

APPENDICES

Forms, words, symptoms, shopping lists and contacts

These pages are for printing and for looking things up. Appendix A has the blank forms for any site (each site pack has the same forms made for its kit). Appendix B explains the words. Appendix C is a way in by symptom. Appendix D has the shopping lists with room for three quotes. Appendix E is the contact sheet and a card of the numbers that never change.

A Forms to print

Print the form, take it to the site, and file it in the customer's folder afterwards. The forms are general: cross out what the site does not have. Each form starts on a new page.

- ☐ A.1 Survey form
- ☐ A.2 Install-day checklist
- ☐ A.3 Commissioning record
- ☐ A.4 Handover record
- ☐ A.5 Rehearsal record

A.1 Survey form

ANY SITE

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Template	☐ Hotel or guesthouse ☐ Solar plant with grid connection ☐ Farm or greenhouse with irrigation ☐ Water tank and pump station
Prices from the bills: electricity per kWh; paid for energy sent to the grid; water per m^3	
Twelve electricity bills and the water bills collected	☐ Yes ☐ Later
Supply: single or three phase; main breaker size (A)	
Inverter: brand, size (kW), AC output cable location; allowed to send to the grid?	
Tank: height (m), water source; pump (kW); starter location; pipe pressure (bar)	
Meters needed (grid, solar, load, pump) and where	
Sensors needed (level, pressure, flow, pump running; soil, weather) and where	
Network: customer's cable or cellular; router brand; who can give an address	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

A.2 Install-day checklist

ANY SITE

The electrician and the project manager work at the same time, in this order. Cross out the rows that do not apply to the kit.

	#	Who	Do this
☐	1	ELECTRICIAN / PROJECT MANAGER	Before anyone touches anything: the signed order form and the data terms are in your hand, the electrician is there, and the lock-out is agreed. Walk the site together. Agree where each meter and sensor goes. Photograph everything before you start.
☐	2	ELECTRICIAN	Apply the customer's lock-out on the board to be worked on.
☐	3	ELECTRICIAN	Fit the meters (grid, solar, load or pump): voltage taps with their fuses or small breakers, current clamps with the arrow towards the site.
☐	4	ELECTRICIAN	Connect the pump's run contact (a spare auxiliary contact, as a dry contact) to input 1 of the pump contact module. Never to a control circuit, and never to the module's relay outputs.
☐	5	ELECTRICIAN	Fit the level, pressure and flow transmitters, with the customer's plumber for the wet parts, and wire their 4 to 20 mA loops with the 24 V supply in series, as the module's manual draws it: level on input 1, pressure on input 2, flow meter on input 3.
☐	6	PROJECT MANAGER	Mount the enclosure, the converter, the two modules, the 24 V supply, the UPS and the box.
☐	7	PROJECT MANAGER	Run the RS-485 data cable: one cable, a daisy chain, no branches. The electrician connects its ends at the meters inside the board.
☐	8	PROJECT MANAGER	Check the number on each device's label against the install card (grid meter 1, solar meter 2, 4 to 20 mA module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7). They were set at the office: do not change them. Photograph each label.
☐	9	PROJECT MANAGER	Check the converter's label: Modbus TCP to RTU gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502. Ask for a fixed address for it and write the address down.
☐	10	PROJECT MANAGER	Connect the network: the customer's router, or the 4G router with its SIM.
☐	11	ELECTRICIAN / PROJECT MANAGER	Photograph before closing: every meter's label, the wiring, the converter's settings. Close the board.
☐	12	ELECTRICIAN	Remove the lock-out only when you both agree the work is closed up.
☐	13	PROJECT MANAGER	Power up: UPS, supply, converter, box.
☐	14	PROJECT MANAGER	Open the customer for setup and go down the setup list: press **Find equipment**, give the ranges, and write down the electrician's name and licence number.
☐	15	ELECTRICIAN	Stay until commissioning is done.

A.3 Commissioning record

ANY SITE

Compare in this order and write the pair of numbers. Do not start, stop or open equipment yourself: the owner's own person does.

Check and how	Pass when	Greecon shows	Real	Result
Every meter's display against Greecon. Read both at the same time.	Within 5 percent, or 1 kW on a small load			☐ Pass ☐ Fail
Grid direction. A known load on, the sun down.	Import rises, export does not			☐ Pass ☐ Fail
Solar at night.	Zero, or a few watts			☐ Pass ☐ Fail
Solar at midday. Against the inverter's display.	Within about 5 percent			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow. A bucket test, or the flow meter's display.	Within 10 percent			☐ Pass ☐ Fail
Pump running. The owner starts the pump.	Shown within 30 seconds, changes both ways			☐ Pass ☐ Fail
Soil probe. In the air, in a glass of water, in the soil.	Three different readings			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	Bottom, middle and top			☐ Pass ☐ Fail
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____ Site: _____

A.4 Handover record

ANY SITE

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Alert limits agreed with the customer	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Anything the site cannot raise (no flow meter, no pump contact), said to the owner	
Prices entered and where they came from	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, Energy now or the site's page, Value, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer's folder	☐ Yes

Project manager: name and signature

Owner: name and signature

A.5 Rehearsal record

ON A DESK

Once before the first real site, once more with the first real kit, and again when anything in the manual changes (section 3.11). An afternoon, a laptop, and the box. Date: _____ Done by: _____

	#	Who	Step
☐	1	PROJECT MANAGER	Run the pretend kit on the laptop: `node scripts/pretend-meter.mjs --kit --port 5020`. It answers as unit 1 grid meter, 2 solar meter, 3 4 to 20 mA module (tank 12 mA, pressure 8.96 mA, flow 5.12 mA) and 5 pump contact module (closed).
☐	2	PROJECT MANAGER	On **Clients** press **Onboard a client** for a client named Rehearsal, with a second address of your own for the owner. Open it for setup and make a site from the **Hotel or guesthouse** template. Write that the papers are signed (**The papers are signed**).
☐	3	PROJECT MANAGER	Add the box, paste the line on the box, and watch **Install the box** turn Done.
☐	4	PROJECT MANAGER	The pretend kit listens on port 5020, not 502, so the box does not find it by itself: open **I know the converter's address**, type the laptop's address on the network (not 127.0.0.1) and port 5020, and press **Look at this address**. The grid meter, the solar meter, the 4 to 20 mA module and the pump contact module are each **Set**.
☐	5	PROJECT MANAGER	Give the ranges: level transmitter 5 and tank 4, pressure transmitter 10, flow meter 200, then **Save the ranges**. Tank level reads 62.5 %, line pressure 3.1 bar, pump flow 14 litres a minute, and Pump running reads 1.
☐	6	PROJECT MANAGER	Watch every step turn Done. Add yourself under **Who is told**, send a test message, write "no electrical work was needed" for the electrician, and confirm the steps that need a person. Commission the rehearsal site.
☐	7	PROJECT MANAGER	Stop the pretend kit. After about five minutes **Today** says, under **Needs a look**, that the readings are out of date. Start it again, and raise the tank's limits above 62.5 (Low below 80, Critical below 70): the tank is critical, an **Act now**. It arrives by email at once and by text ten minutes later unless you press **Acknowledge** or open the *I have seen this* link in the message.
☐	8	PROJECT MANAGER	Print the handover sheet. Download the readings from **Reports** (**Downloads**) and open the file in a spreadsheet.
☐	9	PROJECT MANAGER	Clean up: remove the rehearsal site's equipment and then the site, or suspend the client, so that it does not appear in the morning check. Keep it if you will use it for a sales demonstration.
☐	10	PROJECT MANAGER	With the first real kit: do steps 1 to 6 again with the real converter and devices on the desk in place of the pretend kit, and write down anything that differs from this book.

What did not match the manual, and what I changed

B Words

B.1 Words used in the manual

Word	Meaning
Box	The small computer at the site that reads devices and reports to Greecon
Converter	Puts the RS-485 cable on the network (a Modbus gateway: Modbus TCP to RTU). Set at the office
Modbus	The language most meters and modules speak. A device has a **unit number** and **registers**
Register	A numbered place in a device that holds one value. **Input** registers are read-only, **holding** can be written (Greecon never writes)
Clamp / CT	A ring that goes around a cable and tells the meter how much current flows
4 to 20 mA	A signal wire: 4 mA means the bottom of the transmitter's range, 20 mA the top. The Waveshare module reads it in microamps: 4000 to 20000. A wire that has come off reads 0
Range (of a transmitter)	The two ends printed on a 4 to 20 mA transmitter's label, for example 0 to 5 m or 0 to 10 bar. Greecon asks for it, in the label's words, and works out the scale
Approved equipment	The models a kit is made of, and no others: the Eastron SDM630 and SDM230 meters, the Waveshare Modbus RTU Analog Input 8CH and the Waveshare Modbus RTU IO 8CH. Greecon knows their registers and settings
Find equipment	The button that makes the box look at the site's network and find where each device is. It fills in what is certain and asks you about the rest. It only reads
Install card	The page for each site with what to bring, what each device is set to, the converter's settings, the visit in order, and three messages for the customer
Pump contact module	The module that reads whether the pump is running, from a dry contact on its starter. It only reads
Commissioning	Checking that what the platform shows matches the real world, once, on site
Act now / Check today	The two kinds of alert that reach a person. Everything else is only shown

B.2 More words

Word	Meaning
Unit number	A number that tells the devices on one cable apart. Every device has its own, and in every kit it is the same: grid meter 1, solar meter 2, 4 to 20 mA module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7. They are set at the office; the meters and the two modules are delivered as 1
Gateway mode	The converter's mode (Modbus TCP to RTU) in which it turns the box's requests into requests on the data cable. Not transparent, not raw TCP, not RTU over TCP
Speed, parity	The serial settings of the data cable. Greecon's standard is 9600 baud, 8 data bits, no parity, 1 stop bit, the same on every device and on the converter. They are set at the office
RS-485, A and B	The two-wire data cable the meters and modules share. A and B swapped is the commonest mistake
Daisy chain	One cable from device to device, with no branches
Lock-out	The customer's own procedure for switching a board off and keeping it off while the electrician works in it
CT ratio	On a meter with 5 A current transformers, the ratio set in the meter's menu (for example 250 for 250/5 A) so that it shows the real power
Loop calibrator	A hand-held that sends 4, 12 and 20 mA, to check a 4 to 20 mA channel and the range given for it
Microamps	How the 4 to 20 mA module reports a signal: 4 mA reads 4000, 12 mA reads 12000, 20 mA reads 20000. A wire that has come off reads 0
Dry contact	A switch that only opens and closes and carries no voltage of its own, like the spare contact in a pump's starter. The pump contact module reads whether it is closed
Fixed address, reservation	A network address the router always gives the converter (a "DHCP reservation" on its MAC address), or one set in the converter itself
Heartbeat	The box reports every 30 seconds, so that Greecon knows it is alive
Queue	The readings the box keeps while the internet is down: about three days for a site of up to about thirteen devices
Template	The plan of a kind of site in the platform: the equipment it usually has, with the readings each gives
Setup list	The platform's own list of up to twelve steps for a site. A step that does not apply is left out
Handover sheet	One printable page per site: where everything is, who is told, who commissioned it and when
Install line, code	The line shown once on the setup page that installs the box. Its code works once, for seven days
Owner, Administrator, Operator, Viewer, Auditor	A person's role in the customer's account. An Owner, an Administrator or an Operator can press Acknowledge in the app; a Viewer or an Auditor cannot (anyone a message was sent to can use the link in it)
Morning message	The one email at 07:30 that says whether everything is all right
Quiet hours	22:00 to 07:00: everything is held except Act now

Word	Meaning
Zero-export	A plant that is not allowed to send energy to the grid. Tick the box in the prices
Pretend kit	A program for the laptop that answers like a whole kit (two meters, the 4 to 20 mA module and the pump contact module), for practice

C Symptoms

Start from what you see. Each table says what it probably is and what to do; the longer answers are in Part 7.

C.1 On site and at the box

Short answers; the longer ones are in Part 7.

What you see	Likely cause	What to do
The install line stops with `STOP:`	The sentence says: no internet (or a firewall, or a clock far out), code used or expired	Do what it says. Before step 5 run the same line again; after it, run the line with no code at the end (section 3.6)
The box says Offline	Power, network cable, or the site's internet is down	Is the box's light on? Is the router up? On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`
Find equipment finds nothing, or a device says Not found	The converter is off, or on another network than the box; A and B swapped; the speed or the mode at the converter; a device with no power	Read the sentence on the screen. On the box, `greecon-edge find` says what answers. Part 7, R15
A device says Found, or Not sure which	Found at another address than the one it has (the router gave the converter a new one), at a number that is not the kit's, or as the other model of meter; or two of a kind answer at one number, or every number answers alike	Read the line. Press **Use this address** only if it makes sense. Numbers are not changed on site. Part 7, R16 and Part 7
A device needs attention (on the box: NOT OK)	The converter's address changed, the number on the device's label, speed, A and B swapped, converter not in gateway mode	Press **Find equipment** again; then `greecon-edge modbus-test` against the device; check each in that order
A reading is zero or constant	A wrong register on equipment mapped by hand, the wrong register kind (input vs holding), a clamp on the wrong cable	`modbus-test` raw numbers against the manual's table
A reading is ten or a thousand times too big or small	A wrong scale (W against kW) on equipment mapped by hand, or a range or a tank's height given wrongly on a transmitter	By hand: re-enter the conversion. Transmitter: give the ranges again; check with the calibrator

What you see	Likely cause	What to do
A reading cannot be real (*cannot be a tank level*, below zero, or far above 100)	A wire that has come off (the module reads 0), the range or the tank's height given wrongly, a transmitter on the wrong input	Part 7, R18
Import and export are swapped	The clamps are the wrong way round	Electrician turns them; repeat 3.8.9
The clock is out	The box has no time service	`timedatectl set-ntp true`; the setup page names it ("its clock is N minutes fast")
The test message did not arrive	A wrong address, spam, or the number without its country code	Check **What was sent** under **Who is told** (it shows what happened); try the other way
The test email arrived in spam	The sender is new to that mail provider	Mark it not spam; ask them to add `alerts@greecon.earth` to contacts
A reading went Unknown after a while	The meter or the converter lost power or the cable came loose, or the converter's address changed	The setup page says which device and why; press **Find equipment**

C.2 A reading looks wrong

A value that is zero, never changes, is ten or a thousand times too big or small, is negative where it cannot be, or has import and export swapped.

Symptom	Likely cause	Fix
Zero all the time	A clamp on the wrong cable, a closed breaker upstream, the wrong register, or the inverter is limited to zero export	`modbus-test` against the meter's display; confirm the clamp is on the right phase
Constant	A frozen device, a wrong register kind (input vs holding)	`modbus-test` twice a minute apart
x1000 too big	Watts shown as kilowatts, on a reading mapped by hand (the Eastron meters already scale it)	In the register form: multiply by **0.001**
Tank above 100 percent, or a value that "cannot be a tank level"	The range or the tank's height was given wrongly, a wire has come off (the reading is below zero), or the transmitter is on the wrong input	Give the ranges again (Part 3, the ranges step) and see R18
Import and export swapped	Clamps the wrong way round	The electrician turns them; repeat the direction check. Greecon notices on its own at night or in cloud (a Check today alert, *The grid meter may be fitted the wrong way round*, after five minutes of energy going out of a plant that nothing is producing for); at midday it cannot tell
The pump shows running when stopped	The contact is inverted	Swap the contact's normally-open and normally-closed wires, or use the opposite contact. Until it is right, and if a flow meter is fitted, the platform sees a running pump with no flow and opens the no-flow alert (an Act now) whenever the pump is stopped; when the pump really runs it shows stopped

C.3 A device does not answer the box

The setup list shows the device as **Needs attention**, with *no answer from 192.168.1.50:502 (unit 2)*, and on the box `greecon-edge status` lists it as NOT OK. Look in this order:

1. **The converter's address has changed** (the router gave it a new one): press **Find equipment** again. The box looks for the converter, and each device behind it says **Found** at the new address: press **Use this address**.
2. The **number on the device's label** is not the one on the install card, or two devices carry the same number (Find equipment says **Not sure which**).
3. A wrong **speed or parity** at the converter: it must match every device (9600 baud, 8 data bits, no parity, 1 stop bit).
4. **A and B swapped** on the data cable: swap the two wires at the converter.

5. The converter **not in gateway mode** (Modbus TCP to RTU).
6. The device has **no power**: a meter powered from the grid goes dark in a power cut, and returns with the power.

Nothing answers at all: the converter is off, or it is on another network than the box. Some readings work and others do not: a register address that is wrong for that device (on a probe or a battery mapped by hand), or, on a Waveshare module, a table in Greecon that differs from the real module (no real module has been tried yet, so write down what the box says and report it to Greecon). Test one register from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0
greecon-edge find
```

Could not read: no answer means the device or the cable. *Has no register at that address* means the register is wrong (the device answers). A value means the path works. `greecon-edge find` looks at the network and says what answers, as **Find equipment** does.

D Shopping lists

One list for each kit, with room for three quotes. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes in the shared drive. The estimates are planning figures in EUR, excluding VAT and installation labour, from public shop prices seen in October 2026: they are not quotes. Buy and bench-test one kit before buying five. For each part keep the make and model, the supplier, the date, the serial number or the batch, and the invoice.

H Hotel or guesthouse

Planning total: About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**Three-phase meter**. Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers	2	90 to 180 each			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Pump contact module**. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			

S Solar plant with grid connection

Planning total: About EUR 600 to 1,200.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**Three-phase meter**. Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers	2, or 3	90 to 180 each			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters’ voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			

F Farm or greenhouse with irrigation

Planning total: About EUR 700 to 1,500.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router**. 4G, DIN rail, data SIM	0 or 1	150 to 200			
☐	**Soil moisture probe**. RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each			
☐	**Temperature and humidity sensor**. RS-485 Modbus, with a published register table	1	20 to 80			
☐	**Pump contact module** *(optional)*. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	0 or 1	30 to 60			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			

W Water tank and pump station

Planning total: About EUR 650 to 1,300.

	Item and specification	Qty	Planning estimate (EUR)	Quote 1	Quote 2	Quote 3
☐	**Box**. Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300			
☐	**Converter**. RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)			
☐	**4 to 20 mA input module**. Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60			
☐	**Pump contact module**. Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used	1	30 to 60			
☐	**Level transmitter**. Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400			
☐	**Pressure transmitter**. 4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150			
☐	**Flow meter** *(optional)*. 4 to 20 mA output. Needed for "the pump is running but no water is flowing"	0 or 1	150 to 500			
☐	**24 V DIN power supply**. 30 to 60 W	1	25 to 50			
☐	**UPS**. A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150			
☐	**Enclosure and fittings**. IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150			
☐	**Cable and labels**. Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80			
☐	**Cellular router** *(optional)*. 4G, DIN rail, data SIM	0 or 1	150 to 200			
☐	**Single-phase meter** *(optional)*. Eastron SDM230, on the pump's feed, to measure its electricity	0 or 1	35 to 60			

The spares kit and the tools

One spares kit for each project manager: one of each, so that a failed part is swapped on the day.

- [ ] A converter
- [ ] A 4 to 20 mA module
- [ ] A pump contact module
- [ ] A meter
- [ ] A 24 V supply
- [ ] A transmitter (pressure)
- [ ] A spare box, pre-installed, with no code, ready to take one
- [ ] Network cables
- [ ] A loop calibrator
- [ ] A multimeter
- [ ] Fuses
- [ ] A spare SIM

The project manager’s tools:

- [ ] A laptop
- [ ] A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- [ ] A loop calibrator (or a 4 to 20 mA source)
- [ ] A multimeter
- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

E Contacts and the numbers that never change

E.1 Contacts

Fill in, print, and keep a copy in the password manager.

Who	Phone	Email
Founder		eridon.manuka@greecon.earth
Project manager		
Electrician (first call)		
Electrician (second)		
Railway support		station.railway.com
The lawyer		
The insurer		
Each customer's first person	(on its handover sheet)	

What	Address
The web app (customers, staff, the project manager)	`https://app.greecon.earth`
The platform's own address, which every box reports to. **Never change it once a box exists**	`https://api.greecon.earth`
Where alert emails come from	`alerts@greecon.earth` (replies go to `info@greecon.earth`)

E.2 The numbers that never change

What	Value
Unit numbers (set at the office)	Grid meter 1, solar meter 2, 4 to 20 mA water module 3, load meter or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7. A battery is addressed by hand (for example 8)
The data cable and the converter	9600 baud, 8 data bits, no parity, 1 stop bit; converter on port 502, in Modbus TCP to RTU gateway mode, at a fixed address. Set at the office
Eastron meters	Total active power: input register 52 on the SDM630, 12 on the SDM230; 32-bit floating point, high half first; watts to kW by 0.001
A register counted from zero (equipment mapped by hand)	A manual's 30053 is address 52; 40001 is address 0
The 4 to 20 mA module	Inputs 1, 2 and 3: level, pressure, flow. In its default 0 to 20 mA mode it reads microamps: 4 mA is 4000 and 20 mA is 20000; a wire that has come off reads 0

What	Value
A tank level from a transmitter longer than the tank	Greecon asks for the range on the label (in metres or bar) and how tall the tank is, and shows a percentage of the tank. 20 mA reads range ÷ tank height × 100 (a 0 to 5 m transmitter in a 4 m tank: 125); a full tank reads 100
What Find equipment looks at	Private addresses on the site's own network, port 502 (or the one you give), unit numbers 1 to 10; reads only; about a minute at most
The box needs outside	Outgoing HTTPS (443) to `api.greecon.earth` (and `nodejs.org` at install) and a time service (UDP 123). Nothing needs to come in
The box keeps	About three days of readings, for a site of up to about thirteen devices. It reports every 30 seconds
The install code	Works once, for seven days
A setup session in a customer's account	Four hours
Test messages	Six an hour for the whole account
Standard limits	Tank Critical below 15 percent (Act now), Low below 35 percent (Check today); pump no flow at or below 0.2 litres a minute held for 30 seconds (Act now); battery Low below 25 percent
How alerts reach people	Act now: email at once, text after ten minutes if not acknowledged, the next person at 30 minutes. Check today: one email, held until morning at night. Morning message 07:30. Quiet hours 22:00 to 07:00 hold everything except Act now. Resolved after 15 minutes clear
After install	Day 1, 3 and 7 checks, then the morning check; day 30 call; day 60 decision meeting; day 90 convert or remove within 14 days

```
greecon-edge status
journalctl -u greecon-edge -n 50 --no-pager
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
greecon-edge find
sudo systemctl restart greecon-edge
sudo timedatectl set-ntp true
```